

(45) **Date of Patent:** Sep. 2, 2025

- 20 Claims, 46 Drawing Sheets**



(56)

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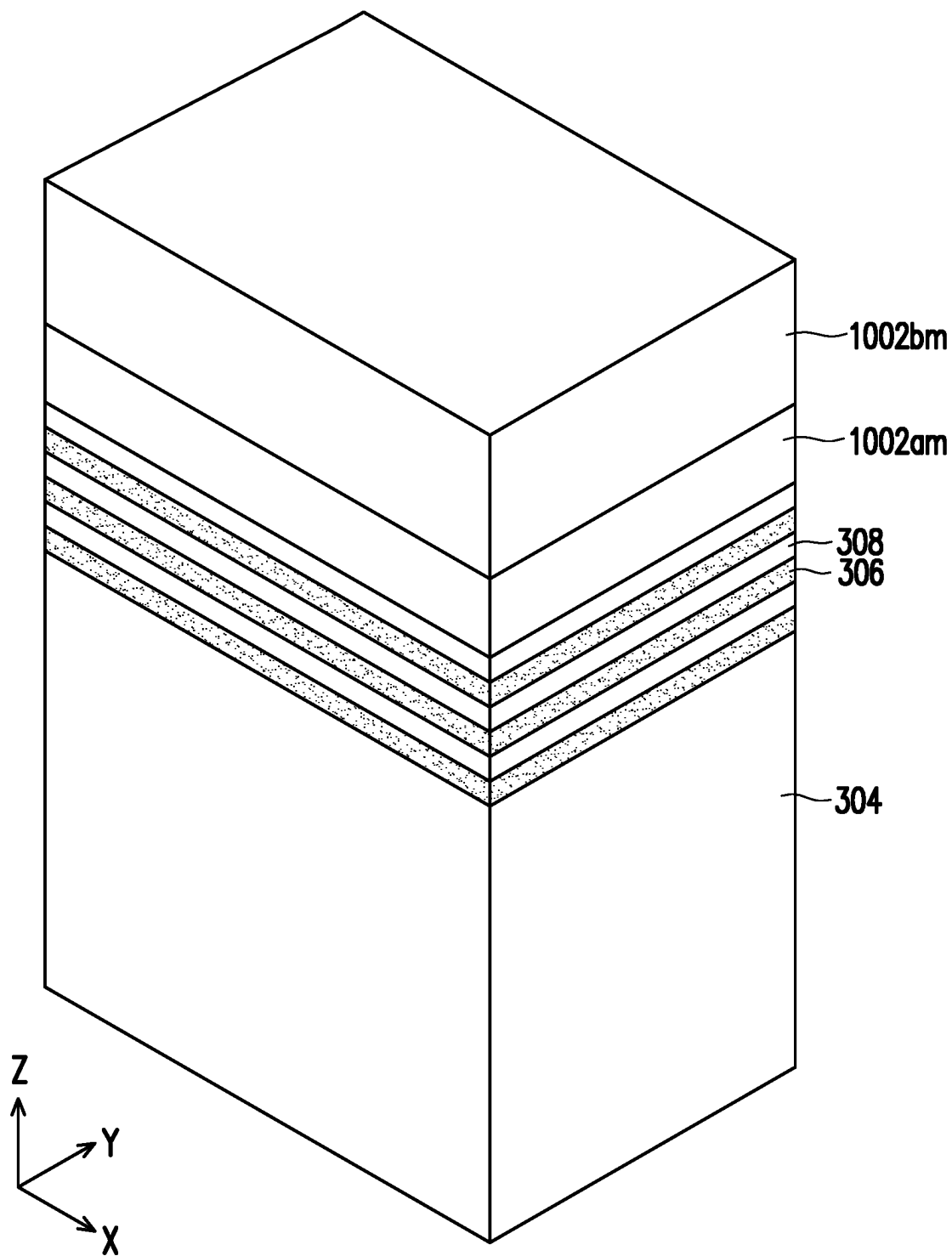


FIG. 1

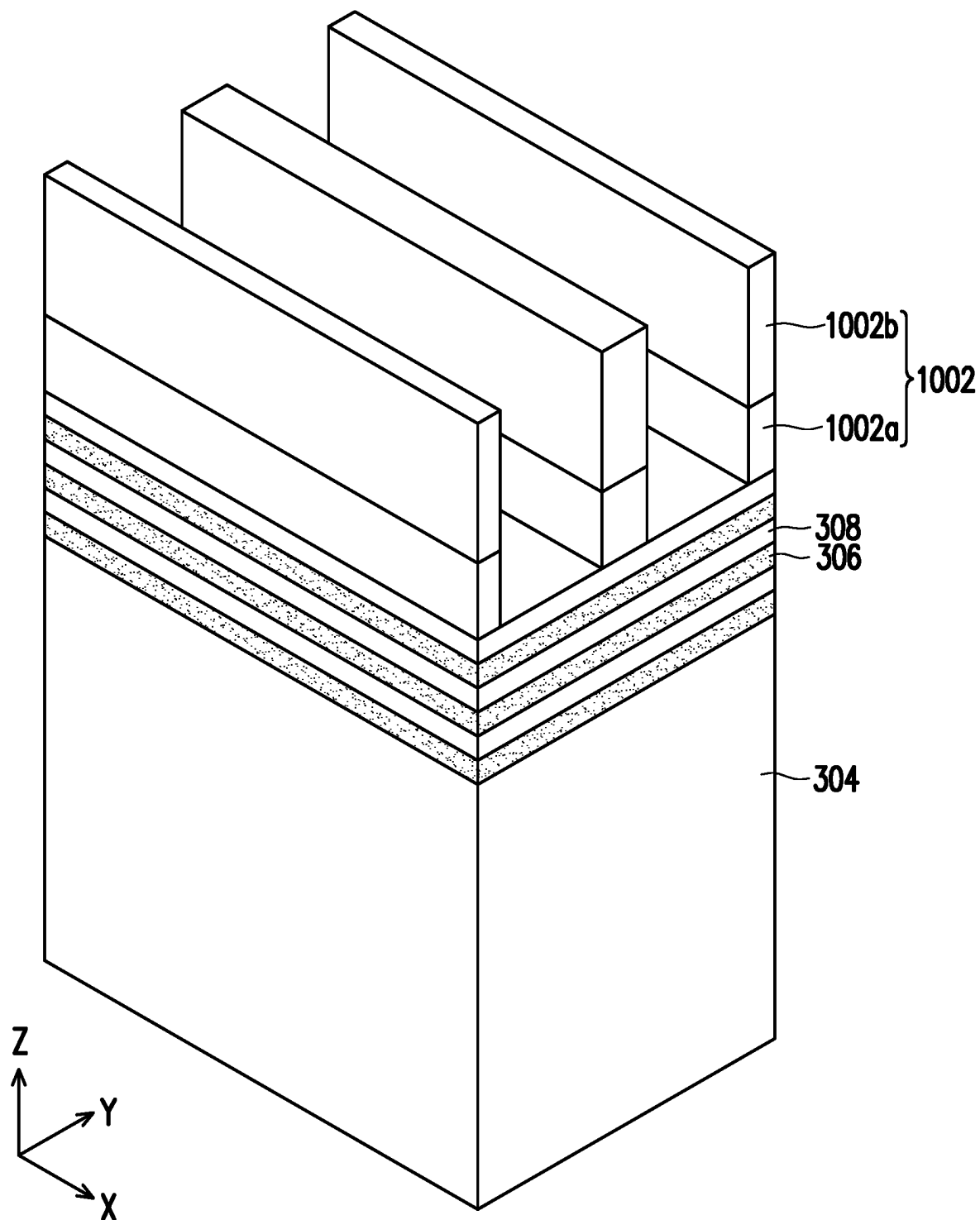


FIG. 2

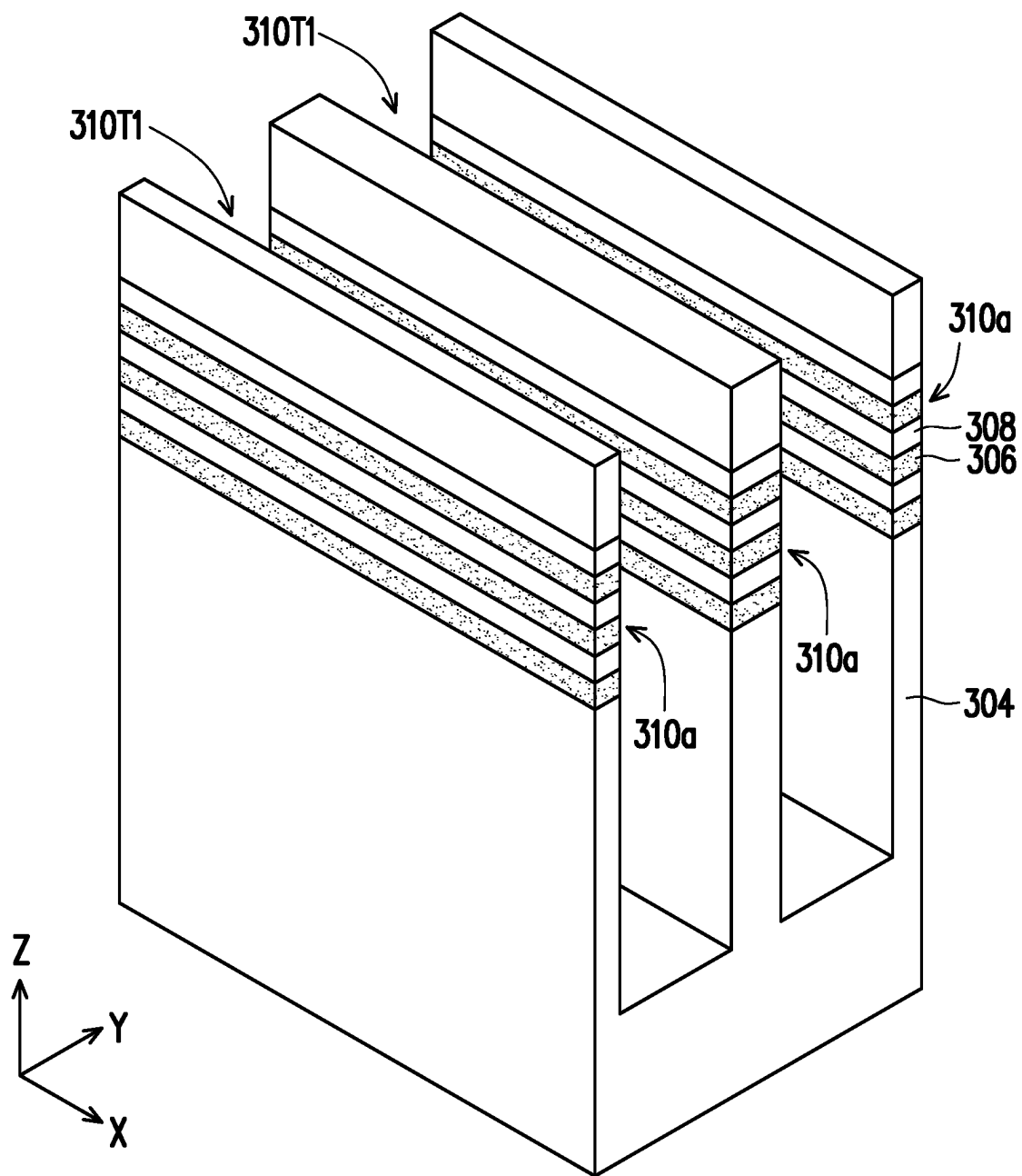


FIG. 3

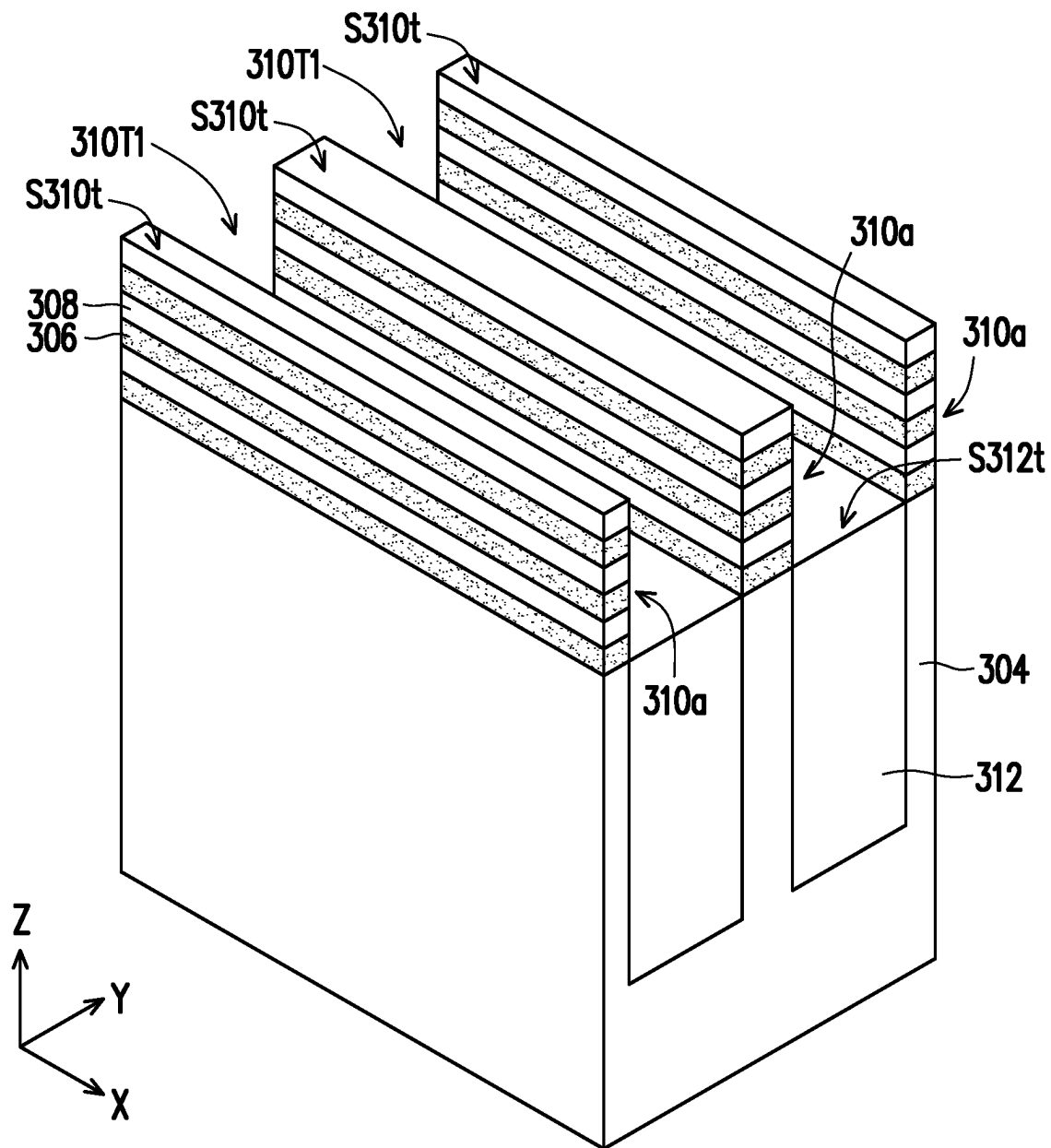


FIG. 4

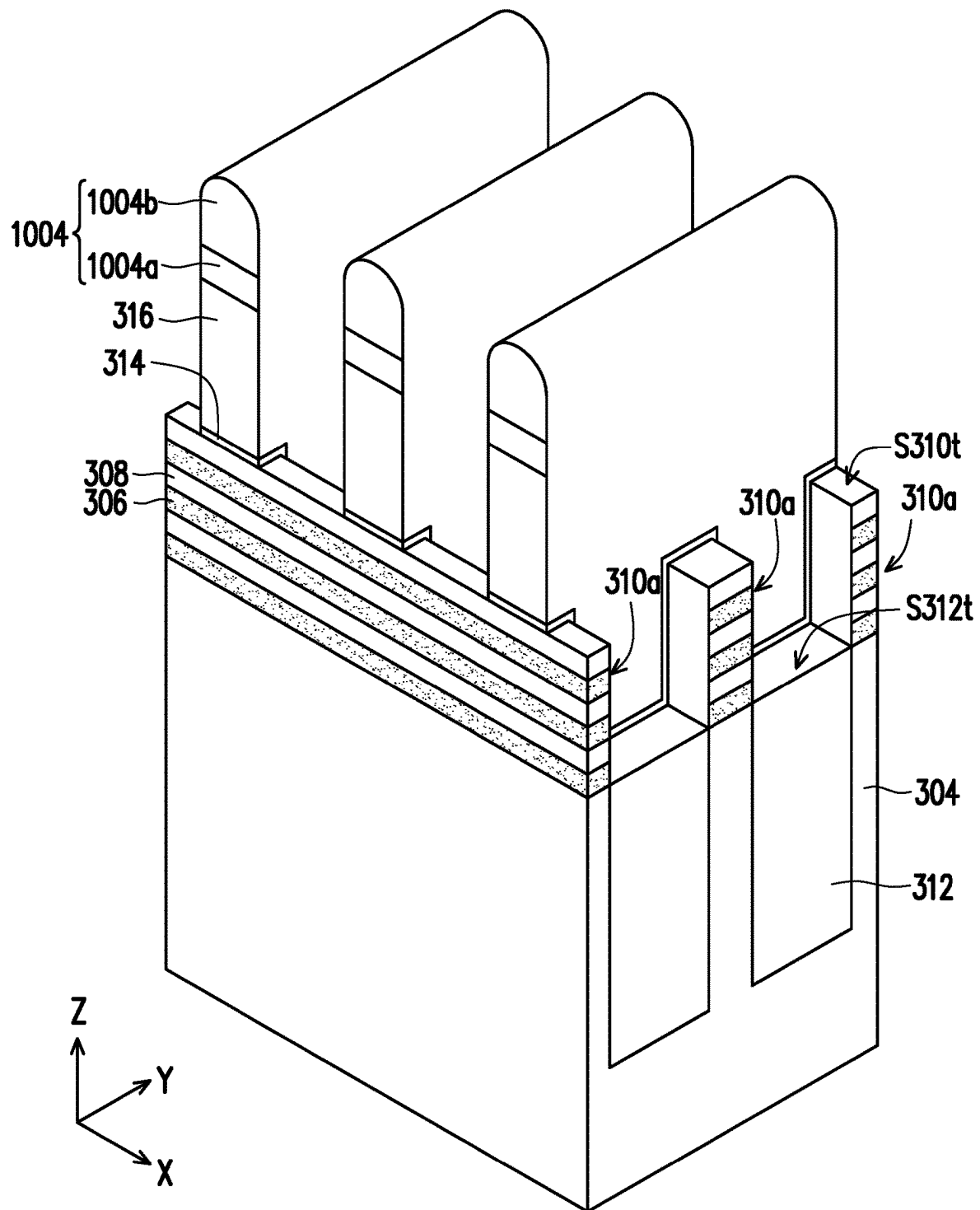


FIG. 5

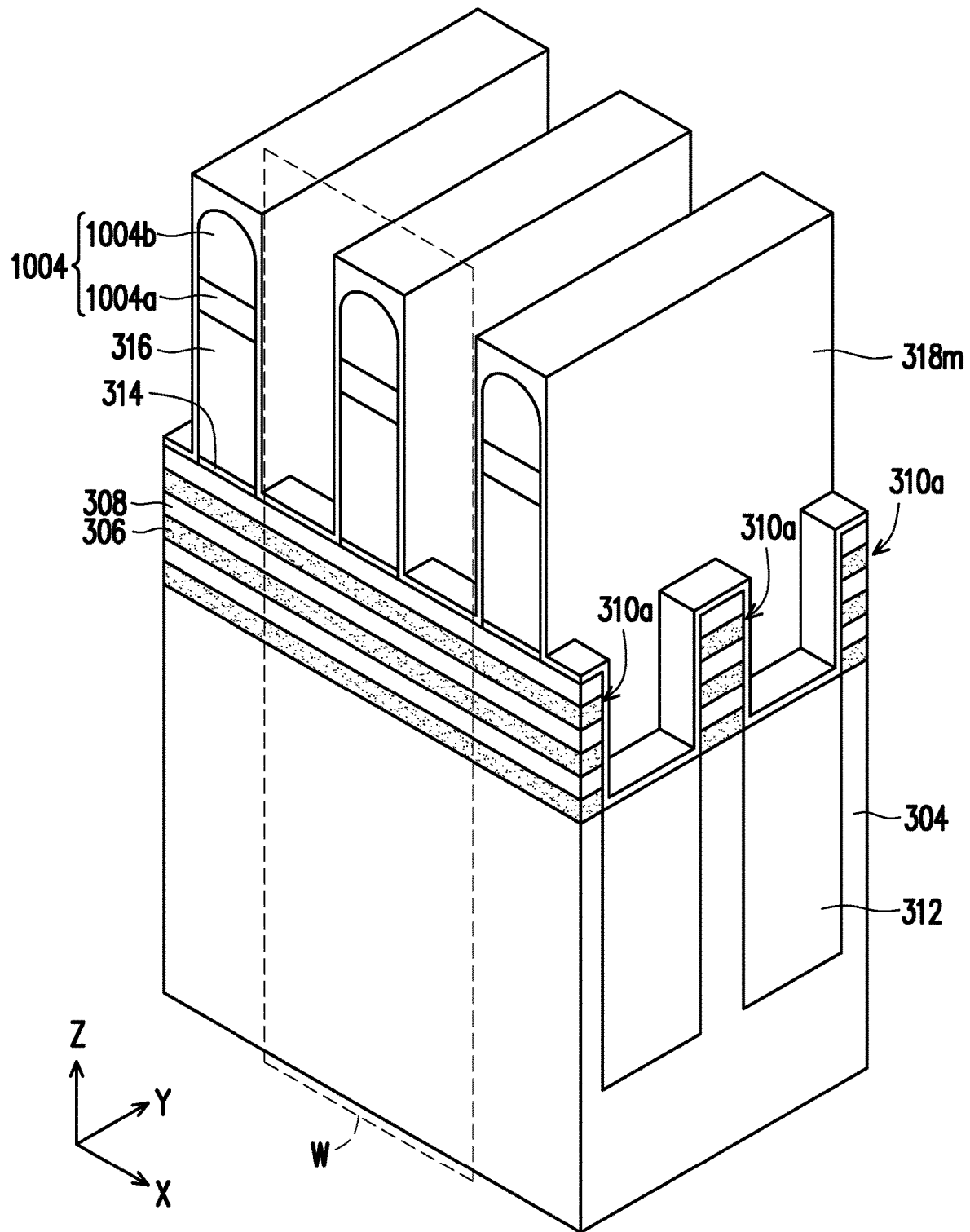


FIG. 6A

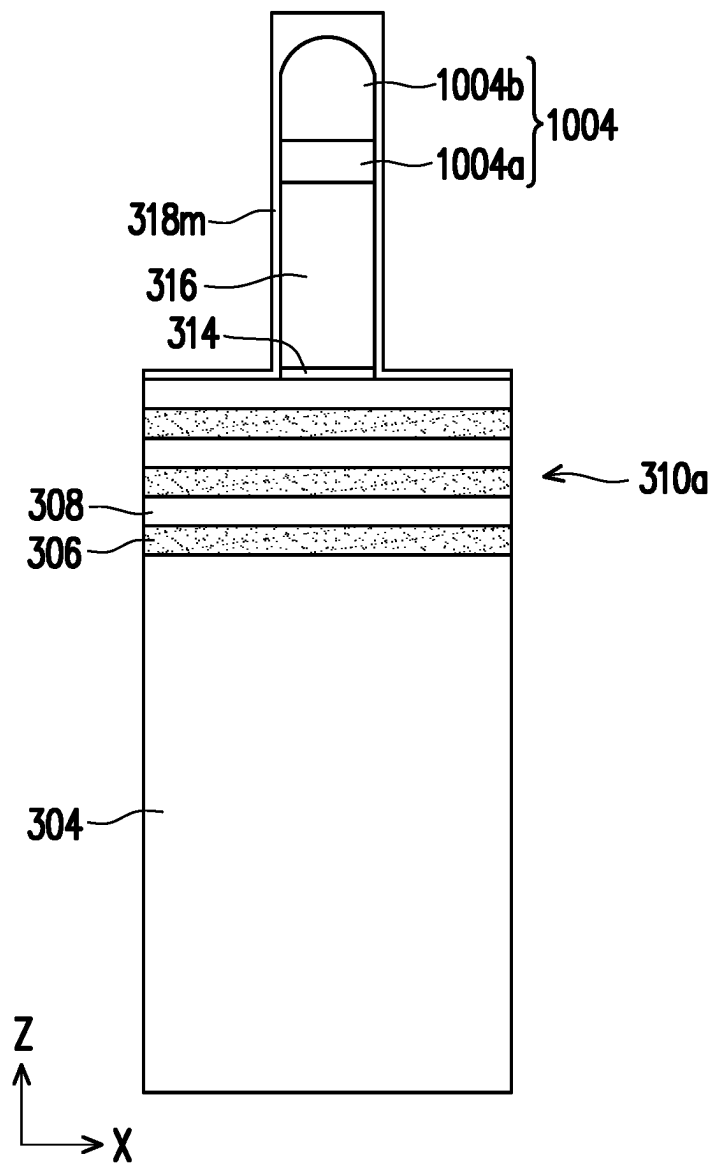


FIG. 6B

FIG. 7A

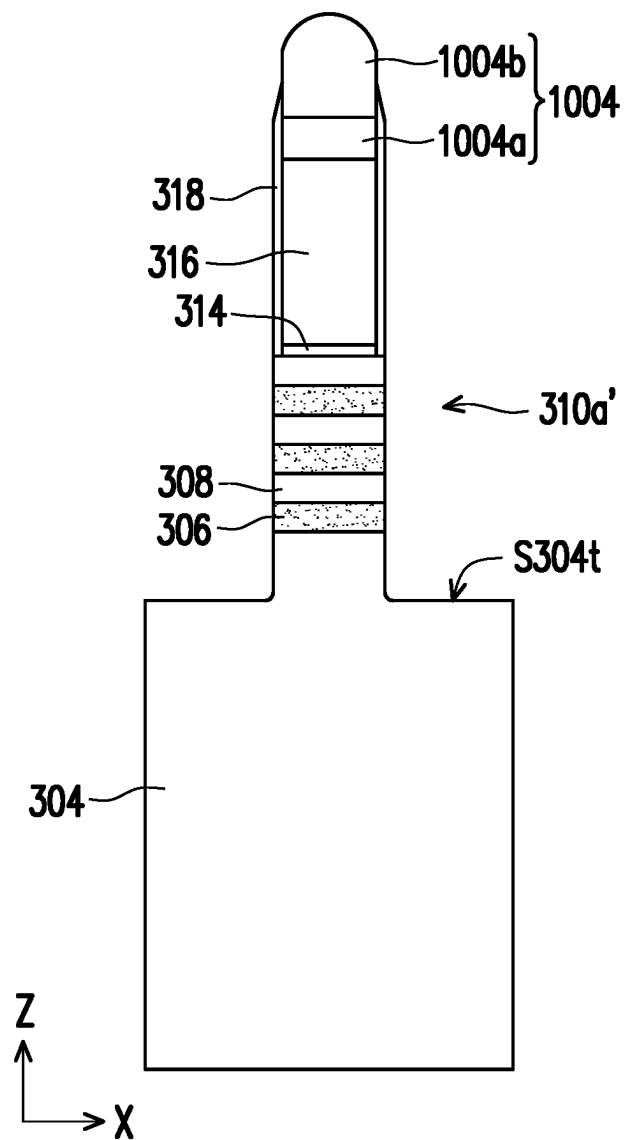


FIG. 7B

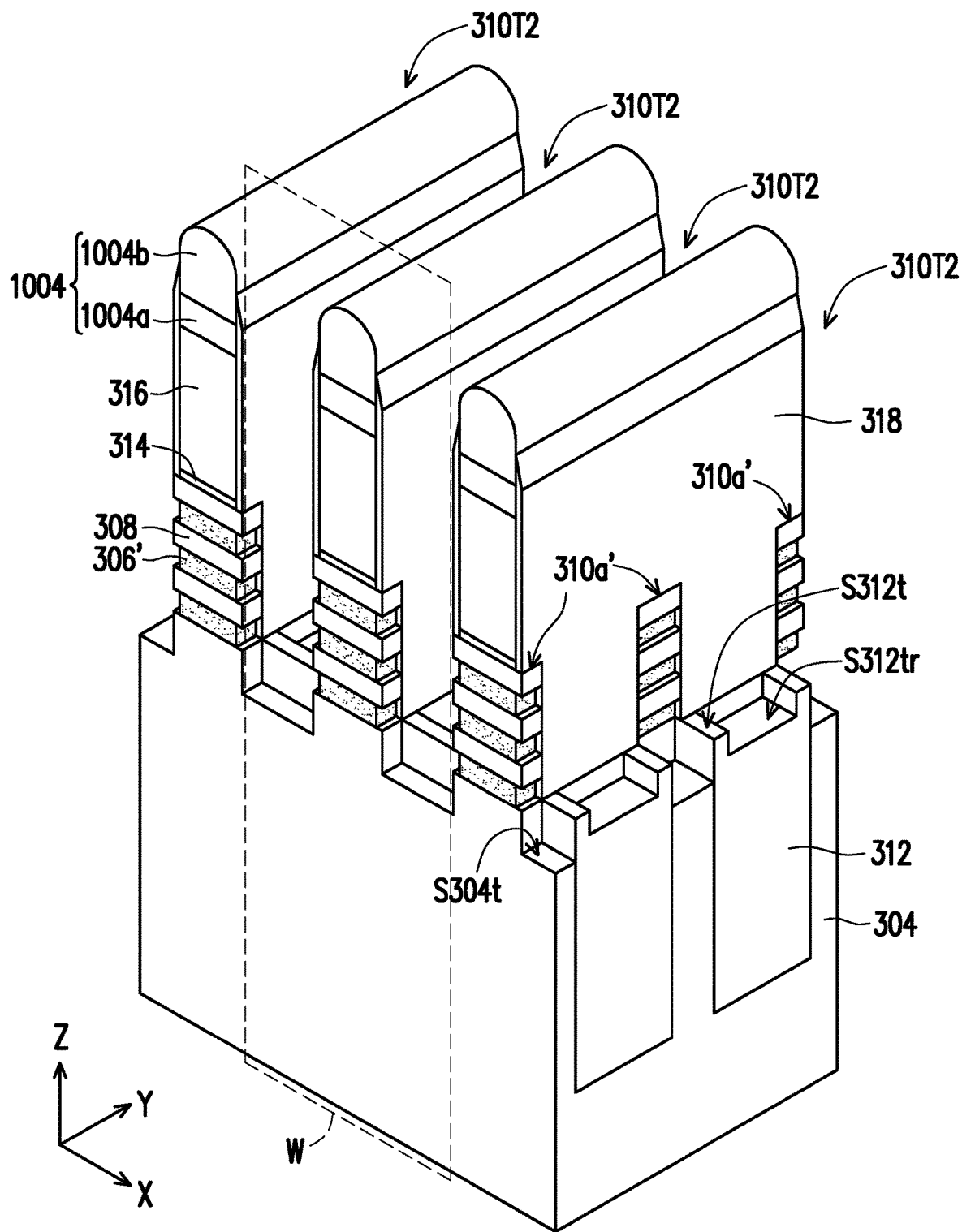


FIG. 8A

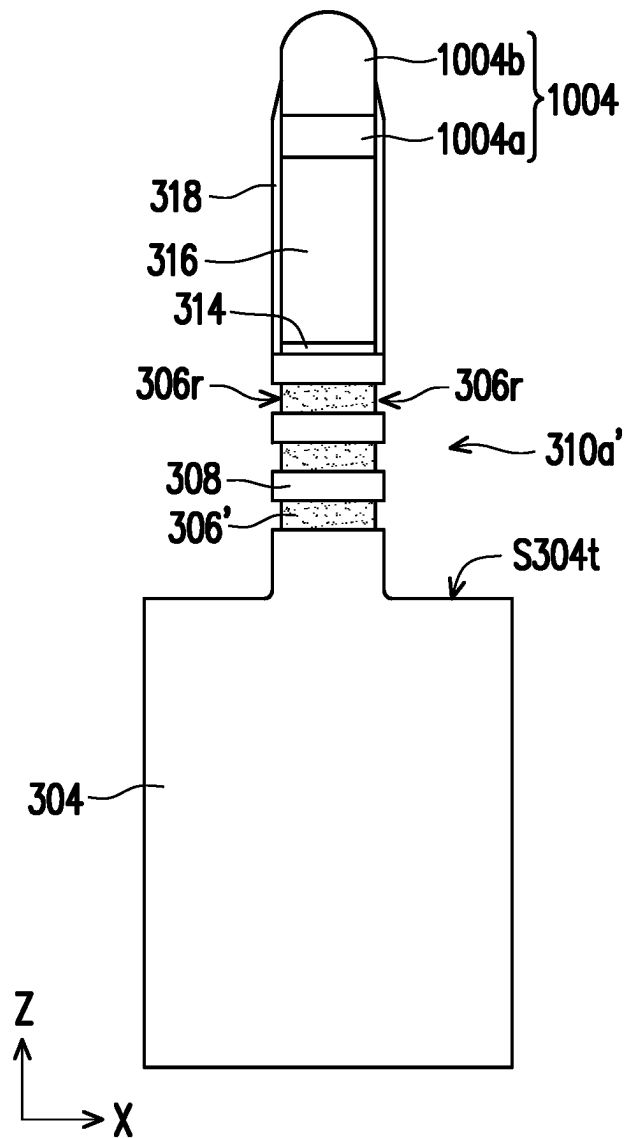


FIG. 8B

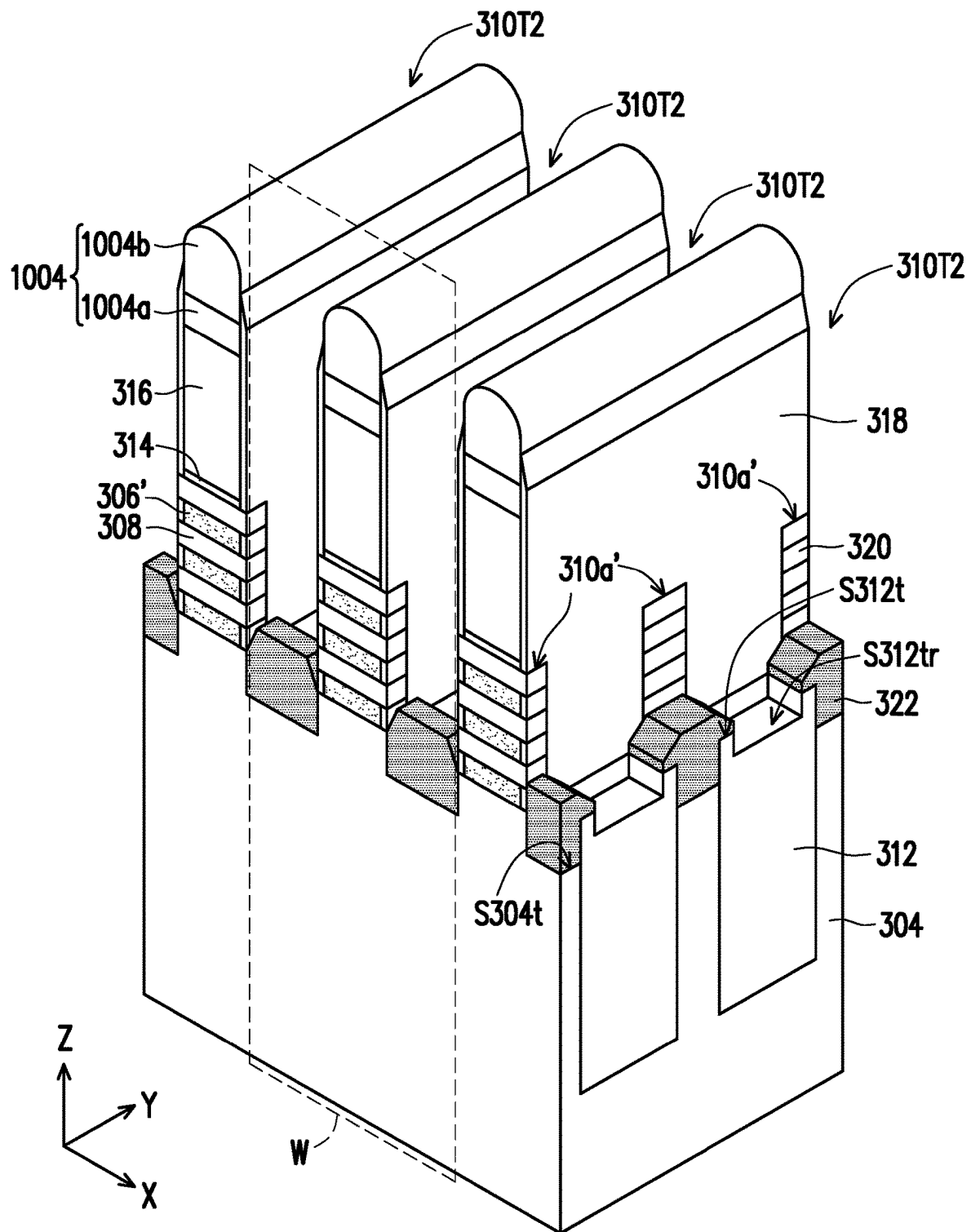


FIG. 9A

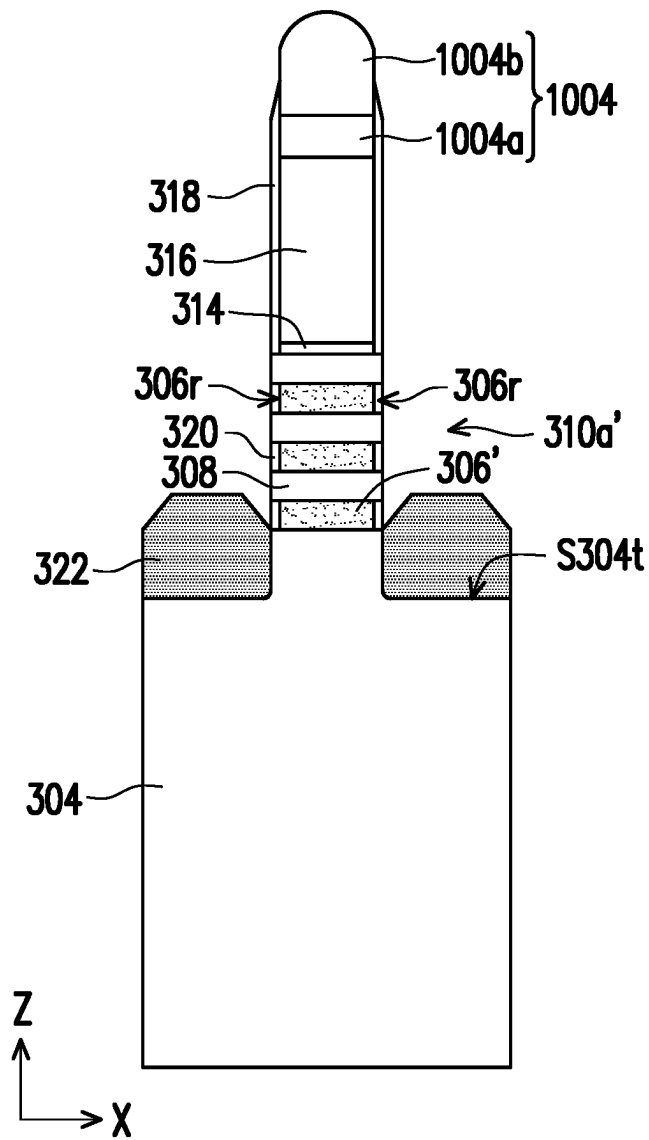


FIG. 9B

FIG. 10

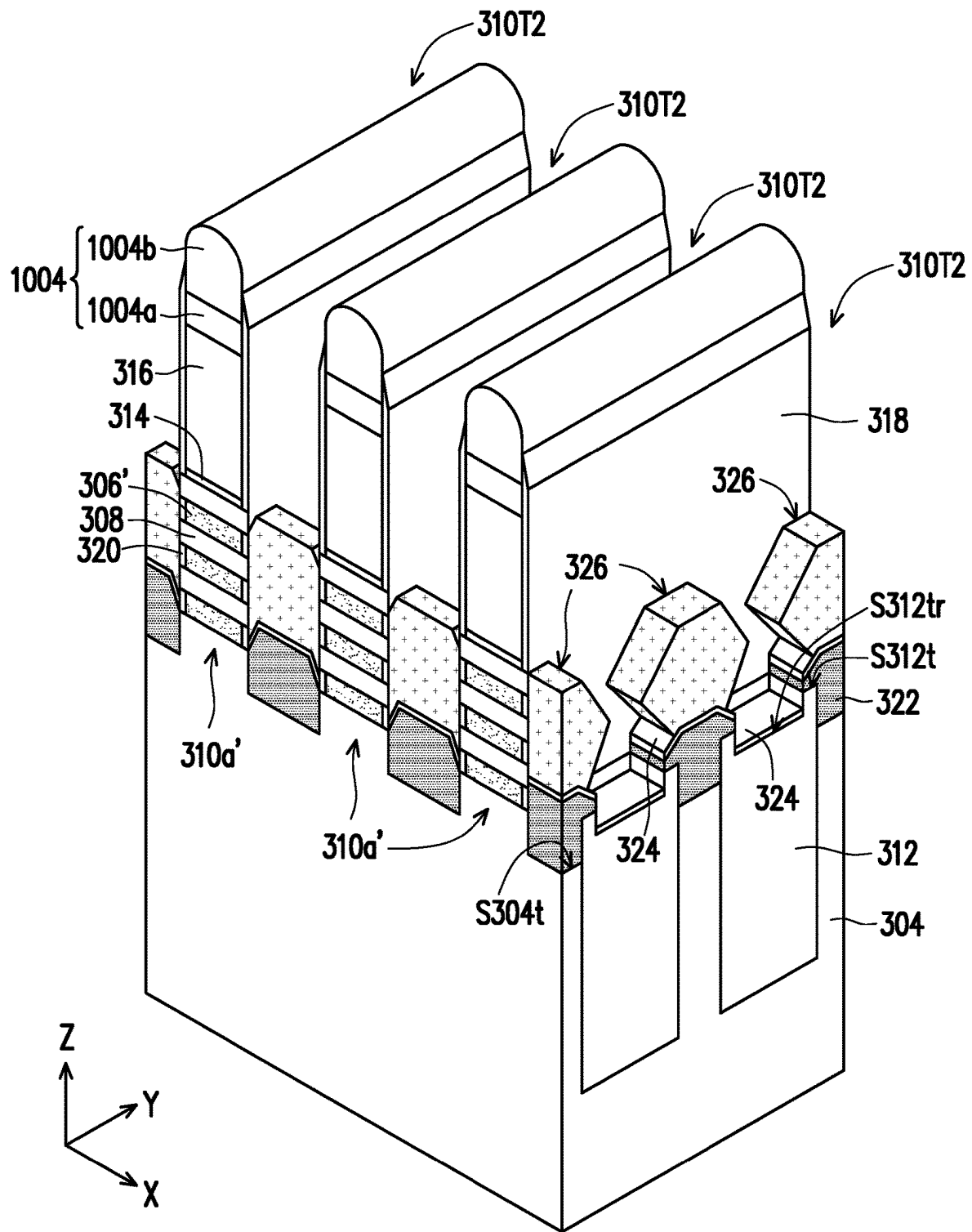


FIG. 11

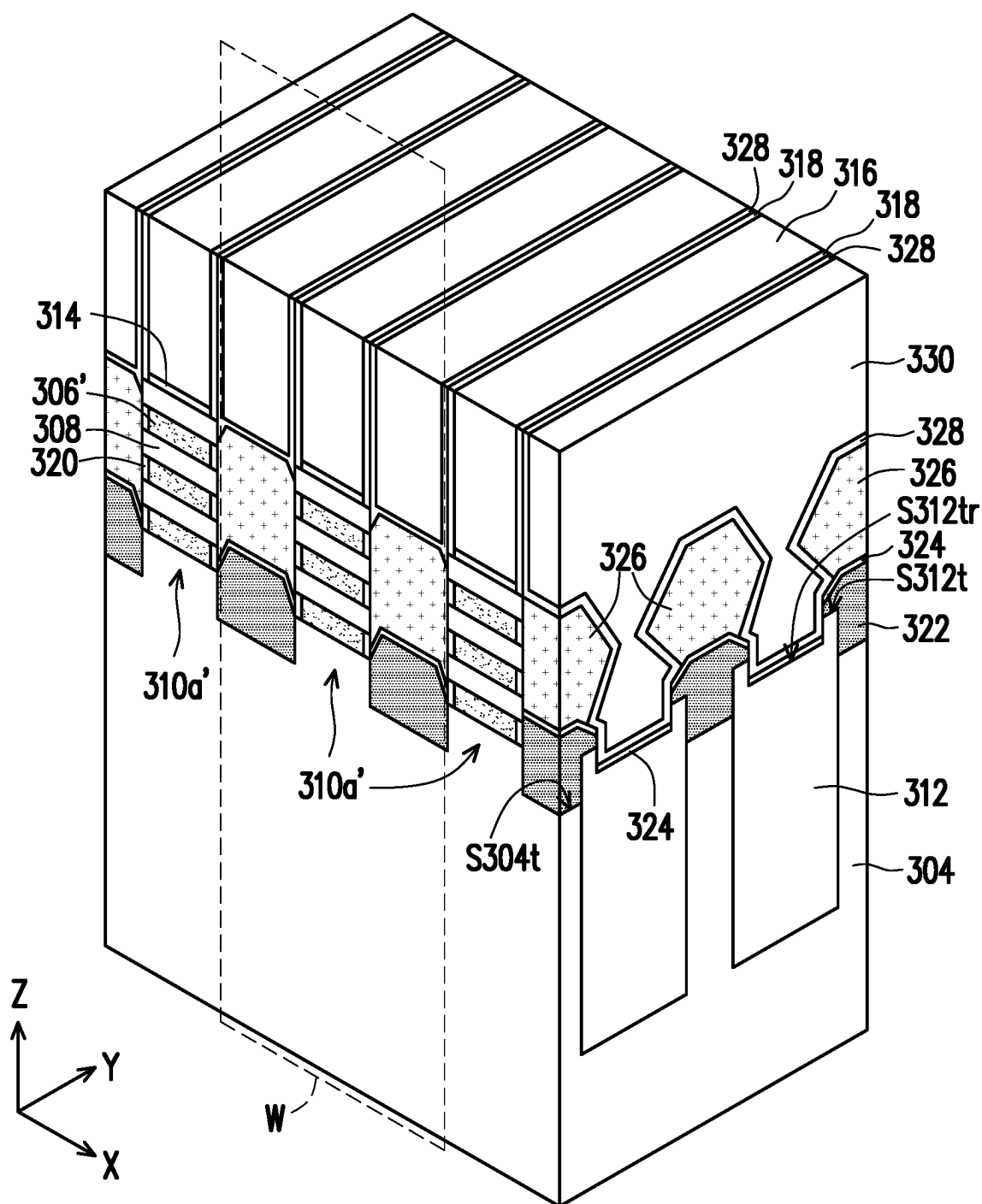


FIG. 12A

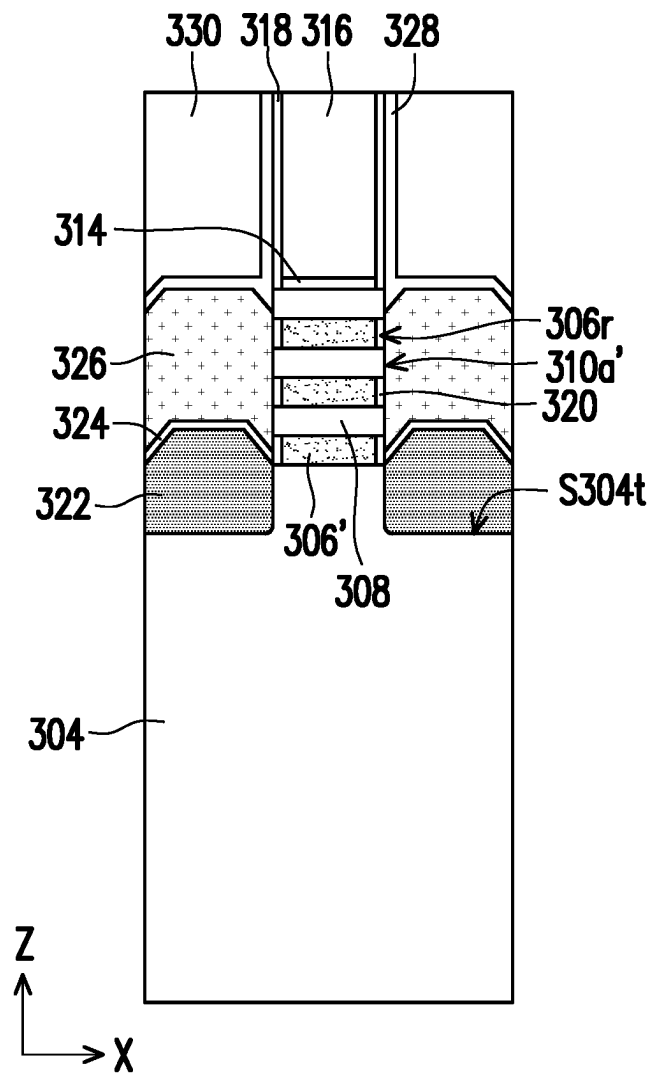


FIG. 12B

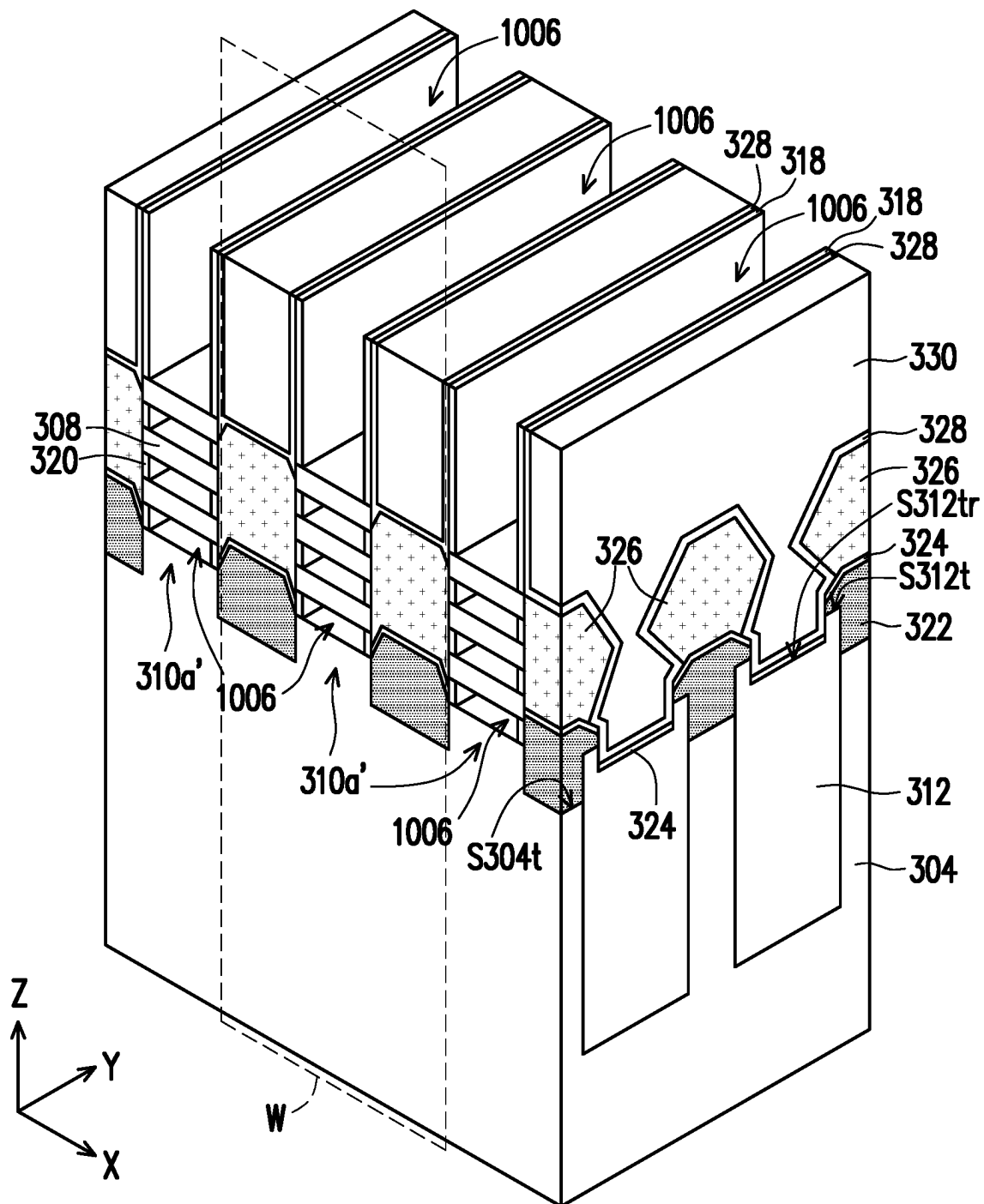


FIG. 13A

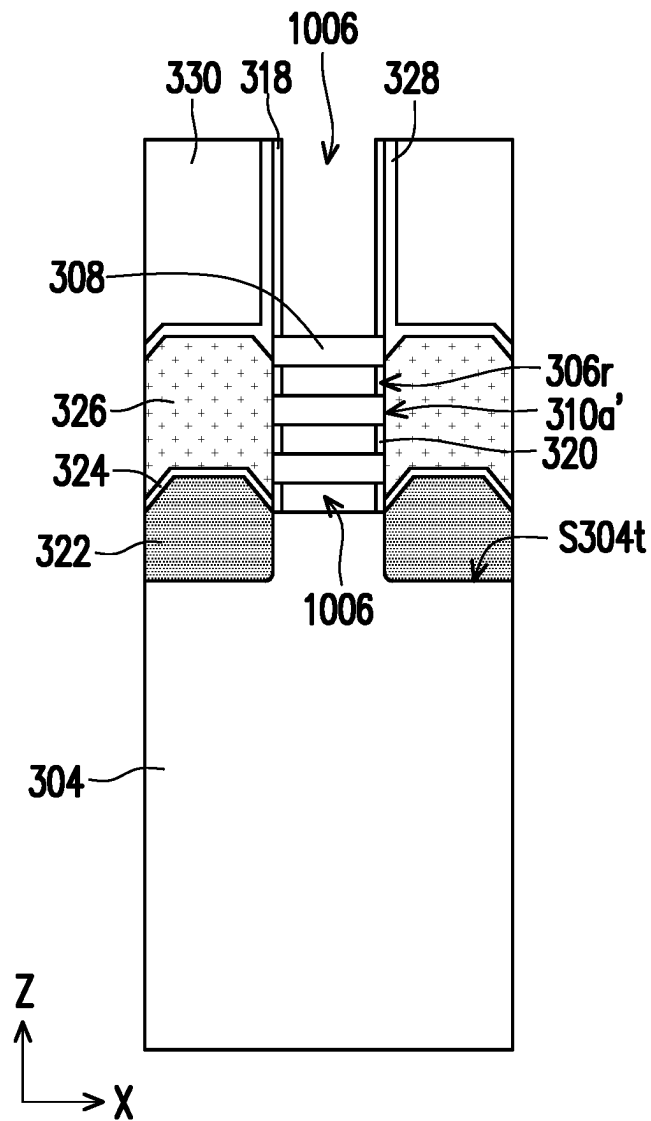


FIG. 13B

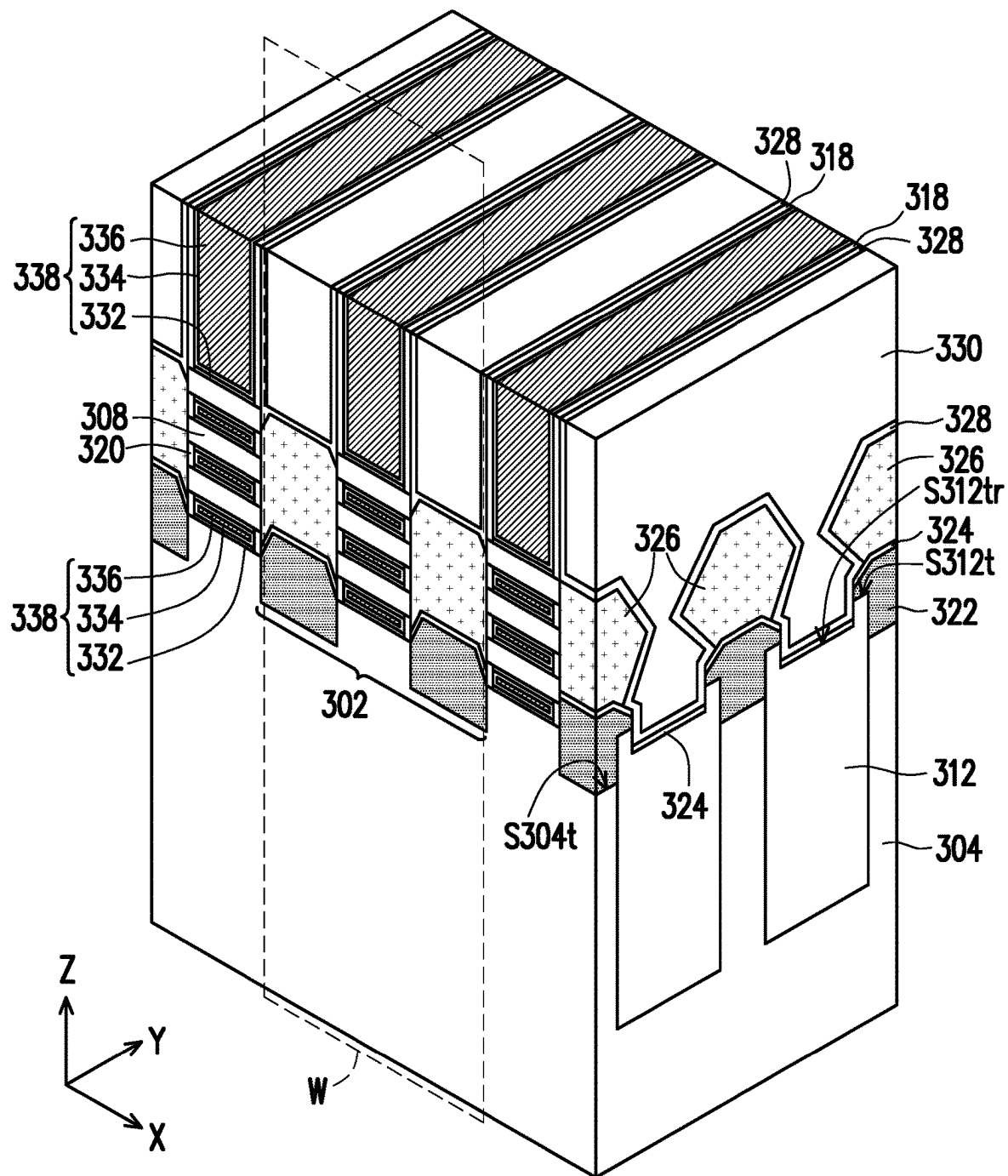


FIG. 14A

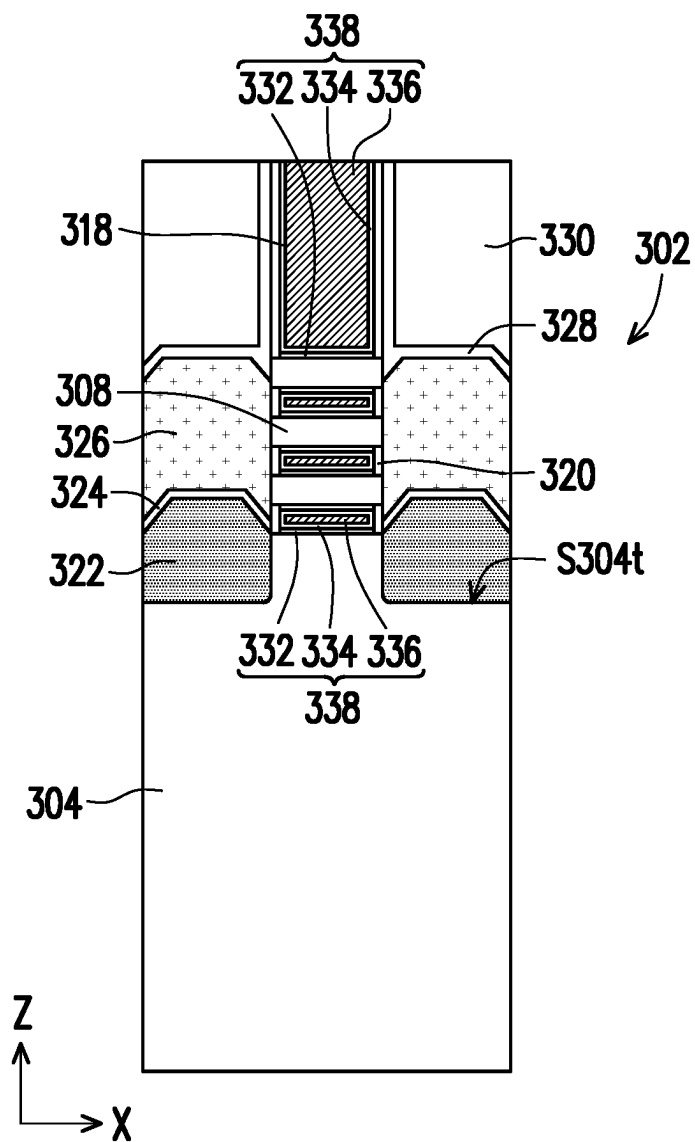


FIG. 14B

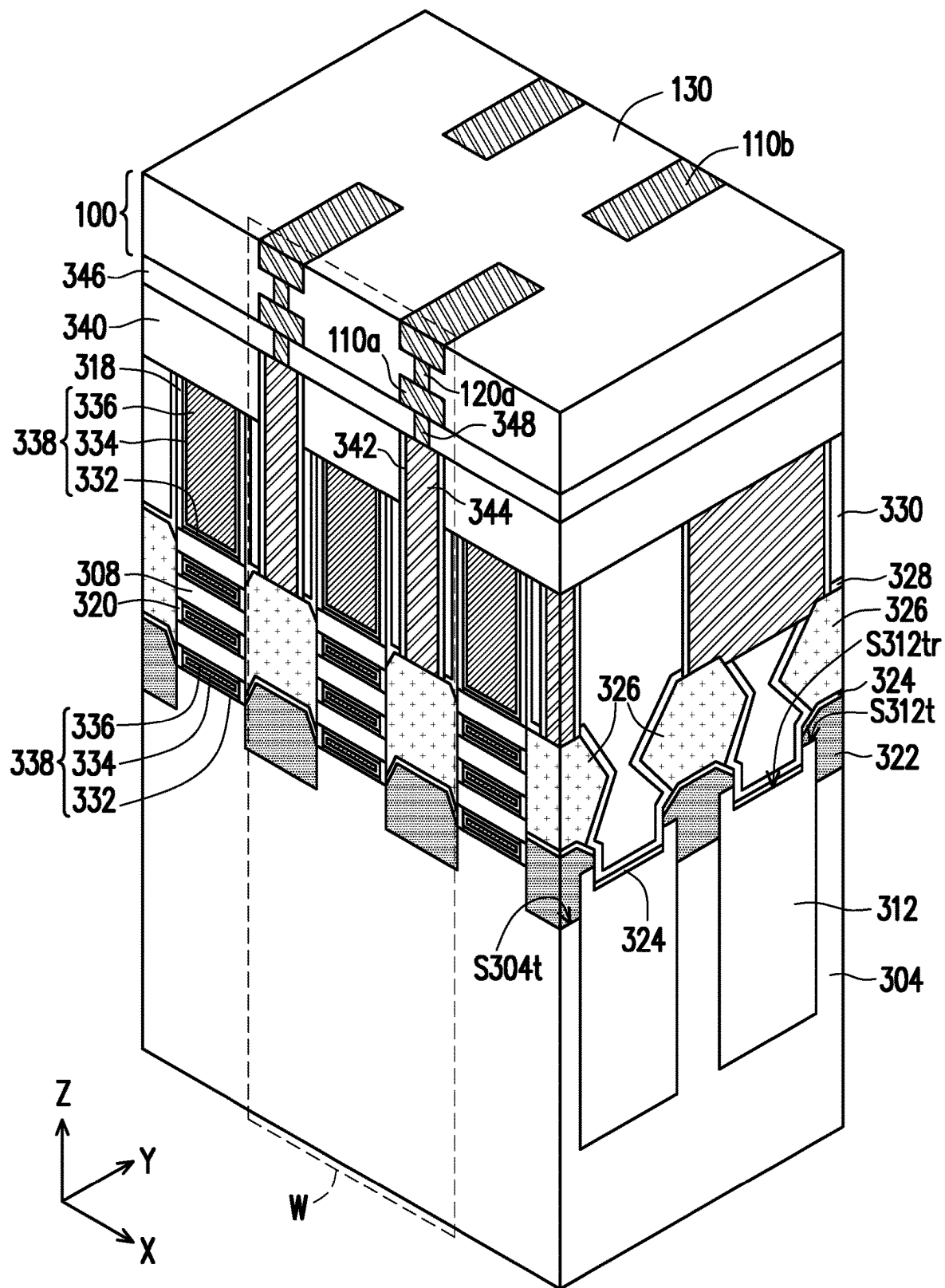


FIG. 15A

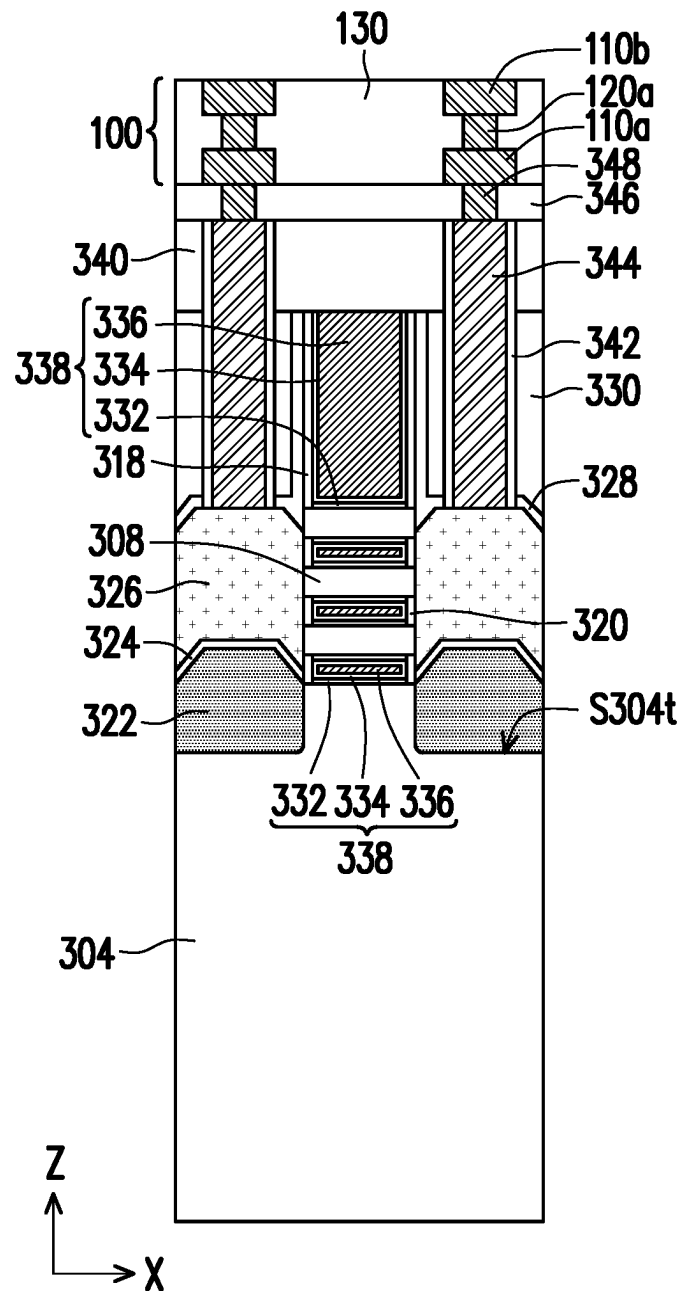


FIG. 15B

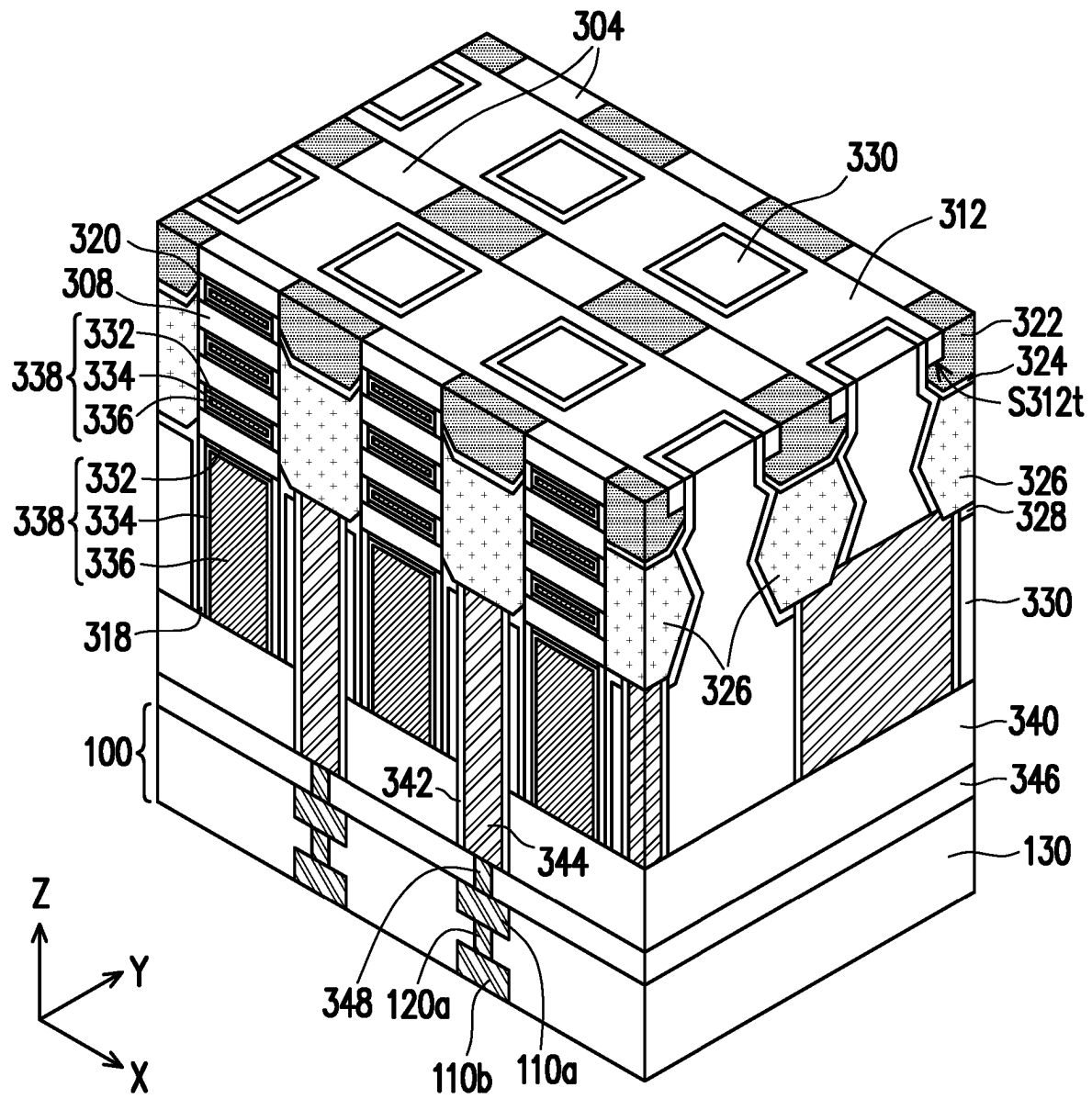


FIG. 16

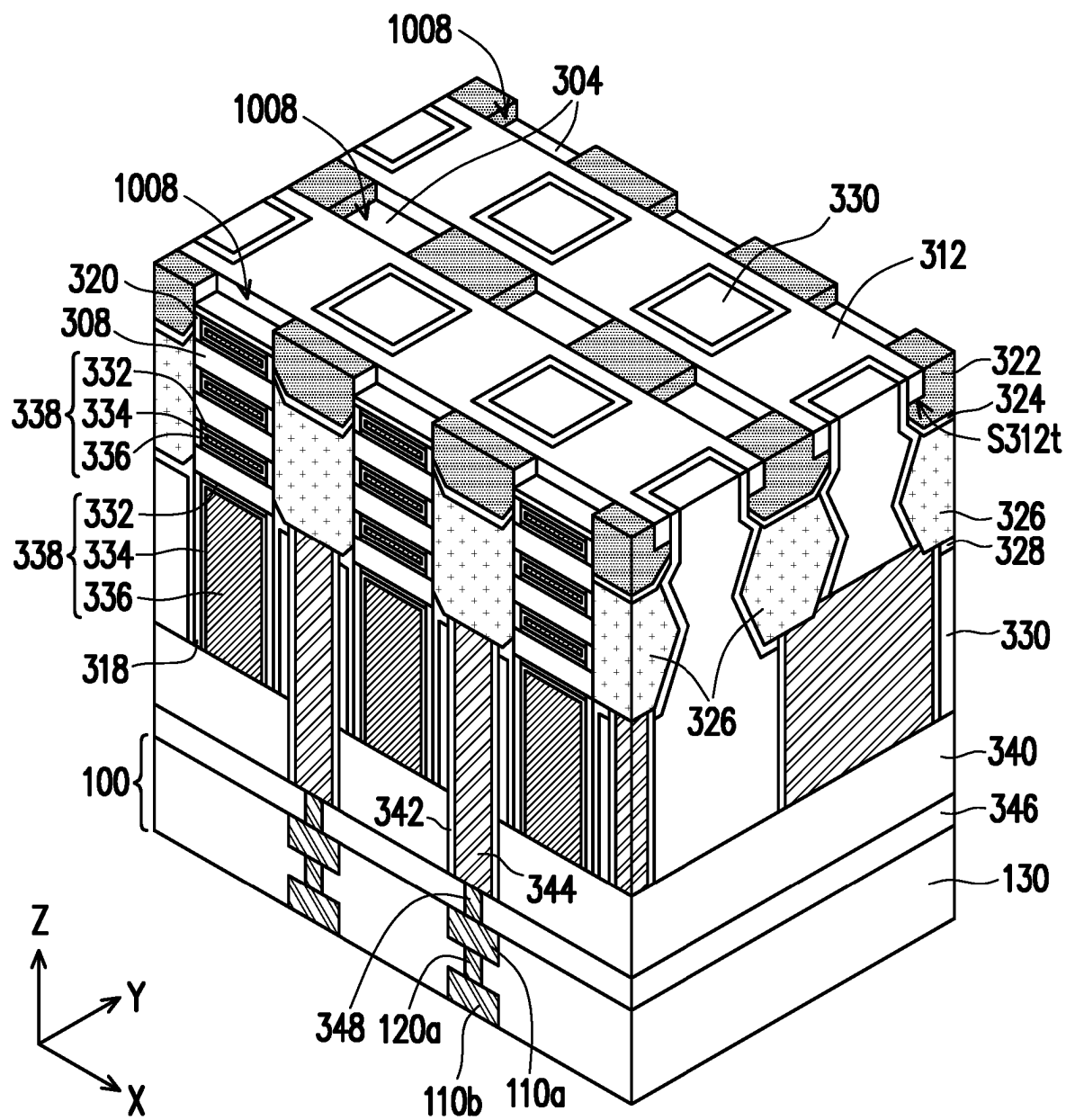


FIG. 17

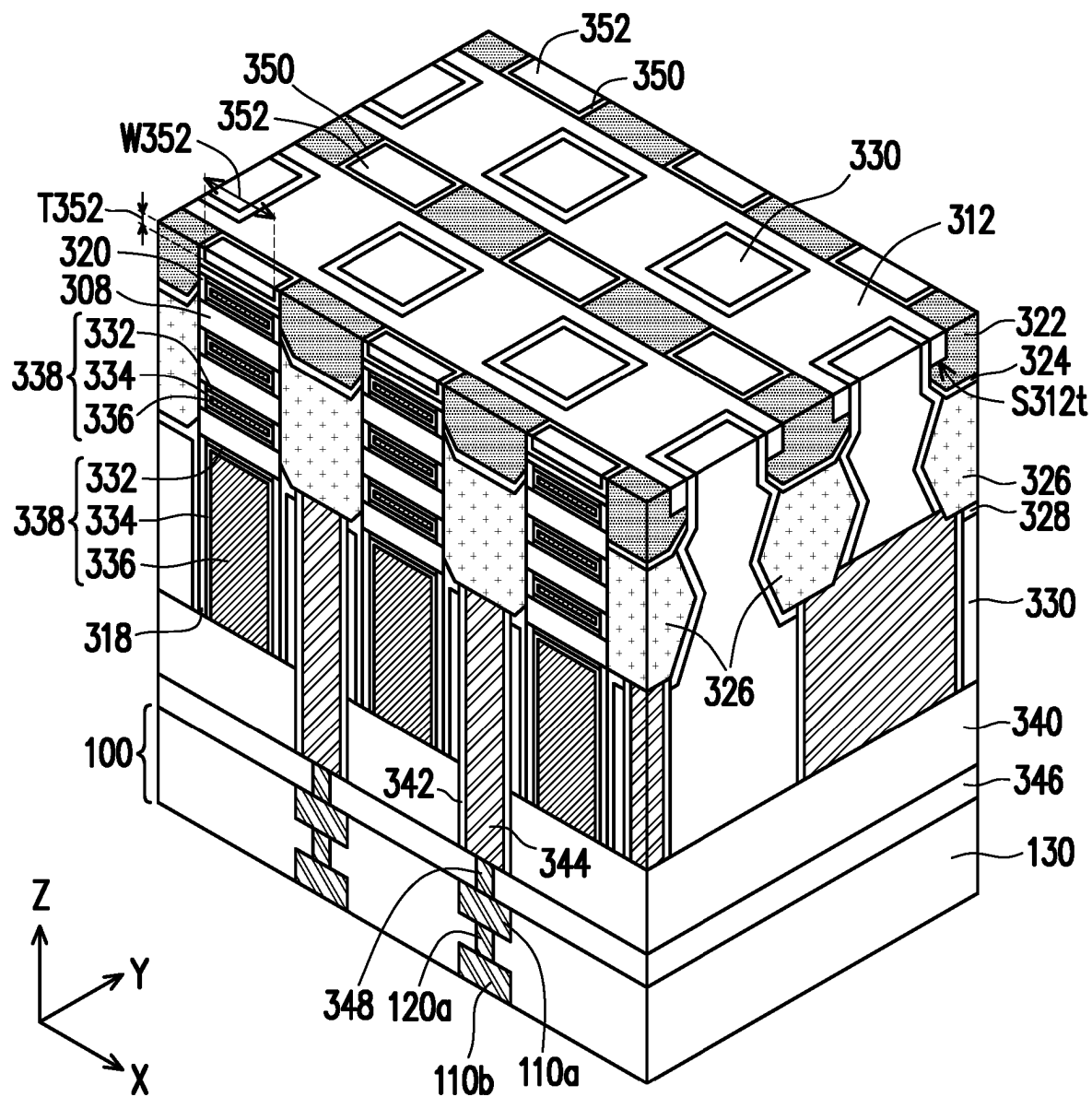


FIG. 18

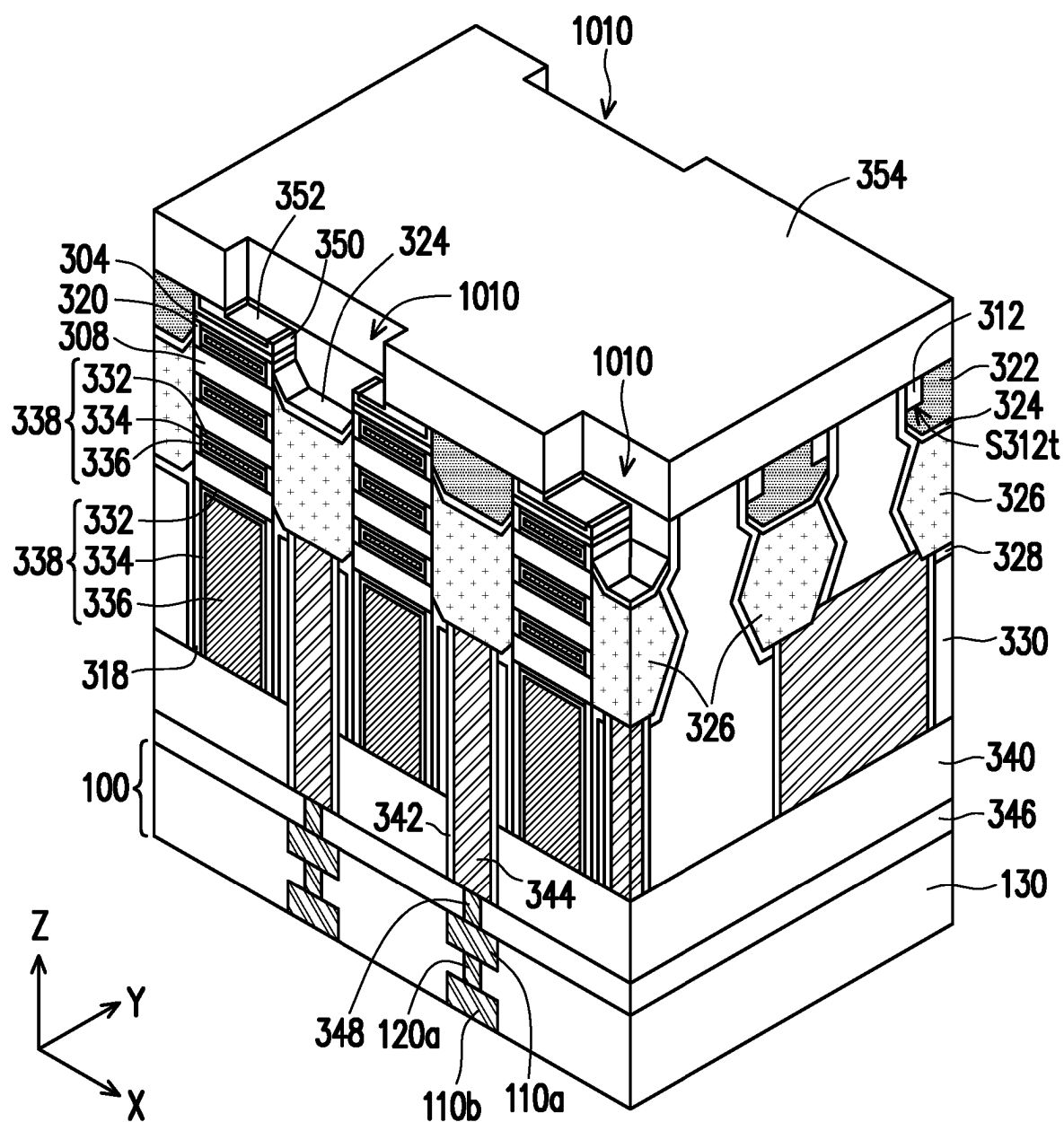


FIG. 19

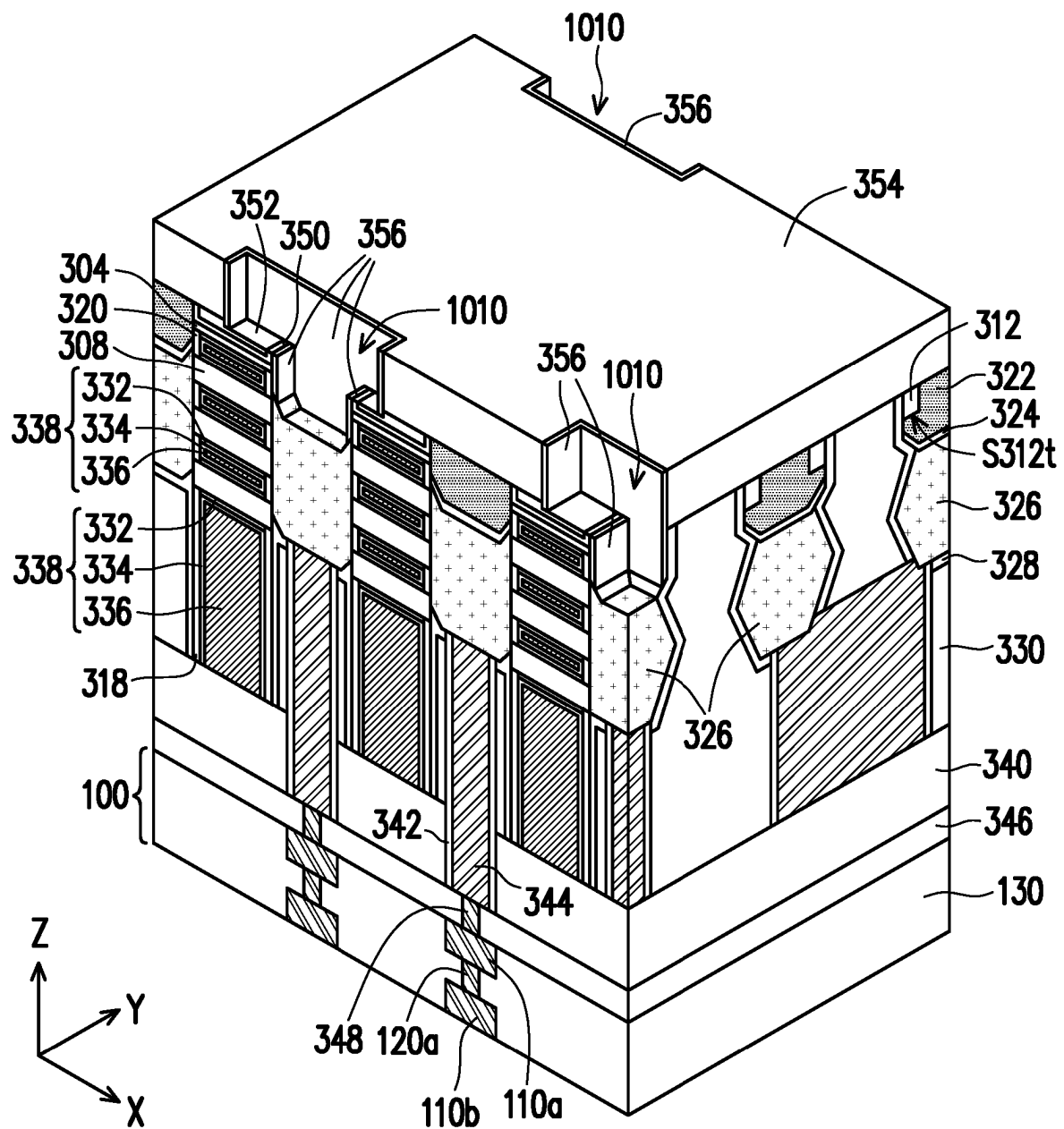


FIG. 20

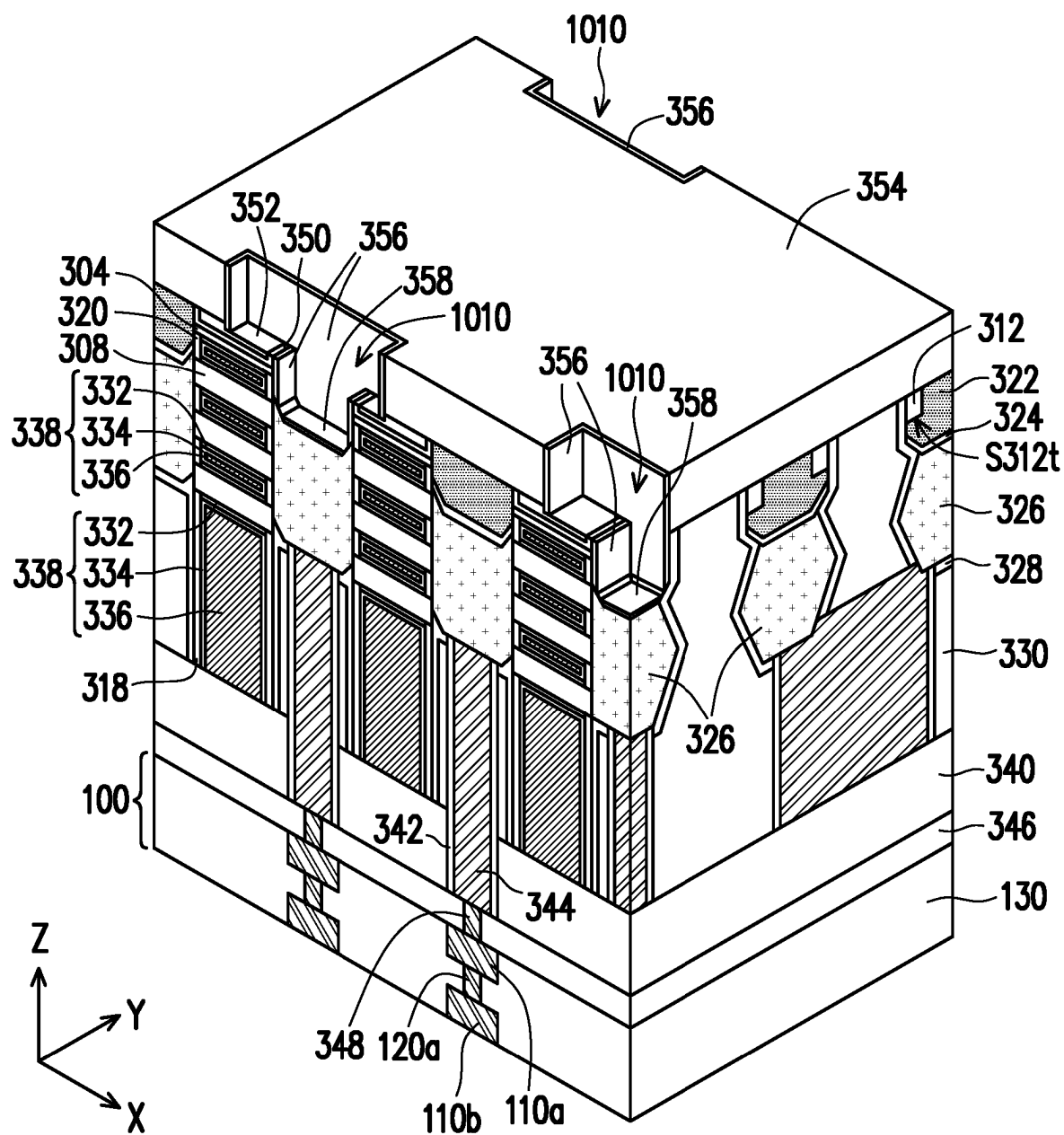


FIG. 21

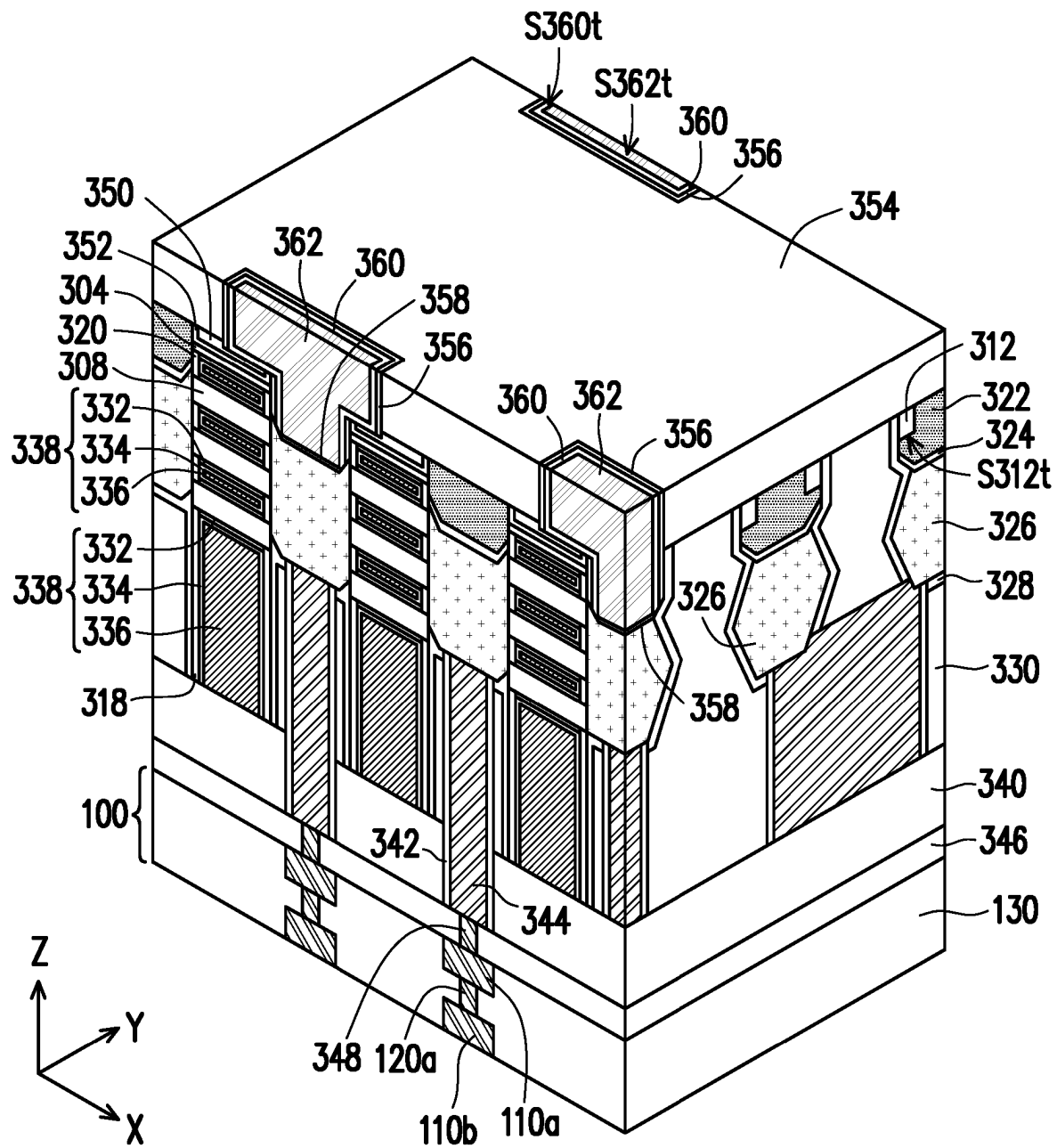


FIG. 22

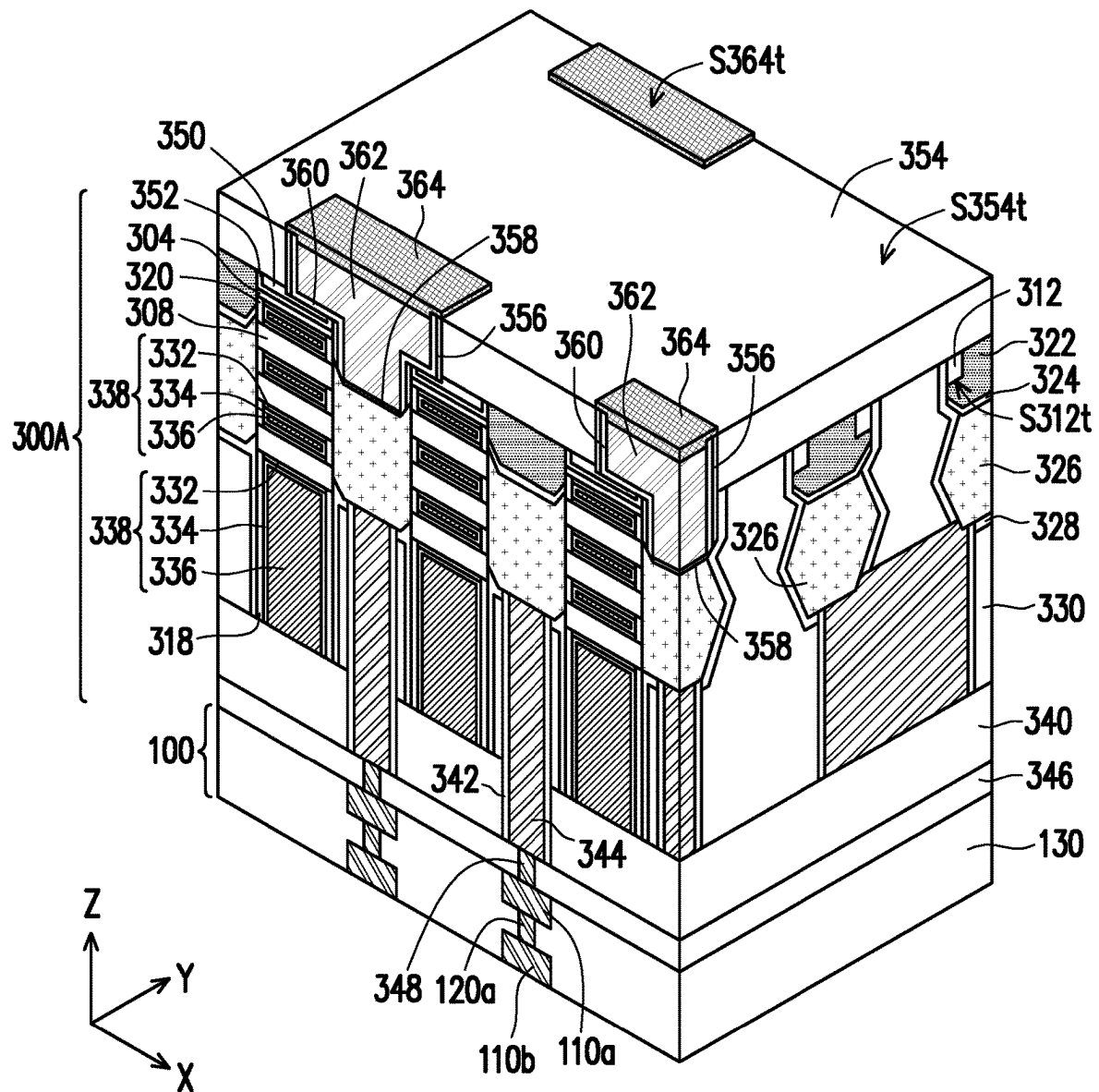


FIG. 23

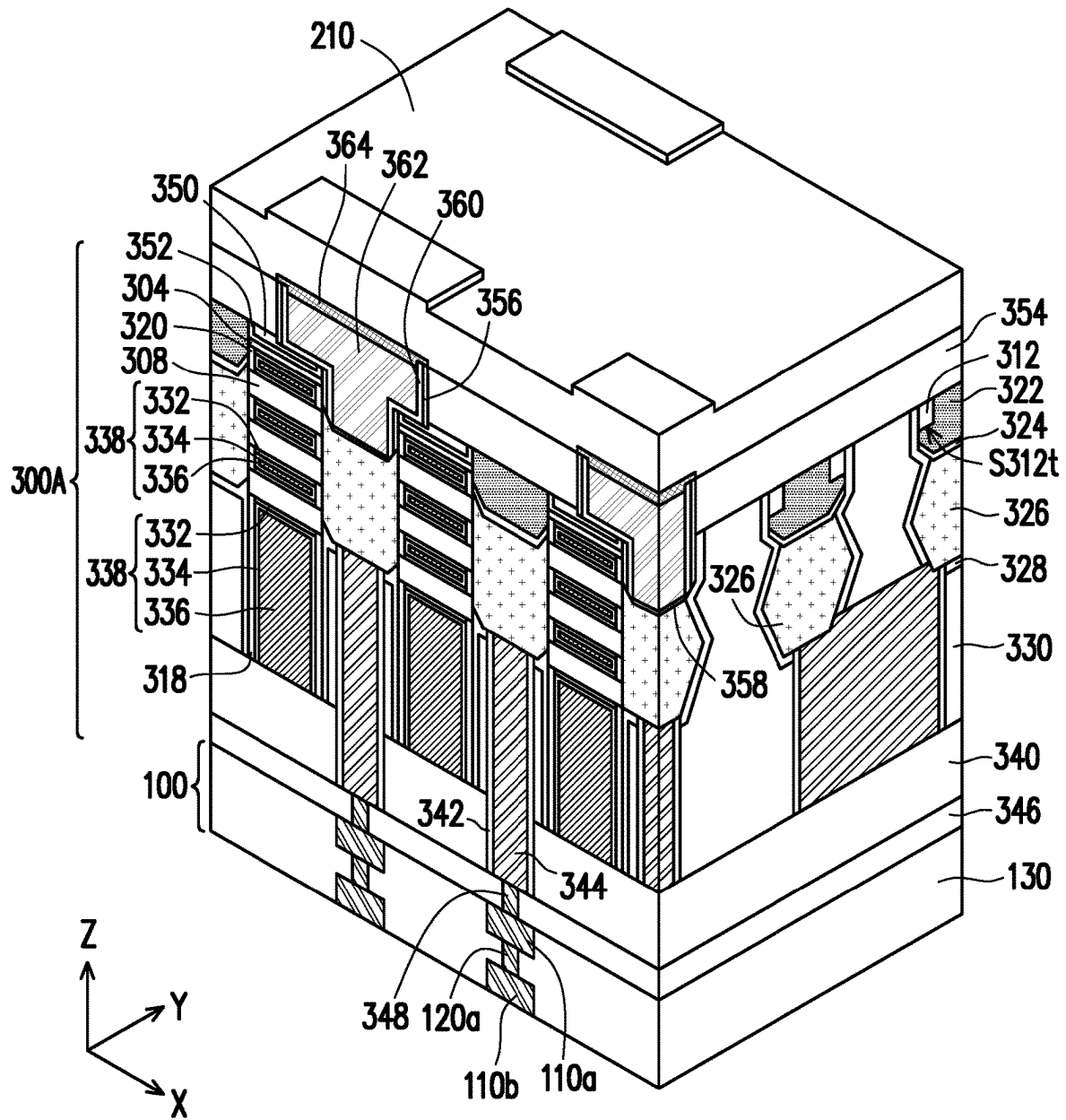


FIG. 24

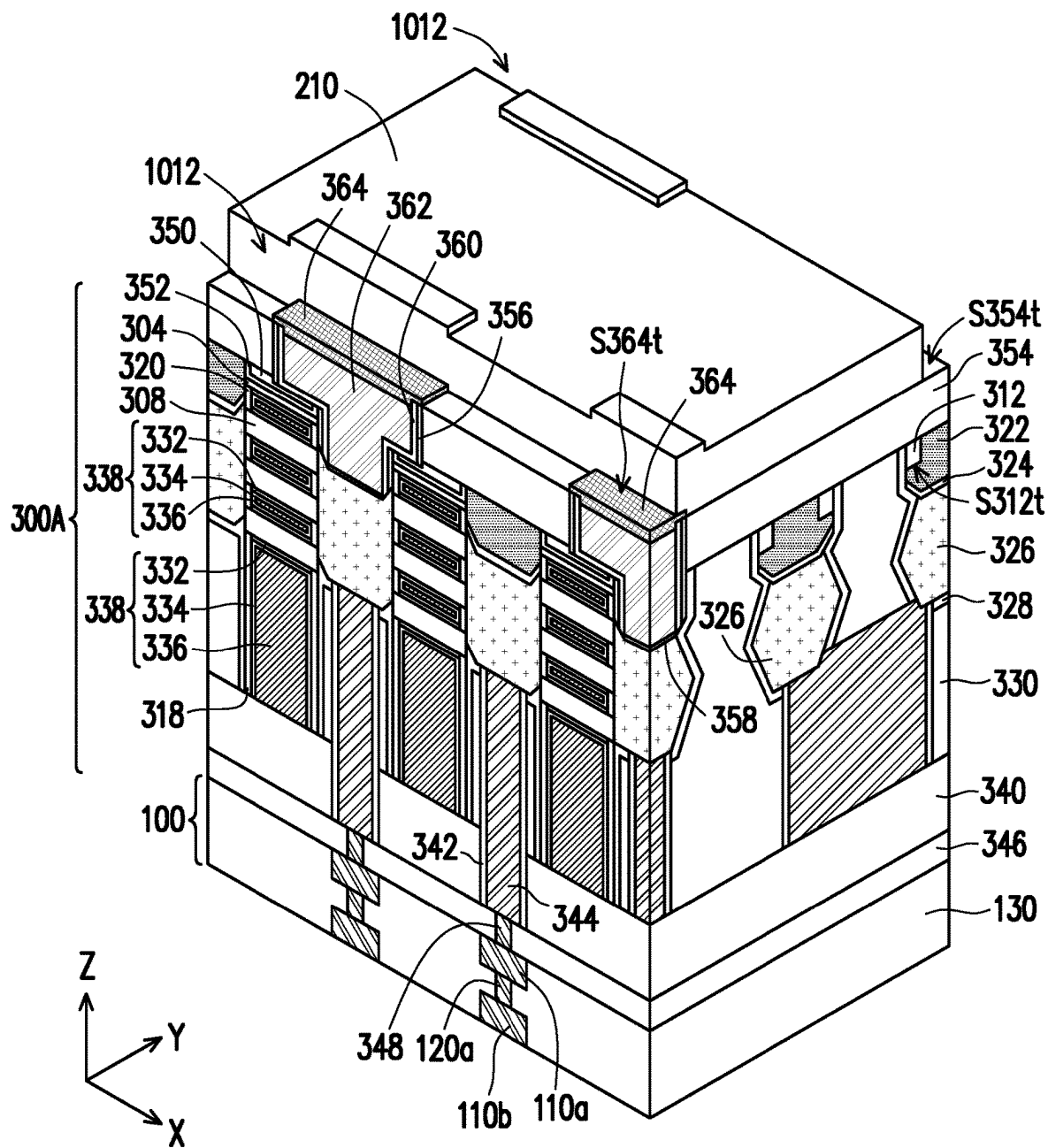


FIG. 25

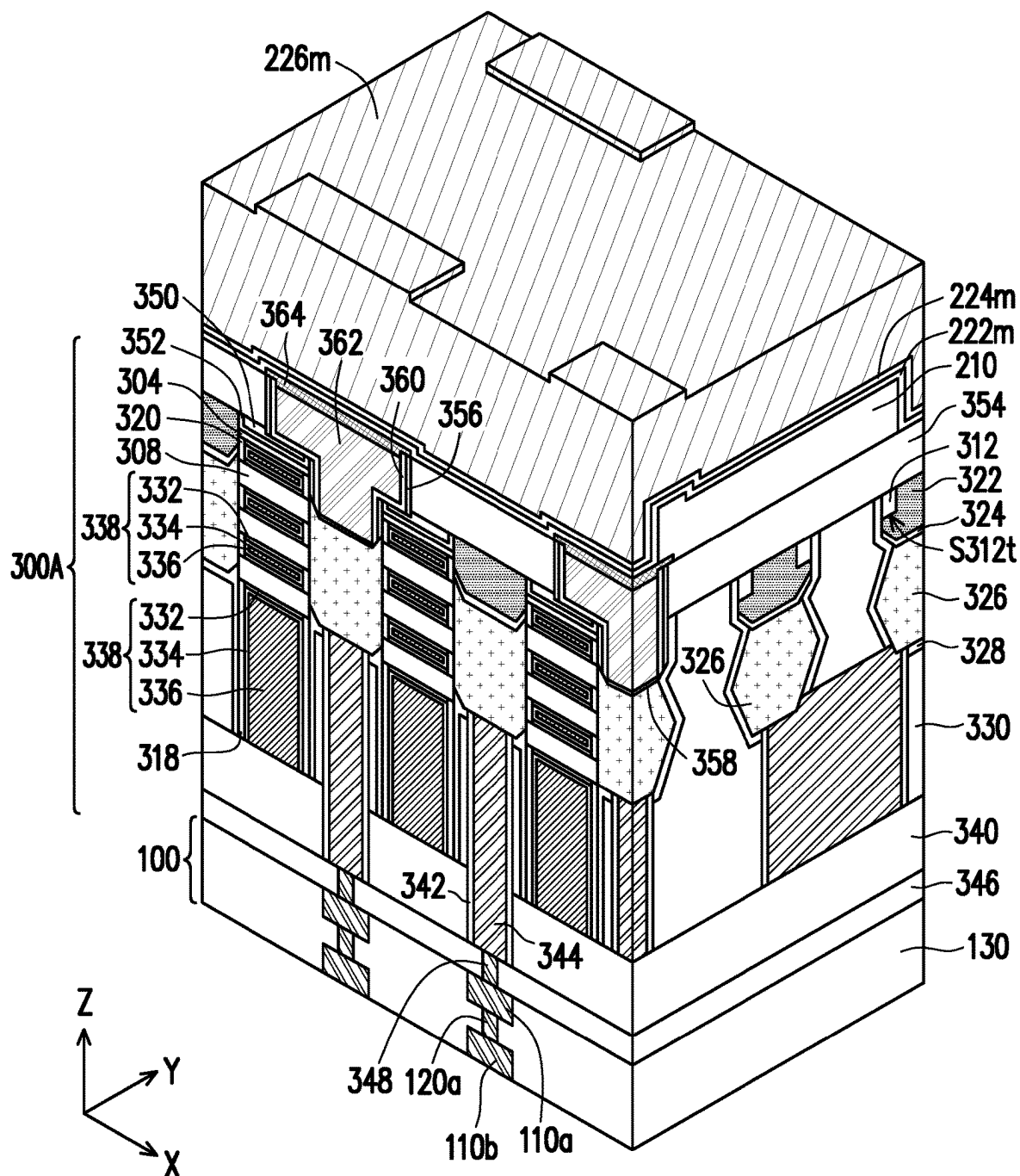
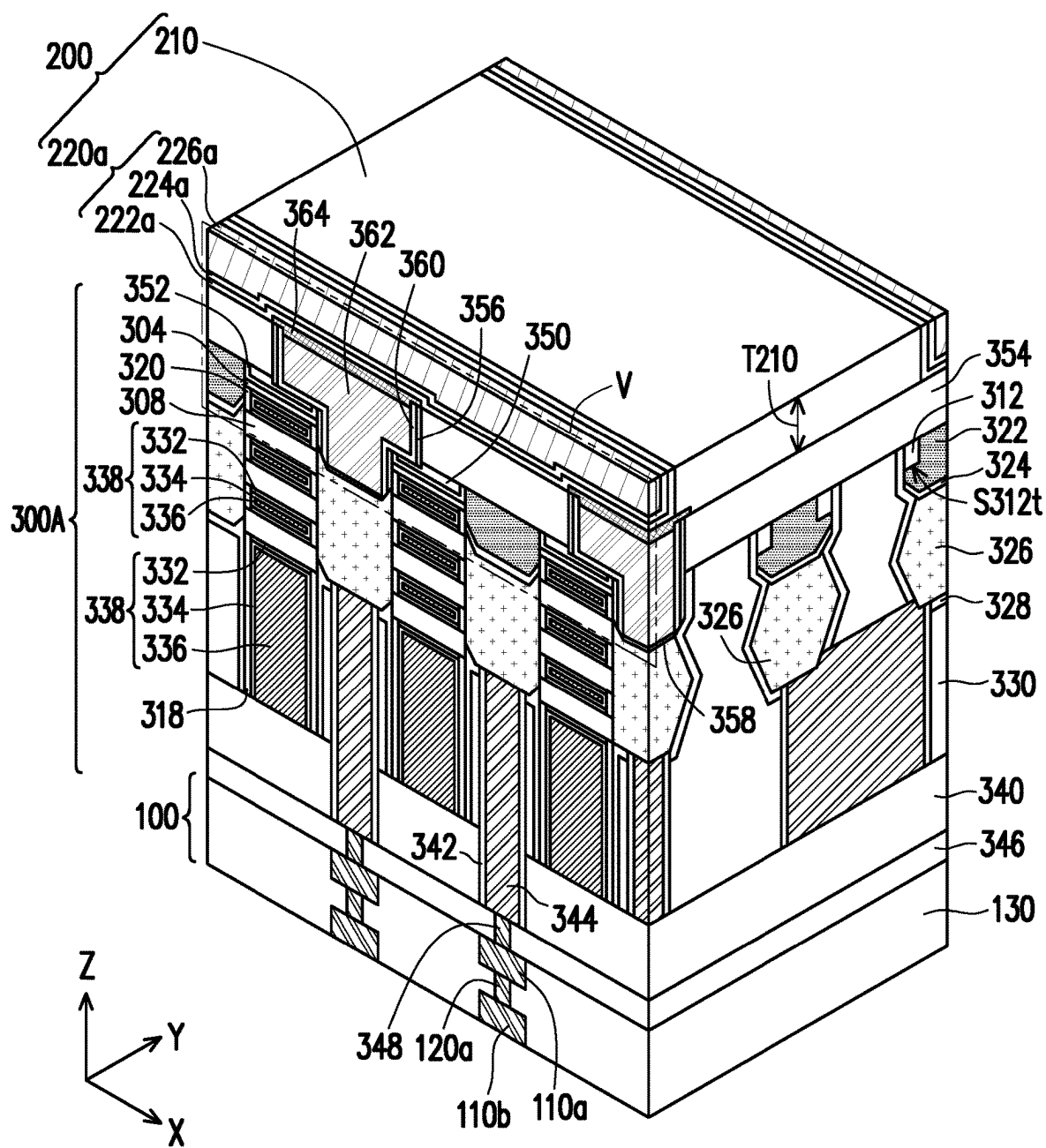


FIG. 26



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FIG. 27A

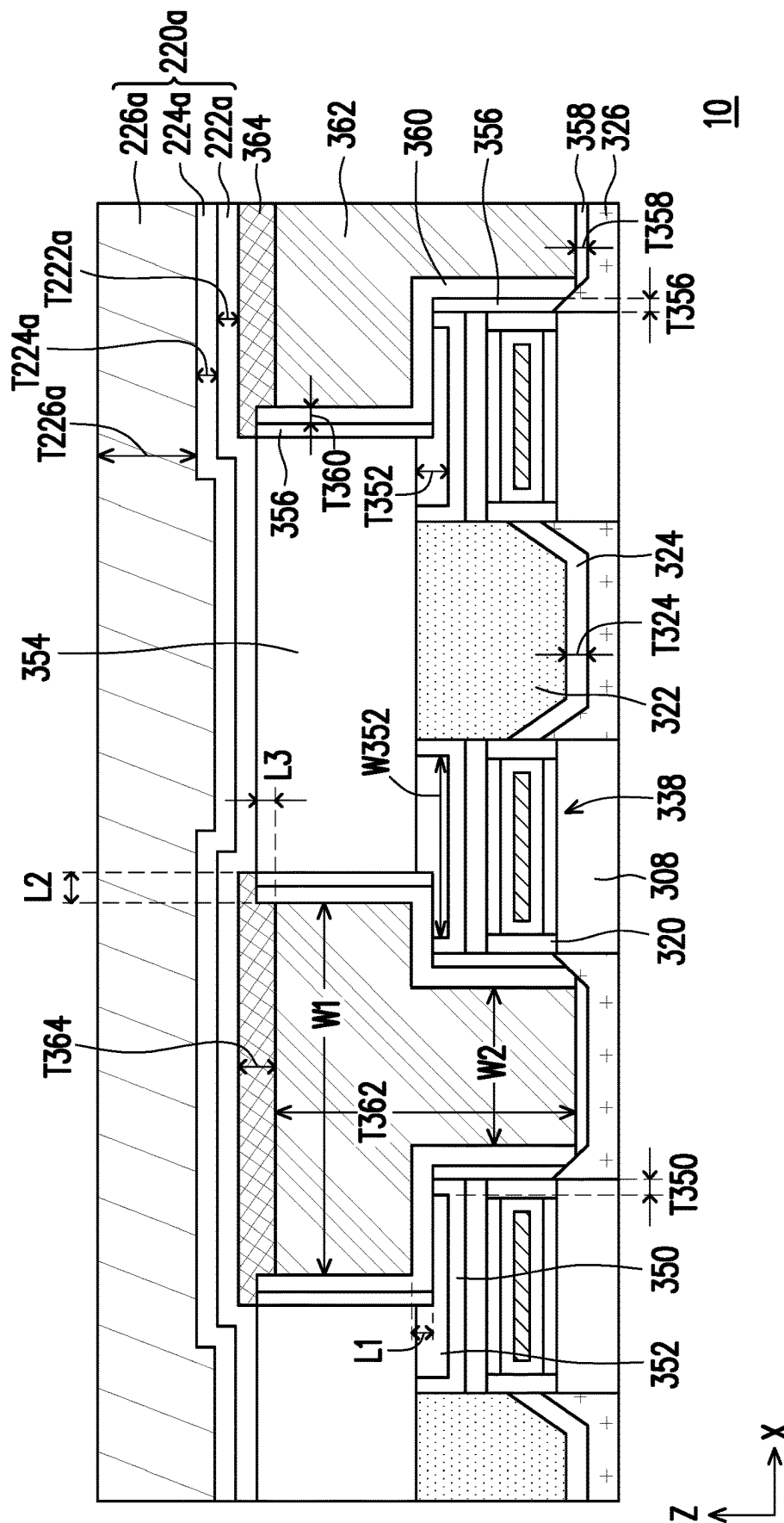


FIG. 27B

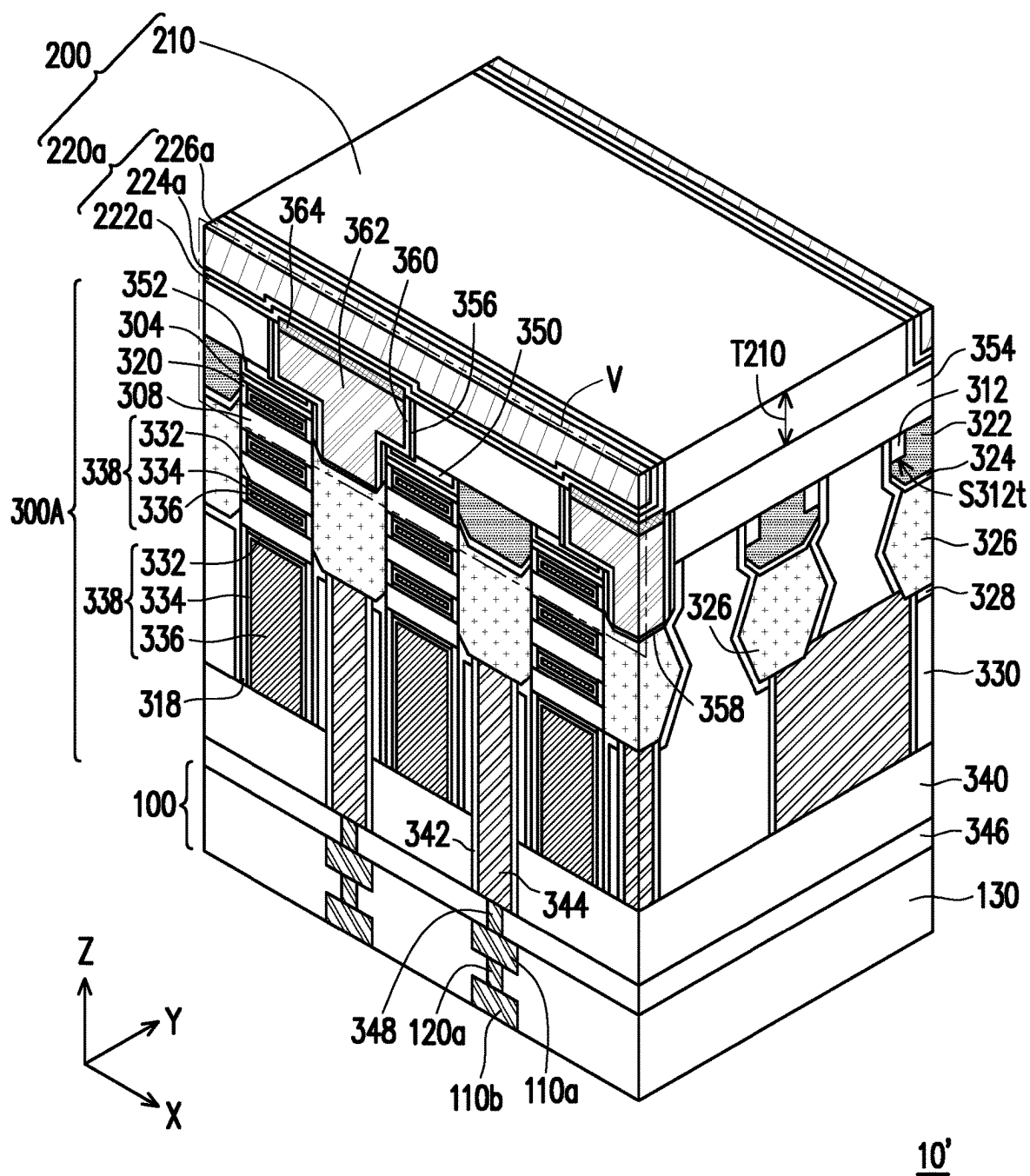


FIG. 28A

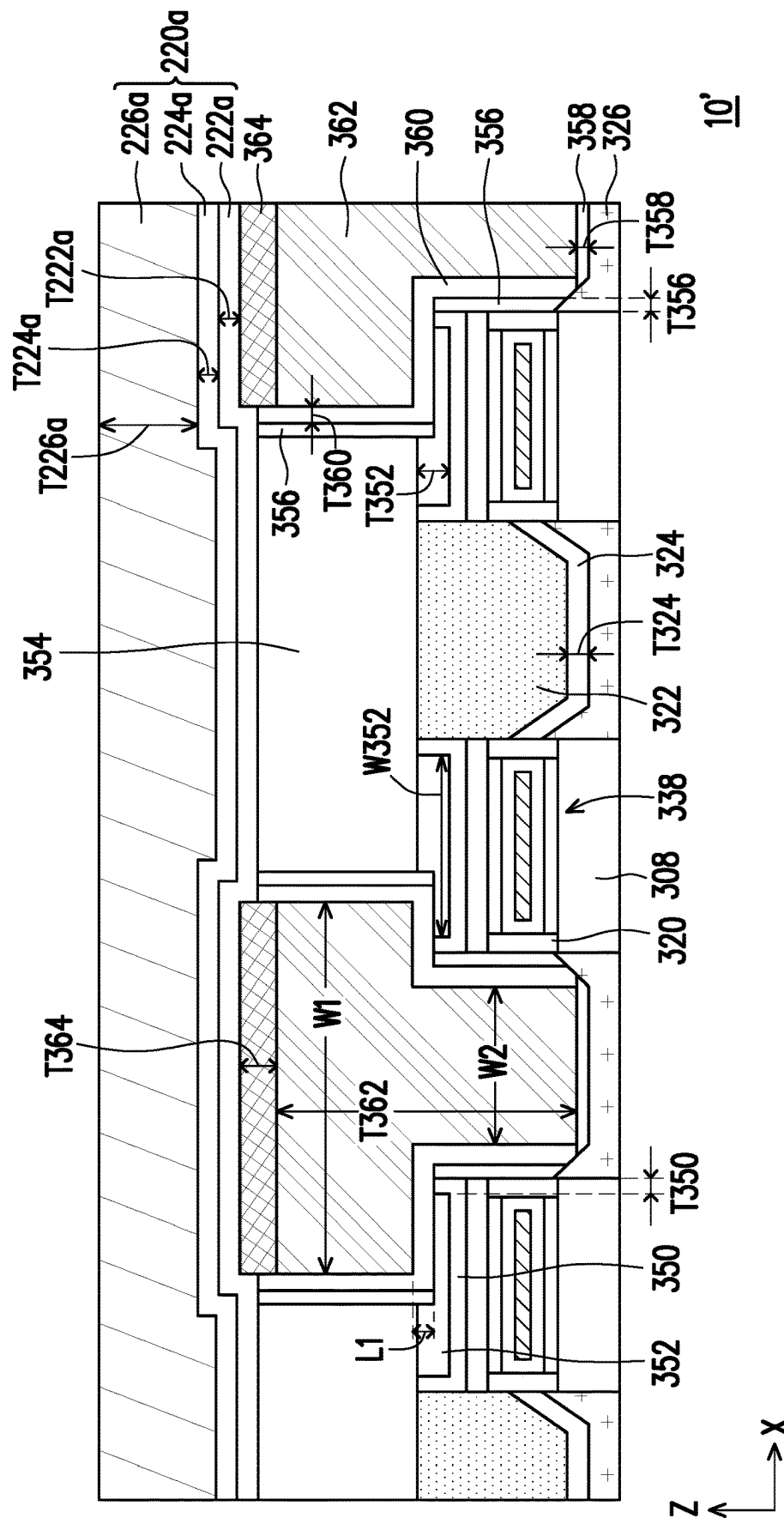


FIG. 28B

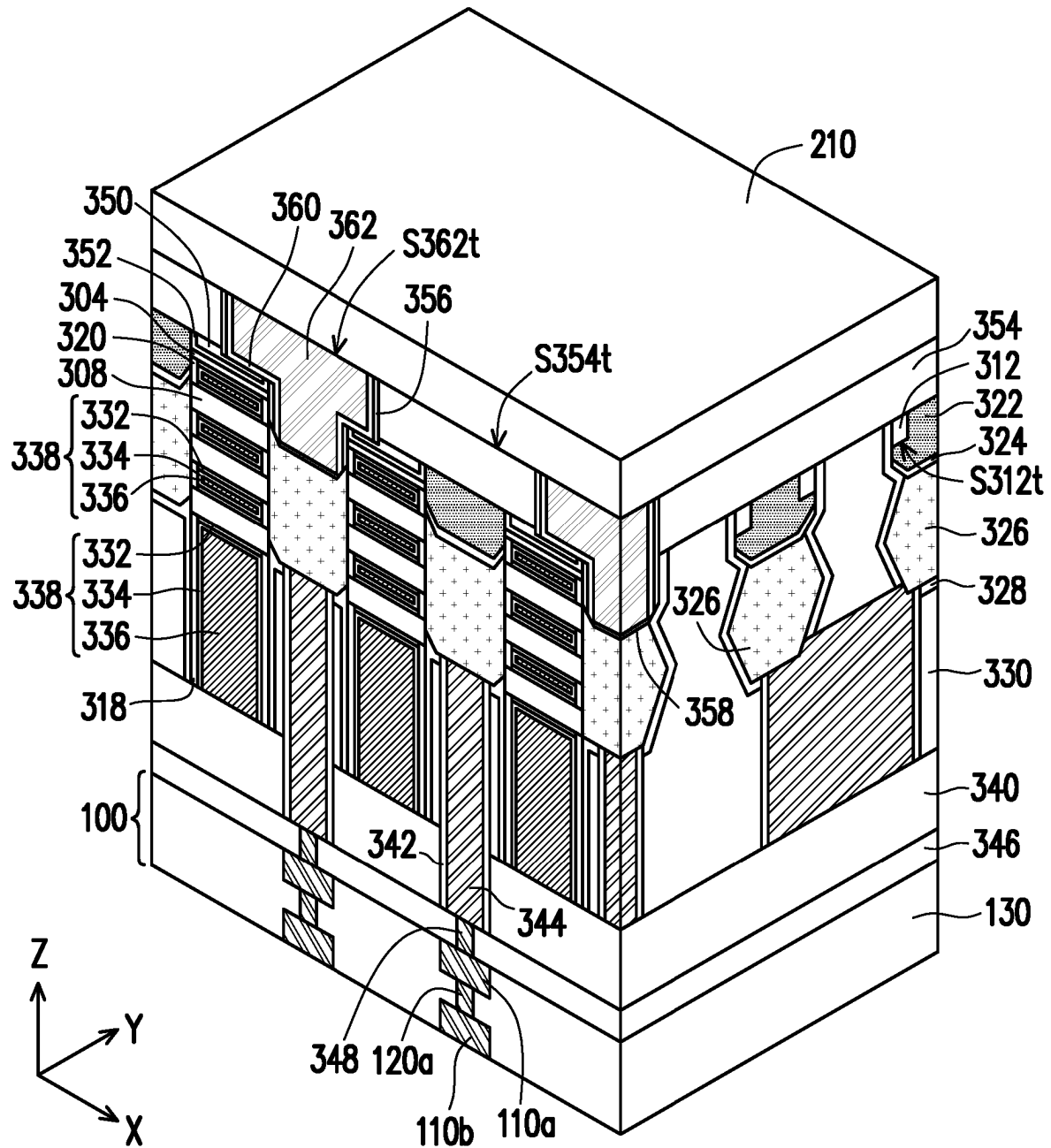


FIG. 29

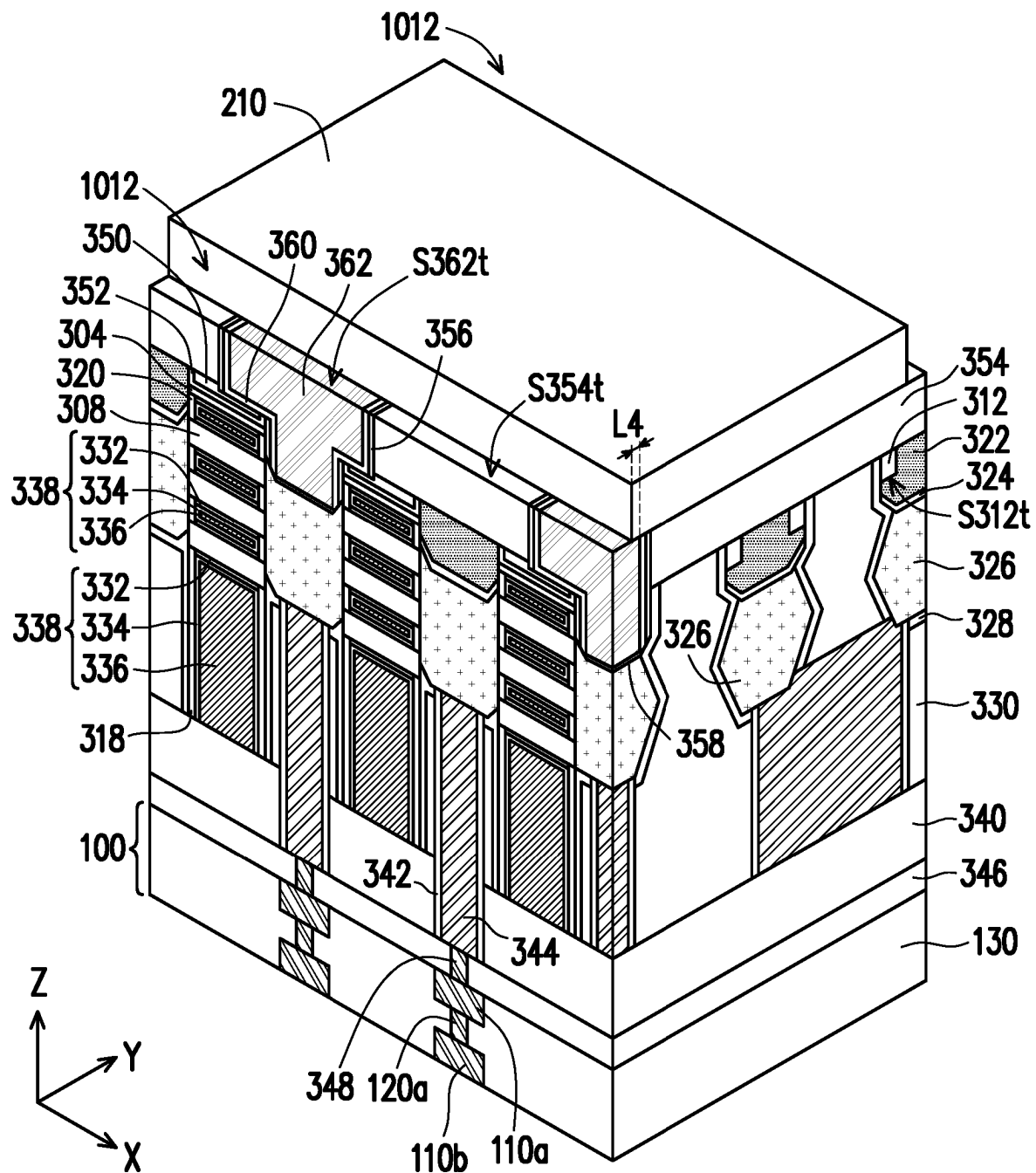


FIG. 30

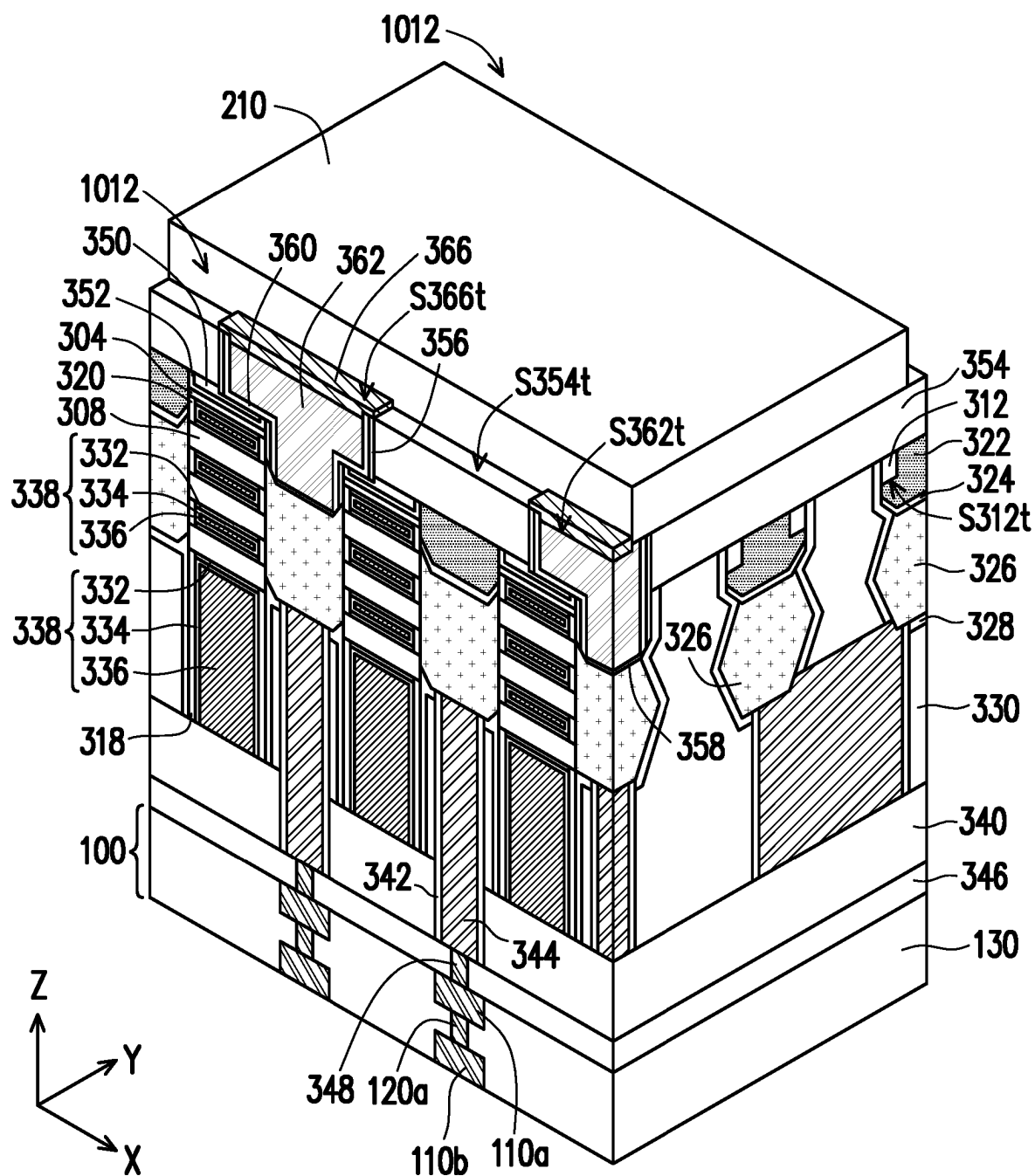


FIG. 31

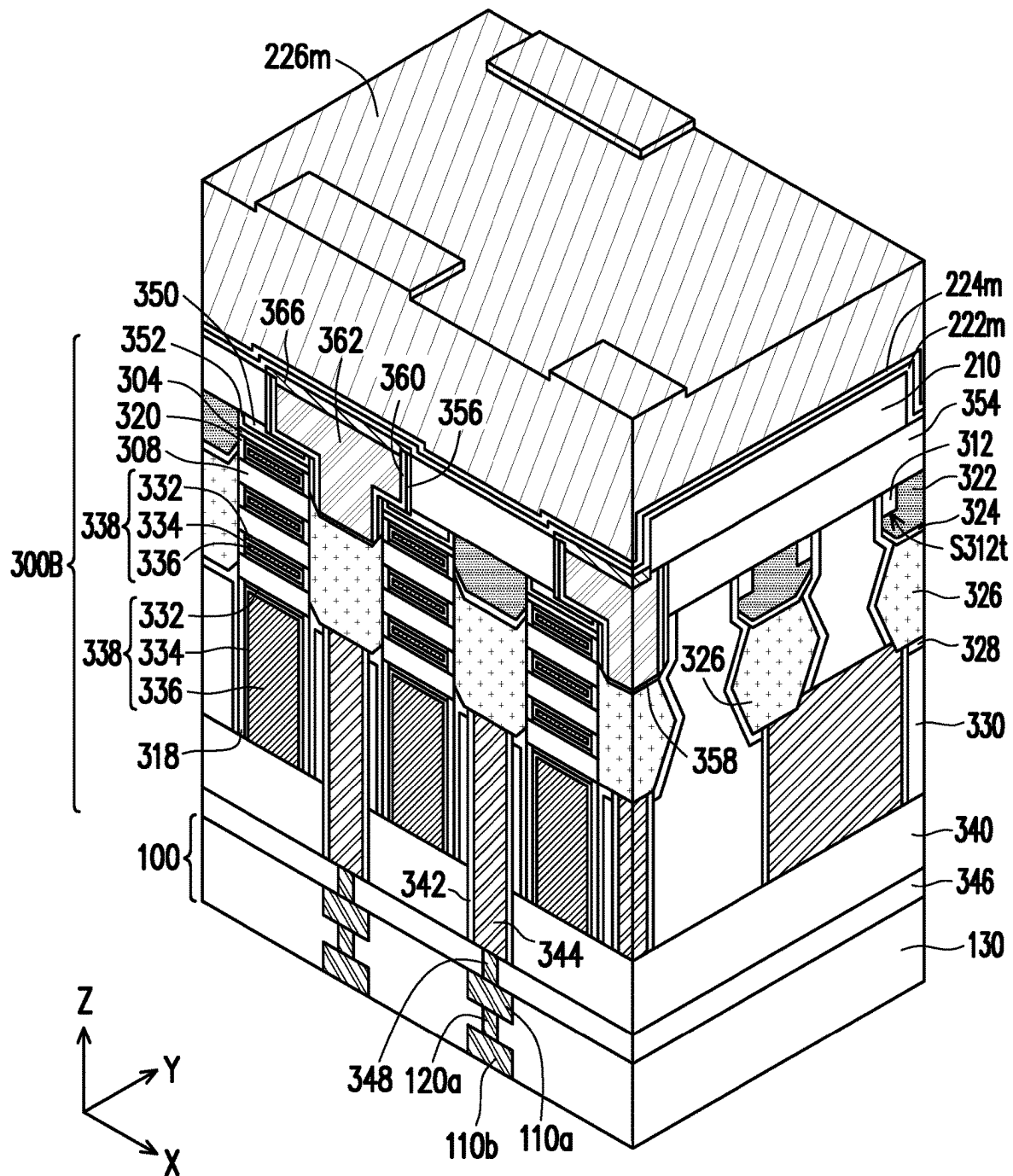
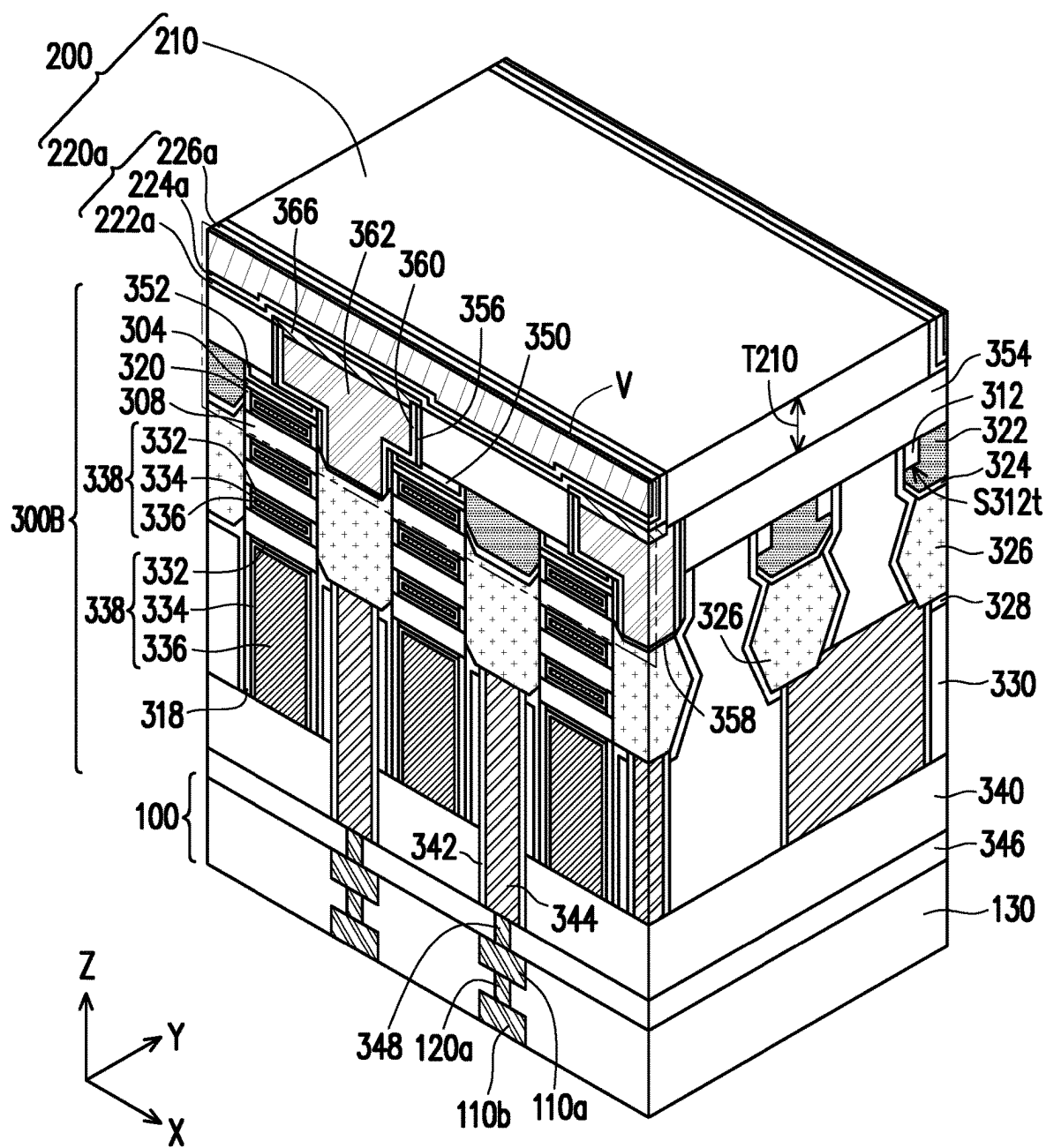


FIG. 32



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FIG. 33A

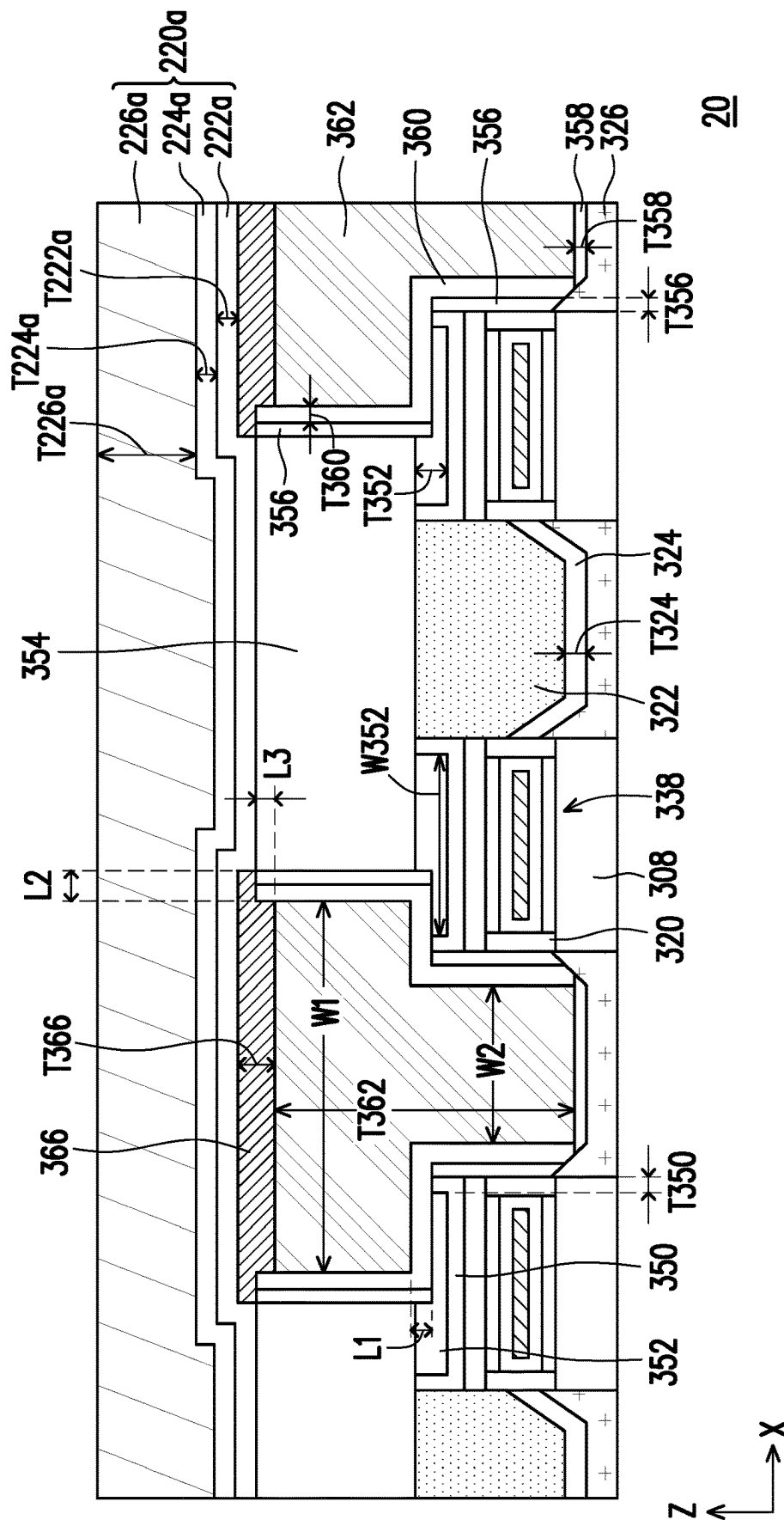


FIG. 33B

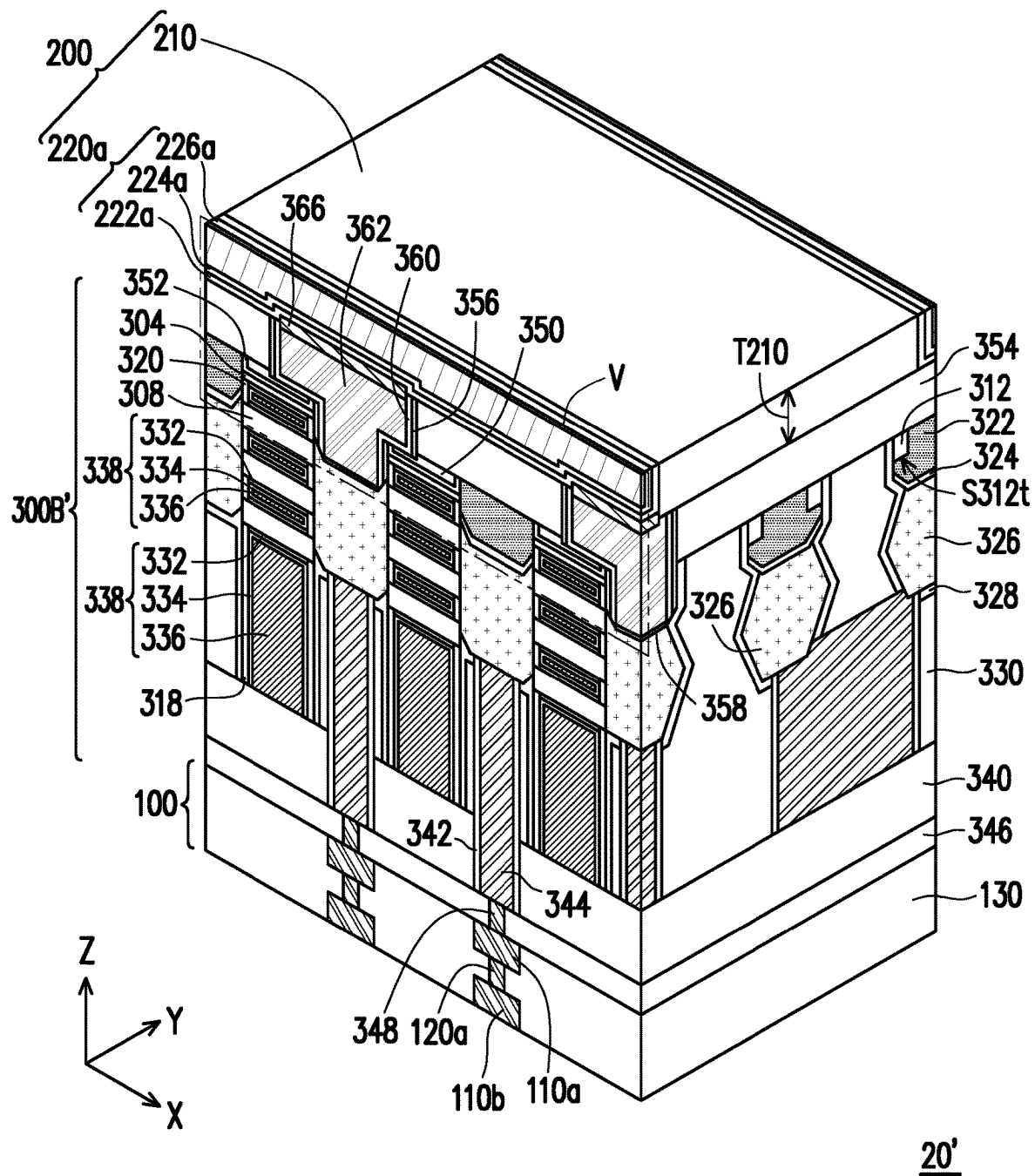


FIG. 34A

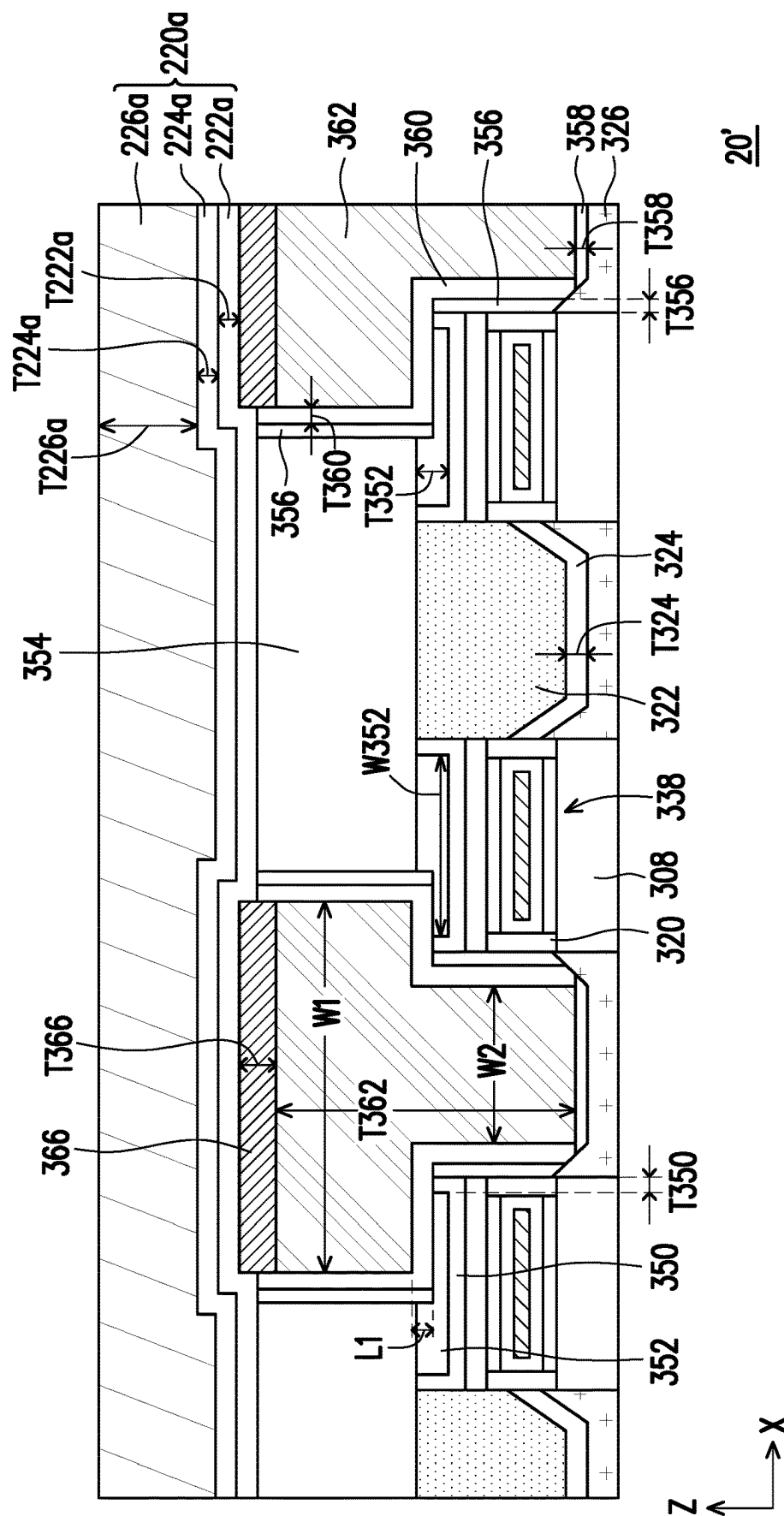


FIG. 34B

SEMICONDUCTOR DEVICE AND MANUFACTURING METHOD THEREOF

BACKGROUND

The semiconductor integrated circuit (IC) industry has experienced a fast-paced growth. Technological advances in IC materials and design have produced generations of ICs where each generation has smaller and more complex circuits than the previous generation. In the course of IC evolution, functional density (i.e., the number of interconnected devices per chip area) has generally increased while geometry size (i.e., the smallest component or line that can be created using a fabrication process) has decreased. This scaling down process generally provides benefits by increasing production efficiency and lowering associated costs. However, such scaling has also introduced increased complexity to the semiconductor manufacturing process. Thus, the realization of continued advances in ICs and devices calls for similar advances in semiconductor manufacturing processes and technology.

BRIEF DESCRIPTION OF THE DRAWINGS

Aspects of the disclosure are best understood from the following detailed description when read with the accompanying figures. It is noted that, in accordance with the standard practice in the industry, various features are not drawn to scale. In fact, the dimensions of the various features may be arbitrarily increased or reduced for clarity of discussion.

FIGS. 1-5, 6A, 7A, 8A, 9A, 10-11, 12A, 13A, 14A, 15A, 16-26, and 27A are schematic three-dimensional views of various stages in a manufacturing method of a semiconductor device in accordance with some embodiments of the disclosure.

FIGS. 6B, 7B, 8B, 9B, 12B, 13B, 14B, and 15B are schematic, enlarged cross-sectional views of a portion of the semiconductor device outlined in a dashed box W as shown in FIGS. 6A, 7A, 8A, 9A, 12A, 13A, 14A, and 15A, respectively.

FIG. 27B is a schematic, enlarged cross-sectional view of a portion of the semiconductor device outlined in a dashed box V as shown in FIG. 27A.

FIG. 28A is a schematic three-dimensional view of a semiconductor device in accordance with some embodiments of the disclosure.

FIG. 28B is a schematic, enlarged cross-sectional view of a portion of the semiconductor device outlined in a dashed box V as shown in FIG. 28A.

FIGS. 29-32 and 33A are schematic three-dimensional views of various stages in a manufacturing method of a semiconductor device in accordance with alternative embodiments of the disclosure.

FIG. 33B is a schematic, enlarged cross-sectional view of a portion of the semiconductor device outlined in a dashed box V as shown in FIG. 33A.

FIG. 34A is a schematic three-dimensional view of a semiconductor device in accordance with some embodiments of the disclosure.

FIG. 34B is a schematic, enlarged cross-sectional view of a portion of the semiconductor device outlined in a dashed box V as shown in FIG. 34A.

DETAILED DESCRIPTION

The following disclosure provides many different embodiments, or examples, for implementing different fea-

tures of the provided subject matter. Specific examples of components, values, operations, materials, arrangements, or the like, are described below to simplify the disclosure. These are, of course, merely examples and are not intended to be limiting. Other components, values, operations, materials, arrangements, or the like, are contemplated. For example, the formation of a first feature over or on a second feature in the description that follows may include embodiments in which the first and second features are formed in direct contact, and may also include embodiments in which additional features may be formed between the first and second features, such that the first and second features may not be in direct contact. In addition, the disclosure may repeat reference numerals and/or letters in the various examples. This repetition is for the purpose of simplicity and clarity and does not in itself dictate a relationship between the various embodiments and/or configurations discussed.

Further, spatially relative terms, such as “beneath,” “below,” “lower,” “above,” “upper” and the like, may be used herein for ease of description to describe one element or feature’s relationship to another element(s) or feature(s) as illustrated in the figures. The spatially relative terms are intended to encompass different orientations of the device in use or operation in addition to the orientation depicted in the figures. The apparatus may be otherwise oriented (rotated 90 degrees or at other orientations) and the spatially relative descriptors used herein may likewise be interpreted accordingly.

In addition, terms, such as “first,” “second,” “third” and the like, may be used herein for ease of description to describe similar or different element(s) or feature(s) as illustrated in the figures, and may be used interchangeably depending on the order of the presence or the contexts of the description.

The gate all around (GAA) transistor structures may be patterned by any suitable method. For example, the structures may be patterned using one or more photolithography processes, including double-patterning or multi-patterning processes. Generally, double-patterning or multi-patterning processes combine photolithography and self-aligned processes, allowing patterns to be created that have, for example, pitches smaller than what is otherwise obtainable using a single, direct photolithography process. For example, in one embodiment, a sacrificial layer is formed over a substrate and patterned using a photolithography process. Spacers are formed alongside the patterned sacrificial layer using a self-aligned process. The sacrificial layer is then removed, and the remaining spacers may then be used to pattern the GAA structure.

It should be appreciated that the following embodiment(s) of the disclosure provides applicable concepts that can be embodied in a wide variety of specific contexts. The specific embodiment(s) discussed herein is merely illustrative and is related to a semiconductor device including a capping layer of nitride between a conductive via(s) and a conductive feature(s) in a backside interconnect, and is not intended to limit the scope of the disclosure. In accordance with some embodiments, the semiconductor device includes a substrate with devices, a frontside interconnect over the substrate and electrically coupled to the devices through a first via(s), and a backside interconnect over the substrate and electrically coupled to the devices through a second via(s), where the substrate is disposed between the frontside interconnect and the backside interconnect, and a nitride capping layer is disposed between a conductive trace/line/wire of the backside interconnect and the second via(s). In the case, a diffusion (e.g., metal atoms) from the second via(s) to the

conductive trace/line/wire of the backside interconnect or from the conductive trace/line/wire of the backside interconnect to the second via(s) can be avoided, thus the reliability of the semiconductor device is enhanced, thereby ensuring the performance of the semiconductor device. Sometimes, such diffusion is also referred to as “miscible”.

In accordance with some embodiments, the semiconductor device is or includes a portion of a nanostructure transistor device. The semiconductor device may include a fin field-effect transistor (finFET) device, which may include a thin (vertical) fin of silicon body on a substrate, and a gate is wrapped around the fin (i.e. the channel region) providing excellent control from three sides of the channel region. The nanostructure transistor device may include a gate-all-around (GAA) transistor device or a nanowire transistor, which may include a gate structure wrapping around the perimeter of one or more nanostructures (i.e. channel regions) for improved control of channel current flow. In some embodiments, the semiconductor device is formed on bulk silicon substrates. Still, the semiconductor device may be formed on a silicon-on-insulator (SOI) substrate, a germanium-on-insulator (GOI) substrate, a silicon germanium substrate, or a Group III-V semiconductor substrate. Also, in accordance with some embodiments, the silicon substrate may include other conductive layers or other semiconductor elements, such as transistors, diodes or the like. The embodiments are not limited in this context. The semiconductor device may be included in microprocessors, memories, and/or other integrated circuits (IC). Accordingly, it is understood that additional processes may be provided before, during, and after the illustrated method, and that some other processes may only be briefly described herein. Also, the structures illustrated in the drawings are simplified for a better understanding of the concepts of the present disclosure. For example, although the figures illustrate the structure of the semiconductor device, it is understood the semiconductor device may be part of an IC that further includes a number of other devices such as resistors, capacitors, inductors, fuses, etc.

FIGS. 1-5, 6A, 7A, 8A, 9A, 10-11, 12A, 13A, 14A, 15A, 16-26, and 27A are schematic three-dimensional views of various stages in a manufacturing method of a semiconductor device 10 in accordance with some embodiments of the disclosure. FIGS. 6B, 7B, 8B, 9B, 12B, 13B, 14B, and 15B are schematic, enlarged cross-sectional views of a portion of the semiconductor device 10 outlined in a dashed box W shown in FIGS. 6A, 7A, 8A, 9A, 12A, 13A, 14A, and 15A, respectively. FIG. 27B is a schematic, enlarged cross-sectional view of a portion of the semiconductor device 10 outlined in a dashed box V as shown in FIG. 27A. FIG. 28A is a schematic three-dimensional view of a semiconductor device 10' in accordance with some embodiments of the disclosure. FIG. 28B is a schematic, enlarged cross-sectional view of a portion of the semiconductor device 10' outlined in a dashed box V as shown in FIG. 28A. In embodiments, the manufacturing method is part of a wafer level packaging process. It is to be noted that the process steps described herein cover a portion of the manufacturing processes used to fabricate the semiconductor device involving a capping layer of nitride between a conductive via(s) and a conductive feature(s) in a backside interconnect. The embodiments are intended to provide further explanations but are not used to limit the scope of the disclosure. In the disclosure, it should be appreciated that the illustration of components throughout all figures is schematic and is not in scale. The elements similar to or substantially the same as the elements described previously will use the same reference numbers, and certain

details or descriptions (e.g., the materials, formation processes, positioning configurations, electrical connections, etc.) of the same elements would not be repeated. For clarity of illustrations, in the drawings are illustrated the orthogonal axes (X, Y and Z) of the Cartesian coordinate system according to which the views are oriented; however, the disclosure is not specifically limited thereto.

Referring to FIG. 1, in some embodiments, a stack of first and second semiconductor layers (306 and 308) may be formed on a semiconductor substrate 304. In some embodiments, the semiconductor substrate 304 includes a crystalline silicon substrate or a bulk silicon substrate (e.g., wafer). In some embodiments, the semiconductor substrate 304 is made of a suitable elemental semiconductor (e.g., germanium), a suitable compound semiconductor (e.g., gallium arsenide, silicon carbide, indium arsenide, or indium phosphide), a suitable alloy semiconductor (e.g., silicon germanium carbide, gallium arsenic phosphide, or gallium indium phosphide), and/or the like. In some embodiments, the semiconductor substrate 304 includes a SOI substrate. The semiconductor substrate 304 may include various doped regions (not individually shown) doped with p-type or n-type dopants, where the doped regions may be configured for an n-type FET, or alternatively, configured for a p-type FET.

The first semiconductor layers 306 and the second semiconductor layers 308 may be alternately stacked upon one another (e.g., along a direction Z) to form a stacking structure over the semiconductor substrate 304. The first semiconductor layers 306 may be considered sacrificial layers in the sense that they are removed in the subsequent process. In some embodiments, the bottommost one of the first semiconductor layers 306 is formed on the semiconductor substrate 304, with the remaining second and first semiconductor layers (308 and 304) alternately stacked on top. However, either the first semiconductor layer 306 or the second semiconductor layer 308 may be the bottommost layer (or the layer most proximate from the semiconductor substrate 304), and either the first semiconductor layer 306 or the second semiconductor layer 308 may be the topmost layer (or the layer most distanced to the semiconductor substrate 304). The disclosure is not limited by the number of stacked semiconductor layers. A thickness (not labeled) of the respective first semiconductor layer 306 measured along the direction Z may be in a range of about 4 nm to about 12 nm. A thickness (not labeled) of the respective second semiconductor layer 308 measured along the direction Z may be in a range of about 6 nm to about 15 nm. Although other values of the thicknesses of the first and second semiconductor layers 306 and 308 are possible depending on product and process requirements.

The first semiconductor layers 306 and the second semiconductor layers 308 may have different materials (or compositions) that may provide for different oxidation rates and/or different etch selectivity between the layers. For example, the second semiconductor layers 308 are formed of the same material as the semiconductor substrate 304, while the first semiconductor layers 306 may be formed of a different material which may be selectively removed with respect to the material of the semiconductor substrate 304 and the second semiconductor layers 308. In some embodiments, the material of the first semiconductor layers 306 includes silicon germanium (SiGe). For example, germanium (Ge) may include about 15% to 35% of the first semiconductor layers 306 of SiGe in molar ratio. In some embodiments, the second semiconductor layers 308 include silicon (Si), where each of the second semiconductor layers

308 may be undoped or substantially dopant-free. A method for forming the first and second semiconductor layers **306** and **308** may include epitaxial processes. The second semiconductor layers **308** may be considered as semiconductor channel layers or channel regions. That is, the second semiconductor layers **308** may be referred to as channels of the semiconductor device **10**. However, the disclosure is not limited thereto, and other suitable material, or other combinations of materials for which selective etching is possible are contemplated within the scope of the disclosure.

Continued on FIG. 1, for example, a hard mask material **1002am** and a hard mask material **1002bm** are subsequently formed on the stacking structure. For example, the hard mask material **1002am** is stacked on the stacking structure, and the hard mask material **1002bm** is stacked on the hard mask material **1002am**. The hard mask material **1002am** and the hard mask material **1002bm** are individually extended along a X-Y plane to cover up the stacking structure. The hard mask material **1002am** and the hard mask material **1002bm** may be made of different insulating materials. For instance, materials of the hard mask material **1002am** and the hard mask material **1002bm** may be selected from a group consisting of silicon oxide, silicon nitride, silicon oxynitride and the like. A method for forming the hard mask material **1002am** and the hard mask material **1002bm** may include one or more deposition processes, such as chemical vapor deposition (CVD) process or the like.

Referring to FIG. 2, in some embodiments, the hard mask material **1002am** and the hard mask material **1002bm** are patterned to form a plurality of hard mask structures **1002** over the stacking structure. In addition, in some embodiments, each hard mask structure **1002** includes a hard mask layer **1002a** and a hard mask layer **1002b** formed over the hard mask layer **1002a**. In some embodiments, the hard mask structures **1002** are arranged along a direction Y, and are extending along a direction X. A method for forming of hard mask structures **1002** may include a self-aligned multiple patterning process (e.g., a self-aligned double patterning (SADP) process or a self-aligned quadruple patterning (SAQP) process). However, the disclosure is not limited thereto; alternatively, the hard mask structures **1002** each may be a single layer structure or include a structure having more than one sublayer by adjusting the number of hard mask materials formed over the stacking structure. The hard mask structures **1002** may be referred to as hard masks or hard mask patterns.

Referring to FIG. 2 and FIG. 3, in some embodiments, a portion of the stack of first and second semiconductor layers (**306** and **308**) and a portion of the semiconductor substrate **304** may be removed to form first trenches (or openings) **310T1**, thereby defining a fin structure **310a** between two adjacent first trenches **310T1**. The first trenches **310T1** may be arranged along the direction Y and continuously extend along the direction X. For example, the critical dimension (or the width measured along the direction Y, not labeled) of the respective first trench **310T1** is in a range of about 25 nm to about 80 nm. The critical dimension (or the width measured along the direction Y, not labeled) of the fin structures **310a** may be in a range of about 5 nm to about 40 nm, depending on the N-type fin or the P-type fin. For example, the critical dimension of the N-type fin may be in a range of about 20 nm to about 40 nm, and the critical dimension of the P-type fin may be in a range of about 5 nm to about 20 nm. Although other values of the critical dimensions are possible depending on various device regions. It should be noted that the disclosure is not limited by the numbers of fin structures **310a** depicted in FIG. 3,

which may be adjusted according to the requirements of the circuit design. When multiple fin structures **310a** are formed, the first trenches **310T1** may be disposed between any adjacent ones of the fin structures **310a**.

The fin structures **310a** may be formed by patterning portions of the stack of first and second semiconductor layers (**306** and **308**) and the semiconductor substrate **304**. A method for patterning the stacking structure to form the fin structures **310a** may include an etching process, such as an anisotropic etching process. The etching process may be stopped when a top portion of the semiconductor substrate **304** may be removed during the etching process as shown in FIG. 3, or be stopped when an illustrated top surface of the semiconductor substrate **304** may be exposed. For example, the hard mask structures **1002** are disposed over the topmost one of the second semiconductor layers **308** (also called the top semiconductor layer **308** herein). The hard mask structures **1002** are used as shadow masks to pattern exposed portions of the stack of first and second semiconductor layers (**306** and **308**) and the semiconductor substrate **304**. In those embodiments where the hard mask structures **1002** are arranged along the direction Y and extending along the direction X, the formed fin structures **310a** are also arranged along the direction Y and extending along the direction X. The fin structures **310a** may be formed by etching trenches (**310T1**) in the stack of first and second semiconductor layers (**306** and **308**) and the semiconductor substrate **304**. In some embodiments, the first trenches **310T1** may be parallel strips (when viewed from the top) elongated along the direction X and distributed along the direction Y.

The hard mask structures **1002** may be optionally removed after forming the fin structures **310a**. In some embodiments, the hard mask layers **1002b** of the hard mask structures **1002** are removed during the etching process (as shown in FIG. 3), and the hard mask layers **1002a** of the hard mask structures **1002** are then removed after forming the fin structures **310a** (see FIG. 4) to expose surfaces **S310r** (e.g., illustrated top surfaces of topmost one of the second semiconductor layers **308**) of the fin structures **310a**. However, the disclosure is not limited thereto; alternatively, the hard mask structures **1002** may be removed during the subsequently-performed etching process.

Referring to FIG. 4, in some embodiments, a plurality of isolation structures **312** (sometimes referred to as shallow trench isolation (STI) structures) may be formed in lower portions of the first trenches **310T1**. For example, the isolation structures **312** extend at opposing sides of a lower portion of the semiconductor substrate **304**. In some embodiments where multiple fin structures **310a** are provided, each of the isolation structures **312** is disposed between adjacent ones of the fin structures **310a** and covers respectively a sidewall of a lower portion of the respective fin structure **310a**. The isolation structures **312** may be formed of an insulation material (e.g., an oxide, a Si-based oxide (e.g., SiOC, SiOCN, or the like), a nitride, the like, any other suitable material, or combinations thereof) which may electrically isolate neighboring fin structures **310a** from each other.

In some embodiments, the isolation structures **312** are formed by initially depositing a layer of insulation material (not shown) in the respective first trench **310T1** and recessing the layer of insulation material using an acceptable etching process, such as one that is selective to the material of the isolation structures **312**. For example, a dry etching process is performed to form the isolation structures **312** having a relatively smooth top surfaces **S312t**. In alternative embodiments, a wet etching process is used. Or alterna-

tively, a dry etching process and wet etching process are both used. The isolation structures **312** may be recessed to where illustrated top surfaces **S312_t** of the isolation structures **312** are substantially coplanar to (e.g., leveled with) the illustrated top surface (not labeled) of the semiconductor substrate **304**, and the fin structures **310a** protrudes from the neighboring isolation structures **312**. The illustrated top surfaces **S312_t** of the isolation structures **312** may be a flat surface, a curved (e.g., convex or concave) surface, or combinations thereof, depending on the etching process. Alternatively, the isolation structures **312** may be recessed to where illustrated top surfaces **S312_t** of the isolation structures **312** are below the illustrated top surface of the semiconductor substrate **304**, and the fin structures **310a** protrudes from the neighboring isolation structures **312**.

Referring to FIG. 5, in some embodiments, sacrificial gate structures are formed on the semiconductor substrate **304**. An extending direction of the sacrificial gate structures is intersected with an extending direction of the fin structures **310a**, and the sacrificial gate structures cover portions of the fin structures **310a** that are overlapped with the sacrificial gate structures. In those embodiments where the fin structures **310a** are arranged along the direction Y and extending along the direction X, the sacrificial gate structures may be arranged along the direction X and extend along the direction Y. The sacrificial gate structures may be referred to as dummy gate structures. In some embodiments, each sacrificial gate structure includes a dummy gate dielectric layer **314** and a dummy gate electrode **316**. The dummy gate dielectric layer **314** is conformally formed on the semiconductor substrate **304** and the fin structures **310a**, whereas the dummy gate electrode **316** covers the dummy gate dielectric layer **314**, and are formed to a height greater than a height of the fin structures **310a**. In some embodiments, each sacrificial gate structure further includes a capping structure **1004** lying on the dummy gate electrode **316**. The capping structure **1004** may include a capping layer **1004a** and a capping layer **1004b** lying above the capping layer **1004a**. In some embodiments, the capping layer **1004b** has rounded top corners.

Materials of the dummy gate dielectric layer **314**, the capping layer **1004a** and the capping layer **1004b** may respectively include silicon oxide, silicon nitride, silicon oxynitride, the like or combinations thereof, whereas a material of the dummy gate electrode **316** may include polysilicon. In addition, methods for forming the dummy gate dielectric layer **314**, the capping layers **1004a**, **1004b** and the dummy gate electrode **316** may respectively include a deposition process, such as a CVD process or an atomic layer deposition (ALD) process. In each sacrificial gate structure, the dummy gate dielectric layer **314** may be referred to as a dummy gate dielectric strip, a sacrificial gate dielectric layer or a sacrificial gate dielectric strip, the dummy gate electrode **316** may be referred to as dummy gate electrode strip **316**, a sacrificial gate electrode **316** or a sacrificial gate electrode strip **316**, the capping structure **1004** may be referred to as a patterned mask structure, and the capping layers **1004a**, **1004b** may be referred to as mask strips or mask patterns.

Referring to FIG. 6A and FIG. 6B, in some embodiments, a gate spacer layer **318m** is formed on the structure depicted in FIG. 5. In some embodiments, the gate spacer layer **318m** is globally formed over the structure as shown in FIG. 5. In these embodiments, the semiconductor substrate **304**, the fin structures **310a** and the sacrificial gate structures (including the dummy gate dielectric layer **314**, the dummy gate electrode **316**, and the capping structure **1004**) may be

conformally covered by the gate spacer layer **318m**. A material of the gate spacer layer **318m** may include silicon oxide, silicon nitride, silicon carbide, silicon oxynitride (SiON), silicon carbide nitride (SiCN), silicon oxycarbide (SiOC), silicon carbide oxynitride (SiOCN), the like or combinations thereof, and a method for forming the gate spacer layer **318m** may include a deposition process, such as a CVD process or an ALD process.

Referring to FIG. 7A and FIG. 7B, in some embodiments, some portions of the fin structures **310a** and the gate spacer layer **318m** are removed to form second trenches (or openings) **310T2**, thereby forming fin structure **310a'** and gate spacers **318**. As shown in FIG. 7A, one sacrificial gate structure and a respective underlying one fin structure **310a'** are located between two adjacent second trenches **310T2**, for example. The second trenches **310T2** may arranged along the direction X and continuously extend along the direction Y. For example, the critical dimension (or the width measured along the direction X, not labeled) of the respective second trench **310T2** is in a range of about 25 nm to about 80 nm. The critical dimension (or the width measured along the direction X, not labeled) of the fin structures **310a'** may be in a range of about 5 nm to about 40 nm, depending on the N-type fin or the P-type fin. For example, the critical dimension of the N-type fin may be in a range of about 20 nm to about 40 nm, and the critical dimension of the P-type fin may be in a range of about 5 nm to about 20 nm. Although other values of the critical dimensions are possible depending on various device regions. It should be noted that the disclosure is not limited by the numbers of fin structures **310a'** depicted in FIG. 7A, which may be adjusted according to the requirements of the circuit design. When multiple fin structures **310a'** are formed, the second trenches **310T2** may be disposed between any adjacent ones of the fin structures **310a'**. In some embodiments, the first trenches **310T1** and the second trenches **310T2** are spatially communicated to each other.

In some embodiments, portions of the gate spacer layer **318m** covering the sidewalls of the sacrificial gate structures are removed, and the rest of the sacrificial gate structures are removed, so to form the gate spacers **318**. On the other hand, portions of the fin structures **310a** not covered by the sacrificial gate structures **316** of the sacrificial gate structures and the gate spacers **318** are accordingly exposed. Thereafter, the exposed portions of the fin structures **310a** are removed, whereas portions of the fin structures **310a** covered by the gate spacers **318** and the sacrificial gate structures are remained to form the fin structure **310a'**. For example, the sacrificial gate structures and the gate spacers **318** together are used as shadow masks to pattern the exposed portions of the fin structures **310a**. In some embodiments, a method for removing these portions of the fin structures **310a** and gate spacer layer **318m** may include one or more etching processes, such as one or more anisotropic etching processes. The etching process may be stopped when a top portion of the semiconductor substrate **304** may be removed during the etching process(es) as shown in FIG. 7A and FIG. 7B, where illustrated top surfaces **S314_t** of the semiconductor substrate **304** exposed by the second trenches **310T2** is lower than the illustrated top surfaces of the semiconductor substrate **304** within the fin structures **310a**. In the case, portions of the isolation structures **312** are removed during the etching process(es), where illustrated top surface **S312_t** of the edges of the isolation structures **312** are above illustrated top surfaces **S312_t** of portions of the isolation structures **312** surrounding by (e.g., enclosed by) the edges of the isolation structures **312** and illustrated top

surface of the semiconductor substrate **304** exposed by the second trenches **310T2**. That is, the isolation structures **312** may include a non-planar top surface. Alternatively, the etching process may be stopped at illustrated top surfaces **S314t** of the semiconductor substrate **304** exposed by the second trenches **310T2** without removing semiconductor substrate **304**. That is, in the alternative embodiment, the isolation structures **312** may include a planar top surface. The disclosure is not limited thereto.

In addition, the illustrated top surface of the semiconductor substrate **304** exposed by the second trenches **310T2** may be lower than the illustrated top surfaces **S312tr** of portions of the isolation structures **312** surrounding by (e.g., enclosed by) the edges of the isolation structures **312**, as shown in FIG. 7A. However, the disclosure is not limited thereto, alternatively, the illustrated top surface of the semiconductor substrate **304** exposed by the second trenches **310T2** may be substantially coplanar to (e.g., leveled with) the illustrated top surfaces **S312tr** of portions of the isolation structures **312** surrounding by (e.g., enclosed by) the edges of the isolation structures **312**.

Referring to FIG. 8A and FIG. 8B, in some embodiments, the first semiconductor layers **306** are laterally recessed from the second semiconductor layers **308** and the gate spacers **318** to form first semiconductor layer **306'** and a plurality of recesses **306r** at opposite sides of the first semiconductor layer **306'**. In the case, the recesses **306r** are formed at sidewalls of the remained portions of the fin structures **310a'**. In some embodiments, the first semiconductor layers **306** are laterally recessed from the second semiconductor layers **308** and the gate spacers **318** by a distance ranging from 0.5 nm to 1 nm. A method for lateral recessing the first semiconductor layers **306** may include an etching process, such as an isotropic etching process. By properly selecting etchants for the etching process and/or by properly selecting the materials of the first semiconductor layers **306** and the second semiconductor layers **308**, the first semiconductor layers **306** can be etched without consuming the second semiconductor layers **308** and other components in the current structure. In some embodiments, the recesses **306r**, the first trenches **310T1** and the second trenches **310T2** are spatially communicated to each other. The recesses **306r** may be referred to as lateral recesses.

Referring to FIG. 9A and FIG. 9B, in some embodiments, a plurality of inner spacers **320** are formed in the recesses **306r** by filling an insulating material in the recesses **306r** at the sidewalls of the fin structures **310a'**. In some embodiments, exposed sidewalls of the inner spacers **320** are substantially coplanar with sidewalls of the second semiconductor layers **308** and sidewalls of the gate spacers **318**. In alternative embodiments, the exposed sidewalls of the inner spacers **320** are dented from the sidewalls of the second semiconductor layers **308** and the sidewalls of the gate spacers **318**. A material of the insulating material for forming the inner spacers **320** may include silicon oxide, silicon nitride, silicon carbide, silicon carbide nitride, silicon oxide carbide, silicon carbide oxynitride, or other suitable dielectric materials or combinations thereof. A method for forming the inner spacers **320** may include initially forming a material layer globally covering the structure shown in FIG. 8A and FIG. 8B, and then removing portions of this blanket layer outside the recesses **306r**. In this way, the remained portions of this material layer form the inner spacers **320**. In some embodiments, the material layer is formed by using a deposition process (e.g., a CVD process or an ALD process), and the portions of the material layer are removed by using an etching process (e.g., an anisotropic

etching process). The inner spacers **320** may be formed from the same or different material as the gate spacers **318**.

Referring to FIG. 10, in some embodiments, a plurality of first layers **322** are formed over the semiconductor substrate **304** between the lower portion of the fin structures **310a'**. In some embodiments, one first layer **322** is disposed between two adjacent isolation structures **312** arranged along the direction Y. On the other hand, each of the fin structures **310a'** is disposed between two adjacent first layers **322** along the direction X. For example, as shown in FIG. 10, the first layers **322** are disposed on (e.g., in physical contact with) the illustrated top surface of the semiconductor substrate **304** exposed by the second trenches **310T2** and further extend onto the illustrated top surfaces **S312t** of the isolation structures **312**. In some embodiments, illustrated top surfaces of the first layers **322** are substantially coplanar to (e.g., leveled with) the illustrated top surface of the semiconductor substrate **304** within the fin structures **310a'**. Alternatively, the illustrated top surfaces of the first layers **322** may be above the illustrated top surface of the semiconductor substrate **304** within the fin structures **310a'**. The first layers **322** may be referred to as a bottom-up epitaxial layers (epi layers) or strained elements. The first layers **322** may include SiGe, which may be epitaxial-grown with a p-type dopant for straining a p-type FET. The p-type dopant includes boron or BF₂, and the strained materials **214** may be epitaxial-grown by LPCVD process with in-situ doping. Alternatively, the first layers **322** may include SiC, which may be epitaxial-grown with an n-type dopant for straining an n-type FET. The n-type dopant includes arsenic and/or phosphorus, and the strained materials **214** may be epitaxial-grown by LPCVD process with in-situ doping.

In some embodiments, the first layers **322** are grown to have substantially identical size. The first layers **322** may be symmetrical to one another, as shown in FIG. 10. However, the disclosure is not limited thereto. Alternatively, the first layers **322** may be grown to have different sizes. In some embodiments, the first layers **322** located at the same side of the fin structures **310a'** along the direction X and arranged along the direction Y may be grown to physically spacing away from each other, which may be considered as discrete pieces, as shown in FIG. 10.

Referring to FIG. 11, in some embodiments, a dielectric layer **324** is formed over the structure depicted in FIG. 10. In some embodiments, the dielectric layer **324** is disposed over the semiconductor substrate **304** to cover (e.g., in physical contact with) the illustrated top surfaces of the first layers **322** and the illustrated top surfaces **S312tr** of portions of the isolation structures **312** surrounding by (e.g., enclosed by) the edges of the isolation structures **312** exposed by the first layers **322**. In the case, sidewalls of the first layers **322**, sidewalls of the edges of the isolation structures **312** exposed by the first layers **322**, the sidewalls of the fin structures **310a'**, and the sidewalls of the gate spacers **318** are free of the dielectric layer **324**. The dielectric layer **324** may be referred to as a coverage dielectric layer or a buffer layer. A material of the dielectric layer **324** may include SiO₂, AlO, AlON, ZrO, HfO, TiO, ZrAlO, ZnO, SiN, SiOCN, SiCN, or other suitable dielectric materials or combinations thereof, and a method for forming the dielectric layer **324** may include a deposition process, such as a CVD process or the like. A thickness (T₃₂₄ in FIG. 27B) of the dielectric layer **324** measured along the direction Z may be in a range of about 1 nm to about 10 nm. Although other value of the thickness of the dielectric layer **324** is possible depending on product and process requirements. Alternatively, the dielectric layer **324** may be optional and can be omitted.

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Referring to FIG. 11, in some embodiments, a plurality of source/drain regions 326 are formed over the first layers 322 and on the dielectric layer 324. In some embodiments, the dielectric layer 324 is disposed between (e.g., in physical contact with) the first layers 322 and the source/drain regions 326. The source/drain regions 326 may be coupled to the exposed surfaces of the second semiconductor layers 308 of the fin structures 310a' (along the Y-direction) and the inner spacers 320. The source/drain regions 326 may each include silicon germanium, indium arsenide, indium gallium arsenide, indium antimonide, germanium arsenide, germanium antimonide, indium aluminum phosphide, indium phosphide, any other suitable material, or combinations thereof. The source/drain regions 326 may be formed using an epitaxial layer growth process on the exposed surfaces of each of the second semiconductor layers 308 and the inner spacers 320. The material of the source/drain regions 326 may be doped with a conductive dopant. For example, a strained material is epitaxially grown with an n-type dopant (or a p-type dopant) for straining the source/drain regions 326 in the n-type region (or the p-type region). That is, the strained material is doped with the n-type dopant (or the p-type dopant) to be the source/drain regions 326 of the p-type FET (or the n-type FET). Owing to the dielectric layer 324, a leakage current from silicon to the source/drain regions 326 can be minimized.

In some embodiments, the source/drain regions 326 are grown to have substantially identical size. The source/drain regions 326 may be symmetrical to one another, as shown in FIG. 11. However, the disclosure is not limited thereto. Alternatively, the source/drain regions 326 may be grown to have different sizes. In some embodiments, the source/drain regions 326 located at the same side of the fin structures 310a' along the direction X and arranged along the direction Y may be grown to physically spacing away from each other, which may be considered as discrete pieces, as shown in FIG. 11. Alternatively, the source/drain regions 326 located at the same side of the fin structures 310a' along the direction X and arranged along the direction Y may be grown to physically connected to each other, which may be together considered as an integral piece. The source/drain regions 326 may be coupled to the exposed surfaces of the second semiconductor layers 308 of the fin structures 310a' (along the Y-direction) and the inner spacers 320.

In some embodiments, one first layer 322, a respective one source/drain region 326 overlying thereto, and the dielectric layer 324 (if any) disposed therebetween together may be referred to as an epitaxial structure (not labeled) of the semiconductor device 10. That is, the epitaxial structures is disposed as a multi-layered structure, with different layers having different degrees of doping. In some embodiments, an illustrated bottom surface of the epitaxial structures (including the first layer 322, the source/drain regions 326, and the dielectric layer 324 disposed therebetween) may be lower than the illustrated top surfaces (e.g., S312tr) of the neighboring isolation structures 312. Alternatively, the illustrated bottom surface of the epitaxial structures may be substantially leveled with the illustrated top surfaces (e.g., S312tr) of the neighboring isolation structures 312. Alternatively, the epitaxial structures may be disposed as a single-layered structure. It should be noted that the epitaxial structures may have other types of configurations, while remaining within the scope of the disclosure. The epitaxial structures may be referred to as source/drain structures or source/drain features.

Referring to FIG. 12A and FIG. 12B, in some embodiments, a dielectric layer 328 is globally formed on the

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structure depicted in FIG. 11. The dielectric layer 328 may be completely disposed on the epitaxial structures, the sacrificial gate structures, the gate spacers 318, and the isolation structures 312 exposed therefrom, as shown in FIG. 12A and FIG. 12B. The dielectric layer 328 includes, for example, a suitable material such as silicon nitride, silicon oxynitride, silicon nitride with oxygen or carbon elements, the like or combinations thereof. In some embodiments, the dielectric layer 328 is deposited by using processes such as CVD (e.g., high density plasma (HDP) CVD or sub-atmospheric CVD (SACVD)), ALD, molecular layer deposition (MLD), or other suitable methods. The dielectric layer 328 functions as a protection layer that effectively blocks water or moisture from penetrating into the elements underlying thereto or damages from the subsequent process(es) such as an etching process. The dielectric layer 328 may be referred to as a protection layer, an etch stop layer or a contact etch stop (CES) layer. A thickness of the dielectric layer 328 measured along the direction Z may be in a range of about 1 nm to about 6 nm. Although other value of the thickness of the dielectric layer 328 is possible depending on product and process requirements.

Thereafter, an interlayer dielectric (ILD) layer 330 is formed over the dielectric layer 328, in some embodiments. For example, the ILD layer 330 is disposed at opposing sides (along the Y-direction) of each sacrificial gate structures to overlay the epitaxial structures and the isolation structures 312 exposed therefrom, with the dielectric layer 328 disposed therebetween. The ILD layer 330 may be formed of a dielectric material such as silicon oxide, silicon nitride, silicon oxynitride, silicon carbonitride, silicon carbide oxynitride, spin-on glass (SOG), tetraethyl-orthosilicate (TEOS) oxide, phosphosilicate glass (PSG), borosilicate glass (BSG), boron-doped phosphosilicate Glass (BPSG), fluoride-doped silicate glass (FSG), undoped silicate glass (USG), carbon doped silicon oxide (e.g., SiCOH), polyimide, and/or a combination thereof. In some alternative embodiments, the ILD layer 330 may include low-K dielectric materials. It should be noted that the low-K dielectric materials are generally dielectric materials having a dielectric constant lower than 3.9. Examples of low-K dielectric materials include BLACK DIAMOND® (Applied Materials of Santa Clara, Calif.), Xerogel, Aerogel, amorphous fluorinated carbon, Parylene, BCB (bis-benzocyclobutenes), Flare, SILK® (Dow Chemical, Midland, Mich.), hydrogen silsesquioxane (HSQ) or fluorinated silicon oxide (SiOF), and/or a combination thereof. It is understood that the ILD layer 330 may include one or more dielectric materials. In some embodiments, the ILD layer 330 is formed to a suitable thickness by CVD such as flowable chemical vapor deposition (FCVD), HDP CVD, SACVD, spin-on, sputtering, or other suitable methods.

In some embodiments, a material layer of the dielectric layer 328 may be conformally formed over the epitaxial structures and the isolation structures 312, the sacrificial gate structures, and the gate spacers 318. Next, a material layer of the ILD layer 330 may be formed over the dielectric layer 328 and fills the second trenches 310T2. Subsequently, a planarization process (e.g., a grinding process, a chemical mechanical polishing (CMP) process, an etching process, or combinations thereof) may be performed to remove excess materials of the dielectric layer 328 and the ILD layer 330. In some embodiments, the planarization process may also remove the capping structures 1004 to expose illustrated top surfaces of the dummy gate electrodes 316 of the sacrificial gate structures. After the planarization process, the illustrated top surfaces of the ILD layer 330 and the dielectric

layer 328 may be substantially leveled with (e.g., coplanar to) illustrated top surfaces of the sacrificial gate structures (e.g., the illustrated top surfaces of the dummy gate electrodes 316) and illustrated top surfaces of the gate spacers 318, within process variations.

In certain cases, parts of top portions of the sacrificial gate structures may also be removed during the planarization process. After the planarization process, a cleaning step may be optionally performed, for example to clean and remove the residue generated from the planarization process.

Referring to FIG. 13A and FIG. 13B, in some embodiments, after the sacrificial gate structures are accessibly revealed, the dummy gate electrodes 316 and the dummy gate dielectric layers 314 of the sacrificial gate structures are removed to form first recesses between a respective pair of the gate spacers 318, and the first semiconductor layer 306' are also removed to second recesses between a respective pair of the inner spacers 320. The removal process may include an etching process (such as a dry etching, a wet etching, or a combination thereof) or any other suitable process. In some embodiments, a method for removing the first semiconductor layer 306' may include an isotropic etching process. By properly selecting etchants for the etching process and/or properly selecting the materials of the first semiconductor layer 306' and the second semiconductor layers 308, the first semiconductor layer 306' can be etched without removing the second semiconductor layers 308 and other components in the current structure. In the case, the isotropic etching process is a selective etching process, while leaving the second semiconductor layers 308 substantially intact. During the removal process, the ILD layer 330 and the dielectric layer 328 may protect the epitaxial structures and the isolation structures 312. In some embodiments the first recesses and the second recesses are spatially communicated to each other to form cavities 1006. That is, the second semiconductor layers 308 of the fin structures 310a' are released. The respective, illustrated bottom surface and illustrated top surface of each of the second semiconductor layers 308 may be exposed (e.g., accessibly revealed) by the cavities 1006, as shown in FIGS. 13A and 13B. In addition, inner sidewalls of the inner spacers 320 previously covered by the first semiconductor layer 306' are currently exposed in the cavities 1006.

Referring to FIG. 14A and FIG. 14B, in some embodiments, interfacial (IL) layers 332, gate dielectric layers 334, and gate electrodes 336 are formed in the cavities 1006 defined between adjacent gate spacers 318 and the inner spacers 320. One interfacial layer 332, one gate dielectric layer 334, and one gate electrode 336 disposed in one cavity 1006 may be collectively referred to as a gate structure 338 of the semiconductor device 10. That is, the previously shown sacrificial gate structures may be regarded as being replaced by the gate structures 338. Up to here, a plurality of transistors 302 included in the semiconductor device 10 is manufactured. The transistors 302 respectively include one of the gate structures 338, the second semiconductor layers 308 in this gate structure 338, and a pair of source/drain structures at opposite sides of this gate structure 338. In addition, the transistors 302 respectively include may further includes the inner spacers 320 in this gate structure 338 and a pair of gate spacer 318 at the opposite sides of this gate structure 338. The number and configurations of the transistors 302 formed in the semiconductor device should not be limited by the embodiments or drawings of this disclosure. It is understood that the number and configurations of the transistors 302 may have different material or configurations depending on product designs.

As shown in FIG. 14A and FIG. 14B, the interfacial layers 332 are lining on the exposed, illustrated top and illustrated bottom surfaces of the second semiconductor layers 308 and the illustrated top surface of the semiconductor substrate 304 underlying the second semiconductor layers 308, for example. In the case, the gate dielectric layers 334 are lining on exposed surfaces of the interfacial layers 332, the inner spacers 320 and the gate spacers 318, and the gate electrodes 336 fill the remainder space in these cavities 1006. The interfacial layers 332 may include a dielectric material such as silicon oxide layer or silicon oxynitride. In some embodiments, the interfacial layers 332 may be formed by a deposition process such as ALD, CVD, and/or other suitable deposition methods. The interfacial layers 332 may be adapted to provide a good interface between the semiconductor surface (i.e., the second semiconductor layers 308) and a gate insulator (i.e., the gate dielectric layers 334) and to suppress the mobility degradation of the channel carrier of the transistors 302. A material of the gate dielectric layer 334 may include a high-k dielectric material. In some embodiments, low-k dielectric materials are generally dielectric materials having a dielectric constant greater than about 4, greater than about 12, greater than about 16, or even greater than about 20. Examples of the high-k dielectric material may include HfO₂, HfSiO, HfSiON, HfTaO, HfTiO, HfZrO, zirconium oxide, aluminum oxide, titanium oxide, hafnium dioxide-alumina (HfO₂—Al₂O₃) alloy, other suitable high-k dielectric materials, and/or combinations thereof. The gate dielectric layer 334 may be one-layer structure or a multi-layer structure of different sublayers. The gate dielectric layer 334 may be referred to as a high-k dielectric layer. A method for forming the gate dielectric layers 334 may include a deposition process, such as a CVD process or an ALD process. A material of the gate electrodes 336 may include polysilicon, aluminum, copper, titanium, tantalum, tungsten, cobalt, molybdenum, tantalum nitride, nickel silicide, cobalt silicide, TiN, WN, TiAl, TiAlN, TaCN, TaC, TaSiN, metal alloys, other suitable materials, and/or combinations thereof. A method for forming the gate electrodes 336 may include a deposition process (e.g., a CVD process or an ALD process), a plating process (e.g., an electrical plating process or an electroless plating process) or a combination thereof. Throughout the description, the term "copper" is intended to include substantially pure elemental copper, copper containing unavoidable impurities, and copper alloys containing minor amounts of elements such as tantalum, indium, tin, zinc, manganese, chromium, titanium, germanium, strontium, platinum, magnesium, aluminum or zirconium, etc.

In some embodiments, one or more work function layer (not shown) is formed between each gate dielectric layer 334 and the overlying gate electrode 336. A material of the work function layer may include p-type work function metals or n-type work function metals. For example, the p-type work function metals include TiN, TaN, Ru, Mo, Al, WN, ZrSi₂, MoSi₂, TaSi₂, NiSi₂, WN, other suitable p-type work function materials, or combinations thereof. For example, the n-type work function metals include Ti, Ag, TaAl, TaAlC, TiAlN, TaC, TaCN, TaSiN, Mn, Zr, other suitable n-type work function materials, or combinations thereof. In some embodiments, the method of forming the work function layer includes performing at least one suitable deposition technique, such as CVD (e.g., PECVD), ALD (e.g., remote plasma atomic layer deposition (RPALD), plasma enhanced atomic layer deposition (PEALD), molecular beam deposi-

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tion (MBD), or the like. The work function layer may serve the purpose of adjusting threshold voltage (V_t) of the transistors 302.

Referring to FIG. 15A and FIG. 15B, in some embodiments, an ILD layer 340 is formed over the transistor 302 and extends onto the illustrated top surfaces of the ILD layer 330, the dielectric layer 328 and the gate spacers 318. The formation and material of the ILD layer 340 is similar to or substantially identical to the formation and material of the ILD layer 330 as described in FIG. 12A and FIG. 12B, and thus are not repeated herein for brevity. Thereafter, a plurality of through openings (not labeled) may be formed in the ILD layer 340 and further extend into the ILD layer 330 and the dielectric layer 328 to expose (e.g., accessibly reveal) portions of the source/drain regions 326, as shown in FIGS. 15A and 15B. The through openings may be formed by patterning the ILD layer 340, the ILD layer 330 and the dielectric layer 328 with lithography and etching processes. The etching process may include a dry etching, a wet etching, or a combination thereof.

In some embodiments, after forming the through openings in the ILD layer 340, the ILD layer 330 and the dielectric layer 328, a plurality of contact plugs 344 are formed in the through openings to couple with the source/drain regions 326. The contact plugs 344 may be referred to as metal contacts or metallic contacts to the source/drain regions 326. For example, the contact plugs 344 electrically coupled to the source/drain regions 326 are referred to as source/drain contacts. In some embodiments, the contact plugs 344 may include ruthenium (Ru), cobalt (Co), copper (Cu), titanium (Ti), titanium nitride (TiN), tantalum (Ta), tantalum nitride (Ta₂N₃), molybdenum (Mo), nickel (Ni), a combination of thereof, or the like. The contact plugs 344 may be formed by, for example, plating such as electroplating or electroless plating, CVD such as PECVD, ALD, and PVD, a combination thereof, or the like. Seed layers (not shown) may be optionally formed before forming the contact plugs 344 to line sidewalls and illustrated bottoms of the through openings. That is, for example, each of the seed layers covers an illustrated bottom surface and sidewalls of a respective one of the contact plugs 344. In some embodiments, each of the seed layers is a metal layer, which may be a single layer or a composite layer comprising a plurality of sub-layers formed of different materials. The seed layers are formed using, for example, PVD or the like. In one embodiment, the seed layers may be omitted.

Barrier layer or adhesive layers 342 may be optionally formed before forming the contact plugs 344. In some embodiments, the barrier layer or adhesive layers 342 may be optionally formed between the contact plugs 344 and the ILD layer 340, between the contact plugs 344 and the ILD layer 330, and between the contact plugs 344 and the dielectric layer 328. Owing to the additional barrier layer or adhesive layers 342, it is able to ensure the adhesion between the contact plugs 344 and the ILD layer 340, between the contact plugs 344 and the ILD layer 330, and between the contact plugs 344 and the dielectric layer 328. As shown in FIG. 15A and FIG. 15B, the barrier layer or adhesive layers 342 line on the sidewalls of the contact plugs 344, where the contact plugs 344 respectively stand on the source/drain regions 326, for example. The additional barrier layer or adhesive layers 342 may include Ti, TiN, Ta, TaN, a combination thereof, a multilayer thereof, or the like, and may be formed using CVD, ALD, PVD, a combination thereof, or the like. In an alternative embodiment of which the seed layer is included, the additional barrier layer or adhesive layer 342 is interposed between the seed layer and

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the ILD layer 340, between the seed layer and the ILD layer 330 and between the seed layer and the dielectric layer 328, where the seed layer is interposed between the contact plugs 344 and the additional barrier layer or adhesive layer 342. In the embodiments of the seed layer is presented, owing to the additional barrier layer or adhesive layer 342, it is able to prevent the seed layer and/or the contact plugs 344 from diffusing to the underlying layers and/or the surrounding layers. In one embodiment, the additional barrier layer or adhesive layer 342 may be omitted. As shown in FIG. 15A, sidewalls of the contact plugs 344 may be substantially vertical sidewalls. The disclosure is not limited thereto, alternatively, the sidewalls of the contact plugs 344 may be slant sidewalls or step-shaped sidewalls.

Continued on FIG. 15A and FIG. 15B, a plurality of conductive vias 348 are formed over the transistor 302 and the contact plugs 344, and a dielectric layer 346 are formed to laterally cover the conductive vias 348. As shown in FIG. 15A and FIG. 15B, illustrated top surface of the conductive vias 348 may be accessibly revealed by the dielectric layer 346 for electrical connection with later-formed elements, such as conductive features in a later-formed interconnect or interconnection structure. The dielectric layer 346 may be referred to as an ILD layer, while the conductive vias 348 may be referred to as contact vias or metallic vias. For example, some of the conductive vias 348 are electrically connected to the contact plugs 344 connected to the source/drain regions 326, and some of the conductive vias 348 are electrically connected to the gate electrodes 336 of the gate structure 338. The conductive vias 348 electrically connected to the contact plugs 344 connected to the source/drain regions 326 may be referred to as source/drain contacts, and the conductive vias 348 electrically connected to the gate electrodes 336 of the gate structure 338 may be referred to as gate contacts.

In some embodiments, a material of the dielectric layer 346 includes polyimide, epoxy resin, acrylic resin, phenol resin, BCB, polybenzoxazole (PBO), or any other suitable polymer-based dielectric material. The dielectric layer 346, for example, may be formed by suitable fabrication techniques such as spin-on coating, CVD, PECVD, or the like. In some embodiments, the conductive vias 348 may include copper (Cu), copper alloys, nickel (Ni), aluminum (Al), manganese (Mn), magnesium (Mg), silver (Ag), gold (Au), tungsten (W), a combination of thereof, or the like. Seed layers (not shown) may be optionally formed before forming the conductive vias 348 to line an illustrated bottom surface and sidewalls of a respective one of the conductive vias 348 or to line an illustrated bottom surface of a respective one of the conductive vias 348. In some embodiments, each of the seed layers is a metal layer, which may be a single layer or a composite layer comprising a plurality of sub-layers formed of different materials. The seed layers are formed using, for example, PVD or the like. In one embodiment, the seed layers may be omitted. The conductive vias 348 may be formed by, for example, plating such as electroplating or electroless plating, CVD such as PECVD, ALD, and PVD, a combination thereof, or the like. In the disclosure, the transistors 302, the contact plugs 344 and the conductive vias 348 are formed in front-end-of-line (FEOL) process.

Thereafter, an interconnection structure 100 may be formed over the conductive vias 348 and the dielectric layer 346. For example, the interconnection structure 100 may include a stack 130 of dielectric layers and interconnections (110a, 120a, and 110b) formed in the stack 130 of dielectric layers. The interconnections (110a, 120a, and 110b) are electrically connected to the transistors 302 through the

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conductive vias **348** and the dielectric layer **346**. As shown in FIG. **15A** and FIG. **15B**, the interconnections includes conductive layers **110a**, **110b** and conductive vias **120a** alternately stacked upon one another (along the direction **Z**), for example. The conductive layers **110a**, **110b** are connected and electrically coupled to each other through the conductive **120a**, and the conductive layer **110a** is connected and electrically coupled to the conductive vias **348**, so to provide routing function to the transistors **302**. The formation and material of each of the conductive layers **110a**, **110b** and the conductive vias **120a** are similar to or substantially identical to the formation and material of the conductive vias **348**, the formation and material of the dielectric layers included in the stack **130** are similar to or substantially identical to the formation and material of the dielectric layer **346**, and thus are not repeated herein for brevity. In the disclosure, the interconnection structure **100** is formed in back-end-of-line (BEOL) process. The interconnection structure **100** may be referred to as a front-side interconnect, a front-side interconnection, or a front-side interconnection structure to provide routing functions to the transistors **302** and/or other devices formed underneath thereto. For illustrative purpose, only two build-up layers (e.g., a first build-up layer including the conductive layer **110a**, conductive vias **120a** and a portion of the stack **130a** laterally covering the conductive layer **110a** and conductive vias **120a**, and a second build-up layer including the conductive layer **110b** and a portion of the stack **130a** laterally covering the conductive layer **110b**) are shown in the interconnection structure **100**, however the disclosure is not limited to the embodiments and/or drawings. The interconnection structure **100** may include one or more than one first build-up layer and one or more than one second build-up layer alternatively stacked along the direction **Z**.

Referring to FIG. **16**, in some embodiments, the structure depicted in FIG. **15A** and FIG. **15B** are flipped (turned upside down) and secured by a holding device (not shown), where portions of the semiconductor substrate **304**, the isolation structures **312**, the first layers **322** and the dielectric layer **324** are removed. The removal process may be performed by a planarization process such as a grinding process, a CMP process, an etching process, or combinations thereof. Such removal process may be referred to a thinning process. After the planarization process, a cleaning step may be optionally performed, for example to clean and remove the residue generated from the planarization process. As shown in FIG. **16**, an illustrated top surface of the current structure is a flat surface includes a surface of the semiconductor substrate **304**, surfaces of the isolation structures **312**, surfaces of the first layers **322**, surfaces of the dielectric layer **328**, surface of the ILD layer **330**, for example. In addition, illustrated top surface of the current structure depicted in FIG. **16** is opposite to an outermost surface (e.g., an exposed surface) of the interconnection structure **100** along the direction **Z**.

Referring to FIG. **17**, in some embodiments, the semiconductor substrate **304** exposed from the illustrated top surface of the current structure are at least partially removed to form a plurality of openings **1008** exposing an illustrated top surface of the remaining semiconductor substrate **304**, sidewalls of the first layers **322**, and sidewalls of the isolation structure **312**. A method for partially removing the semiconductor substrate **304** may include an etching process, such as an isotropic etching process. By properly selecting etchants for the etching process and/or by properly selecting the materials of the semiconductor substrate **304**, the isolation structures **312**, the first layers **322**, the dielectric

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layer **328** and the ILD layer **330**, where the semiconductor substrate **304** can be etched without consuming the isolation structures **312**, the first layers **322**, the dielectric layer **328** and the ILD layer **330** and other components in the current structure.

Referring to FIG. **18**, in some embodiments, dielectric layers **350** are formed to line sidewalls and illustrated bottoms of the openings **1008**, and the dielectric regions **352** are formed to fill the remainder space in these openings **1008**. One dielectric layer **350** and one dielectric region **352** disposed in a respective one opening **1008** may be referred to as a self-aligned dielectric (SAD). A material of the insulating material for forming the dielectric layers **350** may include silicon nitride, silicon carbide nitride, or other suitable dielectric materials without oxygen elements, or combinations thereof. A thickness (**T350** in FIG. **18** and FIG. **27B**) of the dielectric layers **350** measured along the direction **Z** may be in a range of about 0.3 nm to about 3 nm. Although other value of the thickness of the dielectric layers **350** is possible depending on product and process requirements. A material of the insulating material for forming the dielectric regions **352** may include SiO, SiOC, AlO, AlON, ZrO, HfO, TiO, ZrAlO, ZnO, SiOCN, SiCN, or other suitable dielectric materials or combinations thereof. A thickness (**T352** in FIG. **18** and FIG. **27B**) of the dielectric regions **352** measured along the direction **Z** may be in a range of about 5 nm to about 30 nm. A width (**W352** in FIG. **18** and FIG. **27B**) of the dielectric regions **352** measured along the direction **Z** may be in a range of about 5 nm to about 30 nm. Although other values of the thickness and width of the dielectric regions **352** are possible depending on product and process requirements. In some embodiments, the insulating material layer of dielectric layers **350** is formed by using a deposition process (e.g., a CVD process or an ALD process), the insulating material of dielectric regions **352** is formed by using a deposition process (e.g., a CVD process or an ALD process), and the portions of these material layers are removed by using a planarization process such as a grinding process, a CMP process, an etching process, or combinations thereof to form the dielectric layers **350** and the dielectric regions **352** in the openings **1008**.

Referring to FIG. **19**, in some embodiments, a hard mask layer **354** is formed on the structure depicted in FIG. **18**, where the hard mask layer **354** includes a plurality of openings **1010** exposing portions of the structure depicted in FIG. **18**. In some embodiments, the hard mask layer **354** is disposed on the illustrated top surface of the current structure and extends along the X-Y plane. In some embodiments, the hard mask layer **354** includes silicon oxynitride, silicon carbon nitride, or high-K dielectrics. High-K dielectrics includes metal oxides. It should be noted that the high-K dielectric materials are generally dielectric materials having a dielectric constant greater than 4. Examples of metal oxides used for high-K dielectrics include oxides of Li, Be, Mg, Ca, Sr, Sc, Y, Zr, Hf, Al, La, Ce, Pr, Nd, Sm, Eu, Gd, Tb, Dy, Ho, Er, Tm, Yb, Lu, and/or mixtures thereof. The hard mask layer **354** may be formed using a suitable process such as ALD, CVD, PVD, thermal oxidation, UV-ozone oxidation, or combinations thereof.

Thereafter, the first layers **322** exposed by the openings **1010** formed in hard mask layer **354** are removed, in some embodiments, as shown in FIG. **19**. The removal process may be performed by properly selecting etchants for the etching process and/or properly selecting the materials of the first layers **322** and the hard mask layer **354**, where the first layers **322** can be etched without removing the hard mask layer **354** and other components in the current structure. In

the case, the isotropic etching process is a selective etching process, while leaving the hard mask layer **354** substantially intact. After the removal process, a cleaning step may be optionally performed, for example to clean and remove the residue generated from the planarization process. In certain cases, parts of top portions of the self-aligned dielectrics (SADs), which are also exposed by the openings **1010** formed in hard mask layer **354**, may also be removed during the removal process. A loss amount (L1 in FIG. 27B) of the self-aligned dielectrics (SADs) including the dielectric regions **352** and the dielectric layers **350** measured along the direction Z may be in a range of about 0.1 nm to about 10 nm. Although other value of the loss amount (L1 in FIG. 27B) of the self-aligned dielectrics (SADs) including the dielectric regions **352** and the dielectric layers **350** is possible depending on product and process requirements. As shown in FIG. 19, after the removal process, the dielectric layer **324**, sidewalls of the semiconductor substrate **304**, and sidewalls of some of the inner spacer **320** are exposed by the openings **1010** formed in hard mask layer **354**, for example.

Referring to FIG. 20, in some embodiments, liner layers **356** are formed on sidewalls of the openings **1010**, the exposed sidewalls of the semiconductor substrate **304**, and the exposed sidewalls of some of the inner spacer **320**. That is, the liner layers **356** are vertical liners to only line the sidewalls of the openings **1010**, the exposed sidewalls of the semiconductor substrate **304**, and the exposed sidewalls of some of the inner spacer **320**. A material of the liner layers **356** may include SiO, AlO, AlON, ZrO, HfO, TiO, ZrAlO, ZnO, SiN, SiOCN, SiCN, or other suitable dielectric materials or combinations thereof, and a method for forming the liner layers **356** may include a deposition process (such as a CVD process, an ALD process or the like) and a patterning process (such as an etching process or the like). A thickness (T356 in FIG. 27B) of the liner layers **356** measured along the direction Z may be in a range of about 1 nm to about 10 nm. Although other value of the thickness of the liner layers **356** is possible depending on product and process requirements. The liner layers **356** may be formed by, but not limited to, initially forming a material layer of the liner layers **356** globally covering the structure shown in FIG. 19, and then performing an anisotropic etching process to remove a portion of the material layer extending along the X-Y plane, and thus the remaining material layer (extending along the direction Z) forms the liner layers **356** disposed at the sidewalls of the openings **1010**, the exposed sidewalls of the semiconductor substrate **304**, and the exposed sidewalls of some of the inner spacer **320**. During the etching, the dielectric layer **324** is also removed to expose (accessibly reveal) the source/drain regions **326**, as shown in FIG. 20, for example. As shown in FIG. the self-aligned dielectrics (SADs) exposed (e.g., accessibly revealed) by the openings **1010** formed in the hard mask layer **354** have illustrated top surfaces being exposed by the hard mask layer **354** and the liner layers **356** and have sidewalls exposed by the hard mask layer **354** and covered by the liner layers **356**, for example.

Referring to FIG. 21, in some embodiments, a plurality of silicide layers **358** are formed over the source/drain regions **326** exposed by the liner layers **356** and the hard mask layer **354**. In some embodiments, each of the silicide layers **358** are conformally disposed on an exposed surface of a respective one of the source/drain regions **326**. In some embodiments, the silicide layers **358** serves the function of providing better electrical conduction between the previously formed elements (e.g., the exposed source/drain regions **326**) and later-formed elements (e.g., metal contacts) due to

a lower electrical resistance between the silicide layers **358** and the later-formed elements. A material of the silicide layers **358** may include TiSi, MoSi, NiSi, CoSi, WSi, RuSi, or other suitable dielectric materials or combinations thereof, and a method for forming the liner layers **356** may include a deposition process (such as a CVD process, an ALD process or the like) or other suitable process. A thickness (T358 in FIG. 27B) of the silicide layers **358** measured along the direction Z may be in a range of about 2 nm to about 10 nm. Although other value of the thickness of the liner layers **356** is possible depending on product and process requirements.

Referring to FIG. 22, in some embodiments, a plurality of contact plugs **362** are formed in the openings **1010** formed in the hard mask layer **354** and further extend downward (along the direction Z) to be coupled with the source/drain regions **326** through the silicide layers **358**. In some embodiments, the contact plugs **362** may include copper (Cu), tungsten (W), ruthenium (Ru), cobalt (Co), titanium (Ti), titanium nitride (TiN), tantalum (Ta), tantalum nitride (Ta₂N₅), molybdenum (Mo), a combination of thereof, or the like. The contact plugs **362** may be formed by, for example, plating such as electroplating or electroless plating, CVD such as PECVD, ALD, and PVD, a combination thereof, or the like. A thickness (T362 in FIG. 27B) of the contact plugs **362** measured along the direction Z may be in a range of about 10 nm to about 50 nm. Although other value of the thickness of the contact plugs **362** is possible depending on product and process requirements. The material of the contact plugs **362** may be the same as or different from the material of the contact plugs **344**.

As shown in FIG. 22, for example, the contact plugs **362** have a step-shape sidewall profile, where each of the contact plugs **362** has a bottom portion and a top portion connected to and disposed on the bottom portion, and the bottom portion is disposed at a location closer to the source/drain regions **326** than the top portion is (along the direction Z). In some embodiments, as measured along the direction X or Y, a width W1 of the top portion of each of the contact plugs **362** is greater than a width W2 of the bottom portion of each of the contact plugs **362**. However, the disclosure is not limited thereto; alternatively, the width W1 of the top portion of each of the contact plugs **362** may substantially equal to the width W2 of the bottom portion of each of the contact plugs **362**. The width (W2 in FIG. 27B) of the bottom portions of the contact plugs **362** measured along the direction X or Y may be in a range of about 5 nm to about 30 nm. Although other value of the width of the bottom portions of the contact plugs **362** is possible depending on product and process requirements.

Seed layers (not shown) may be optionally formed before forming the contact plugs **362** to line exposed sidewalls of the liner layers **356**, the exposed surfaces of the self-aligned dielectrics (SADs), and exposed surfaces of the silicide layers **358**. That is, for example, each of the seed layers covers an illustrated bottom surface and sidewalls of a respective one of the contact plugs **362**. In some embodiments, each of the seed layers is a metal layer, which may be a single layer or a composite layer comprising a plurality of sub-layers formed of different materials. The seed layers are formed using, for example, PVD or the like. In one embodiment, the seed layers may be omitted.

Barrier layer or adhesive layers **360** may be optionally formed before forming the contact plugs **362**. In some embodiments, the barrier layer or adhesive layers **360** may be optionally formed between the contact plugs **362** and the liner layers **356**, between the contact plugs **362** and the

self-aligned dielectrics (SADs), and between the contact plugs 362 and the silicide layers 358. Owing to the additional barrier layer or adhesive layers 360, it is able to ensure the adhesion between the contact plugs 362 and the liner layers 356, between the contact plugs 362 and the self-aligned dielectrics (SADs), and between the contact plugs 362 and the silicide layers 358. As shown in FIG. 22, the barrier layer or adhesive layers 360 line on the sidewalls of the contact plugs 362, where the contact plugs 362 respectively stand on the source/drain regions 326, for example. The additional barrier layer or adhesive layers 360 may include Ti, TiN, Ta, TaN, a combination thereof, a multilayer thereof, or the like, and may be formed using CVD, ALD, PVD, a combination thereof, or the like. A thickness (T360 in FIG. 27B) of the barrier layer or adhesive layers 360 measured along the direction Z may be in a range of about 0.3 nm to about 3 nm. Although other value of the thickness of the liner layers 356 is possible depending on product and process requirements. Alternatively, the barrier layer or adhesive layers 360 may be optional and can be omitted.

In an alternative embodiment of which the seed layer is included, the additional barrier layer or adhesive layer 360 is interposed between the seed layer and the liner layers 356, between the seed layer and the self-aligned dielectrics (SADs) and between the seed layer and the silicide layers 358, where the seed layer is interposed between the contact plugs 362 and the additional barrier layer or adhesive layer 360. In the embodiments of the seed layer is presented, owing to the additional barrier layer or adhesive layer 360, it is able to prevent the seed layer and/or the contact plugs 362 from diffusing to the underlying layers and/or the surrounding layers. In one embodiment, the additional barrier layer or adhesive layer 360 may be omitted. As shown in FIG. 22, the illustrated surface of the current structure is a flat surface including an illustrated top surface S354 of the hard mask layer 354, illustrated top surfaces S360 of the barrier layer or adhesive layer 360, and illustrated top surfaces S362 of the contact plugs 362, for example.

Referring to FIG. 23, in some embodiments, capping patterns 364 are formed on the illustrated top surface S362 of the contact plugs 362. In some embodiments, the contact plugs 362 are completely covered by the capping patterns 364. A thickness (T364 in FIG. 27B) of the capping patterns 364 measured along the direction Z may be in a range of about nm to about 4 nm. Although other value of the thickness of the capping patterns 364 is possible depending on product and process requirements. In certain case, the capping patterns 364 may further extend onto the liner layers 356 or extend onto the liner layers 356 and the hard mask layer 354. An extended length (L2 in FIG. 27B) of the capping patterns 364 measured along the direction X or Y may be in a range of about 0.2 nm to about 3 nm. Although other value of the extended length of the capping patterns 364 is possible depending on product and process requirements. However, the disclosure is not limited thereto. In alternative embodiments, the extended length L2 of the capping patterns 364 measured along the direction X or Y may be zero. In other words, the capping patterns 364 do not further extend onto the liner layers 356 or extend onto the liner layers 356 and the hard mask layer 354, see a device structure 300A' of a semiconductor device 10' depicted in FIG. 28A and FIG. 28B.

In some embodiments, the capping patterns 364 may include W_xN_y ($x, y > 1$), Ru_xN_y ($x, y > 1$), Co_xN_y ($x, y > 1$), Mo_xN_y ($x, y > 1$), TiN, Ta_xN_y ($x, y > 1$), a combination of thereof, or the like. The capping patterns 364 may be formed

by, for example, a nitridation treatment by thermal process or a nitridation treatment by implantation process.

In some embodiments, the nitridation treatment by thermal process is performed at a pressure in a range from about 0.05 Torr to about 10 Torr, at a temperature in a range from about 400° C. to about 500° C., at a RF power in a range from about 30 watts (W) to about 100 watts, with a gas flow including from about 5 standard cubic centimeters per minute (sccm) to about 1000 sccm of NH₃ (serving as a precursor), a gas flow including from about 5 sccm to about 1000 sccm of NH₃ (serving as a precursor) and from about 5 sccm to about 1000 sccm of H₂ as a carrier gas, or a gas flow including from about 5 sccm to about 1000 sccm of N₂ (serving as a precursor) and from about 5 sccm to about 1000 sccm of H₂ as a carrier gas. However, the disclosure is not limited thereto. For example, the carrier gas includes Ar or the like.

In alternative embodiments, the nitridation treatment by implantation process is performed at an ion beam energy less than or substantially equal to about 1 keV and tilt angle substantially equal to about 0 degree, with an implantation dose greater than about 1×10^{15} atom/cm² of N₂ (serving as a dopant species).

During the formation of the capping patterns 364, portions of the contact plugs 362 with the capping patterns 364 disposed thereon are consumed. For example, a consumption amount (L3 in FIG. 27B) of the capping patterns 364 measured along the direction Z may be in a range of about 0.2 nm to about 3 nm. Although other value of the consumption amount of the capping patterns 364 is possible depending on product and process requirements. In alternative embodiments, there is no contact plugs 362 being consumed during the formation of capping patterns 364; that is, the consumption amount L3 measured along the direction Z is zero. As shown in FIG. 23, the illustrated top surface of the current structure is a non-planar surface including illustrated top surfaces of the capping patterns 364 and the illustrated top surface S354 of the hard mask layer 354, where the illustrated top surfaces of the capping patterns 364 are above the illustrated top surface S354 of the hard mask layer 354, for example. The capping patterns 364 may be referred to as a nitride capping layer, nitride patterns, or nitride capping patterns. Up to here, a device structure 300A is manufactured, where the device structure 300A is disposed over and electrically coupled to the interconnection structure 100.

Referring to FIG. 24, in some embodiments, a hard mask layer 210 is disposed over the illustrated top surface of device structure 300A to cover up the hard mask layer 354 and the capping patterns 364. A material of the hard mask layer 210 may include SiO, SiOC, AlO, AlON, ZrO, HfO, TiO, ZrAlO, ZnO, SiOCN, SiCN, or other suitable dielectric materials or combinations thereof. A thickness (T210 in FIG. 27B) of the hard mask layer 210 measured along the direction Z may be in a range of about 5 nm to about 100 nm. Although other value of the thickness of the hard mask layer 210 is possible depending on product and process requirements. In some embodiments, the hard mask layer 210 is formed by using a deposition process, such as CVD.

Referring to FIG. 25, in some embodiments, the hard mask layer 210 is patterned to form a plurality of openings 1012 exposing portions of the device structure 300A. For example, the illustrated top surface S364 of the capping patterns 364 and the illustrated top surface S354 of the hard mask layer 354 are partially exposed by the openings 1012. The patterning may include lithography and etching process. The etching process may include a dry

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etching, a wet etching, a combination thereof. After the etching process, a cleaning step may be optionally performed, for example to clean and remove the residue generated from the etching process.

Referring to FIG. 26, in some embodiments, a first barrier material **222m**, a second barrier material **224m**, and a conductive material **226m** are sequentially formed over the structure depicted in FIG. 25. A material of the first barrier material **222m** may include TiN, TaN, or other suitable dielectric materials or combinations thereof, and may be formed by using a deposition process, such as CVD or ALD. A material of the second barrier material **224m** may include Co, Ru, W, Mo, or other suitable dielectric materials or combinations thereof, and may be formed by using a deposition process, such as CVD or ALD. A material of the conductive material **226m** may include Cu, Co, Ru, W, Mo, or other suitable dielectric materials or combinations thereof, and may be formed by using a deposition process, such as CVD or PVD. For example, as shown in FIG. 26, sidewalls and bottom surfaces of the openings formed in the hard mask layer **210** are lined by the first barrier material **222m**. That is, the illustrated top surface surfaces **S364** of the capping patterns **364** and the illustrated top surface **S354t** of the hard mask layer **354** exposed by the openings **1012** formed in the hard mask layer **210** are covered by the first barrier material **222m**.

Referring to FIG. 27A and FIG. 27B, in some embodiments, a planarization process is performed to remove portions of the first barrier material **222m**, the second barrier material **224m** and the conductive material **226m** above the openings **1012** to form first barrier layers **222a**, second barrier layers **224a**, and conductive contacts **226a** in the openings **1012** formed in the hard mask layer **210**. One first barrier layer **222a**, one second barrier layer **224a** and one conductive contact **226a** disposed in a respective one opening **1012** may together be referred to as a conductive feature, and the hard mask layer **210** may be referred to as a dielectric layer laterally covering the conductive features. A thickness (**T222a** in FIG. 27B) of the first barrier layers **222a** measured along the direction **Z** may be in a range of about 0.5 nm to about 10 nm. Although other value of the thickness of the first barrier layers **222a** is possible depending on product and process requirements. A thickness (**T224a** in FIG. 27B) of the second barrier layers **224a** measured along the direction **Z** may be in a range of about 0.5 nm to about 10 nm. Although other value of the thickness of the second barrier layers **224a** is possible depending on product and process requirements. Up to here, an interconnection structure **200** is manufactured.

In some embodiments, the conductive features collectively are considered as a conductive layer **220a**, where the conductive layer **220a** and the hard mask layer **210** may serve as a build-up layer of the interconnection structure **200** for providing interconnections between the transistors **302** and/or other devices or providing electrical connection with external devices or later-formed feature to the transistors **302** and/or other devices. In the disclosure, the interconnection structure **200** is also formed in BEO process. The interconnection structure **200** may be referred to as a back-side interconnect, a back-side interconnection, or a back-side interconnection structure to provide routing functions to the transistors **302** and/or other devices formed underneath thereto. For illustrative purpose, only one build-up layer is shown in the interconnection structure **200**, however the disclosure is not limited to the embodiments and/or drawings. For example, the processes described in FIG. 24

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through FIG. 27A and FIG. 27B can be repeated multiple times to obtain multiple build-up layers for the interconnection structure **200**.

Up to here, the semiconductor device **10** is manufactured, where the semiconductor device **10** includes the interconnection structure **100**, the interconnection structure **200**, and the device structure **300A**, and the interconnection structure **100** is disposed on the front side of the device structure **300A** and the interconnection structure **200** is disposed on the back side of the device structure **300A**. In some embodiments, the interconnection structure **100** is electrically coupled to the transistor **302** and other devices formed in the device structure **300A** for providing routing functions thereto, and the interconnection structure **200** is electrically coupled to the transistor **302** and other devices formed in the device structure **300A** for providing routing functions thereto. Owing to the capping patterns **364** disposed between the contact plugs **362** of the device structure **300A** and the conductive layer **220a** of the interconnection structure **200**, the metal diffusion between the contact plugs **362** of the device structure **300A** and the conductive layer **220a** of the interconnection structure **200** (e.g., a miscible phenomenon between Ru and Co or between Mo and W) can be greatly mitigate or greatly suppress, thereby improving the reliability of the semiconductor device. Due to the metal diffusion between the contact plugs **362** of the device structure **300A** and the conductive layer **220a** of the interconnection structure **200** can be prevented by the capping patterns **364**, a great, various selection in the material of forming the contact plugs **362** is promised.

FIGS. 29-32 and 33A are schematic three-dimensional views of various stages in a manufacturing method of a semiconductor device **20** in accordance with alternative embodiments of the disclosure. FIG. 33B is a schematic, enlarged cross-sectional view of a portion of the semiconductor device **20** outlined in a dashed box **V** as shown in FIG. 33A. FIG. 34A is a schematic three-dimensional view of a semiconductor device **20'** in accordance with some embodiments of the disclosure. FIG. 34B is a schematic, enlarged cross-sectional view of a portion of the semiconductor device **20'** outlined in a dashed box **V** as shown in FIG. 34A. The elements similar to or substantially the same as the elements described previously will use the same reference numbers, and certain details or descriptions (e.g., the materials, formation processes, positioning configurations, electrical connections, etc.) of the same elements would not be repeated herein.

Referring to FIG. 29, in some embodiments, a hard mask layer **210** is formed on the hard mask layer **354**, the dielectric layer **360** and the contact plugs **362**, following the process as described in FIG. 22. The formation and material of the hard mask layer **210** has been previously described in FIG. 24, and thus are not repeated herein. For example, an illustrated top surface of the current structure is a flat surface, as shown in FIG. 29.

Referring to FIG. 30, in some embodiments, the hard mask layer **210** is patterned to form a plurality of openings **1012** exposing portions of the contact plugs **362** and portions of the hard mask layer **354**. For example, the illustrated top surface surfaces **S362** of the contact plugs **362** and the illustrated top surface **S354t** of the hard mask layer **354** are partially exposed by the openings **1012**. The patterning may include lithography and etching process. The etching process may include a dry etching, a wet etching, a combination thereof. After the etching process, a cleaning step may be optionally performed, for example to clean and remove the residue generated from the etching process. A length (**L4** in

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FIG. 30 and FIG. 33A) of the portion of the contact plugs 362 (covered by the hard mask layer 210) measured along the direction X may be in a range of about 2 nm to about 10 nm. Although other value of the length (L4 in FIG. 30 and FIG. 33A) of the portion of the contact plugs 362 being covered by the hard mask layer 210 is possible depending on product and process requirements.

Referring to FIG. 31, in some embodiments, capping patterns 366 are formed on the illustrated top surface S362 of the contact plugs 362 partially exposed by the openings 1012 formed in the hard mask layer 210. In some embodiments, the contact plugs 362 are partially covered by the capping patterns 366. The formation and material of the capping patterns 366 are similar to or substantially identical to the formation and material of the capping patterns 366 as described in FIG. 23, and thus are not repeated herein. A thickness (T366 in FIG. 33B) of the capping patterns 366 measured along the direction Z may be in a range of about 0.5 nm to about 4 nm. Although other value of the thickness of the capping patterns 366 is possible depending on product and process requirements. In certain case, the capping patterns 366 may further extend onto the liner layers 356 or extend onto the liner layers 356 and the hard mask layer 354. An extended length (L2 in FIG. 33B) of the capping patterns 366 measured along the direction X or Y may be in a range of about 0.2 nm to about 3 nm. Although other value of the extended length of the capping patterns 366 is possible depending on product and process requirements.

During the formation of the capping patterns 366, portions of the contact plugs 362 with the capping patterns 366 disposed thereon are consumed. For example, a consumption amount (L3 in FIG. 33B) of the capping patterns 366 measured along the direction Z may be in a range of about 0.2 nm to about 3 nm. Although other value of the consumption amount of the capping patterns 366 is possible depending on product and process requirements. In alternative embodiments, there is no contact plugs 362 being consumed during the formation of capping patterns 366; that is, the consumption amount L3 measured along the direction Z is zero. As shown in FIG. 31, the illustrated top surfaces of the capping patterns 366 are above the illustrate top surface S354 of the hard mask layer 354, for example. The capping patterns 366 may be referred to as a nitride capping layer, nitride patterns, or nitride capping patterns. Up to here, a device structure 300B is manufacture, where the device structure 300B is disposed over and electrically coupled to the interconnection structure 100.

However, the disclosure is not limited thereto. In alternative embodiments, the extended length L2 of the capping patterns 366 measured along the direction X or Y may be zero. In other words, the capping patterns 366 do not further extend onto the liner layers 356 or extend onto the liner layers 356 and the hard mask layer 354, see a device structure 300B' of a semiconductor device 20' depicted in FIG. 34A and FIG. 34B.

Referring to FIG. 32, in some embodiments, a first barrier material 222m, a second barrier material 224m, and a conductive material 226m are sequentially formed over the structure depicted in FIG. 31. The formation and material of each of the first barrier material 222m, the second barrier material 224m, and the conductive material 226m have been previously described in FIG. 26, and thus are not repeated herein.

Referring to FIG. 33A and FIG. 33B, in some embodiments, a planarization process is performed to remove portions of the first barrier material 222m, the second barrier material 224m and the conductive material 226m above the

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openings 1012 to form first barrier layers 222a, second barrier layers 224a, and conductive contacts 226a in the openings 1012 formed in the hard mask layer 210. Up to here, an interconnection structure 200 is manufactured. The details of the interconnection structure 200 including the hard mask layer 210 and a conductive layer 220a (including the first barrier layers 222a, the second barrier layers 224a, and the conductive contacts 226a) laterally covered by the hard mask layer 210 has been described in FIG. 27A and FIG. 27B, and thus are not repeated herein, for brevity.

After the formation of the interconnection structure 200, the semiconductor device 20 is manufactured, where the semiconductor device 20 includes the interconnection structure 100, the interconnection structure 200, and the device structure 300B, and the interconnection structure 100 is disposed on the front side of the device structure 300B and the interconnection structure 200 is disposed on the back side of the device structure 300B. In some embodiments, the interconnection structure 100 is electrically coupled to the transistor 302 and other devices formed in the device structure 300B for providing routing functions thereto, and the interconnection structure 200 is electrically coupled to the transistor 302 and other devices formed in the device structure 300B for providing routing functions thereto. Owing to the capping patterns 366 disposed between the contact plugs 362 of the device structure 300B and the conductive layer 220a of the interconnection structure 200, the metal diffusion between the contact plugs 362 of the device structure 300B and the conductive layer 220a of the interconnection structure 200 (e.g., a miscible phenomenon between Ru and Co or between Mo and W) can be greatly mitigate or greatly suppress, thereby improving the reliability of the semiconductor device 20. Due to the metal diffusion between the contact plugs 362 of the device structure 300B and the conductive layer 220a of the interconnection structure 200 can be prevented by the capping patterns 366, a great, various selection in the material of forming the contact plugs 362 is promised.

In accordance with some embodiments, a semiconductor device includes two source/drain features, a gate structure, a first contact plug, a second contact plug, a conductive line, and a nitride capping layer. The two source/drain features are laterally arranged to each other. The one or more channel layers connects the two source/drain features. The gate structure engages the one or more channel layers and interposes the two source/drain features. The first contact plug extends from above a first source/drain feature of the two source/drain features to the first source/drain feature. The second contact plug extends from below a second source/drain feature of the two source/drain features to the second source/drain feature. The conductive line is disposed underneath the second contact plug and electrically coupled to the second contact plug. The nitride capping layer is disposed between the second contact plug and the conductive line.

In accordance with some embodiments, a semiconductor device includes a device structure, a first interconnection structure, and a second interconnection structure. The device structure has a front side and a rear side opposite to the front side, where the device structure includes at least one transistor, a first contact connected to the at least one transistor and exposed by the front side, a second contact connected to the at least one transistor and exposed by the rear side, and a capping layer disposed on the rear side and covering the second contact exposed by the rear side. The first interconnection structure is disposed on the front side of the device structure and includes a first conductive feature electrically

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coupled to the at least one transistor through the first contact. The second interconnection structure is disposed on the rear side of the device structure and includes a second conductive feature electrically coupled to the at least one transistor through the second contact, where the capping layer is interposed between the second contact and the second conductive feature.

In accordance with some embodiments, a method of method of manufacturing a semiconductor device includes the following steps: providing a device structure with a front side and a rear side opposite to the front side, the device structure comprising at least one transistor, a first contact connected to the at least one transistor and exposed by the front side, a second contact connected to the at least one transistor and exposed by the rear side, and a capping layer disposed on the rear side and covering the second contact exposed by the rear side; disposing a first interconnection structure on the front side of the device structure, the first interconnection structure comprising a first conductive feature electrically coupled to the at least one transistor through the first contact; and disposing a second interconnection structure on the rear side of the device structure, the second interconnection structure comprising a second conductive feature electrically coupled to the at least one transistor through the second contact, where the capping layer is interposed between the second contact and the second conductive feature.

The foregoing outlines features of several embodiments so that those skilled in the art may better understand the aspects of the disclosure. Those skilled in the art should appreciate that they may readily use the disclosure as a basis for designing or modifying other processes and structures for carrying out the same purposes and/or achieving the same advantages of the embodiments introduced herein. Those skilled in the art should also realize that such equivalent constructions do not depart from the spirit and scope of the disclosure, and that they may make various changes, substitutions, and alterations herein without departing from the spirit and scope of the disclosure.

What is claimed is:

1. A semiconductor device, comprising:

two source/drain features, laterally arranged to each other; one or more channel layers, connecting the two source/drain features;

a gate structure, engaging the one or more channel layers and interposing the two source/drain features;

a first contact plug, extending from above a first source/drain feature of the two source/drain features to the first source/drain feature;

a second contact plug, extending from below a second source/drain feature of the two source/drain features to the second source/drain feature;

a conductive line, disposed underneath the second contact plug and electrically coupled to the second contact plug; and

a nitride capping layer, disposed between the second contact plug and the conductive line, wherein sidewalls of the conductive line are free of the nitride capping layer.

2. The semiconductor device of claim 1, wherein each of the source/drain features comprises a first source/drain region and a second source/drain region disposed on the first source/drain region, wherein the first source/drain region is overlapped with the second contact plug in a direction perpendicular to a stacking direction of the first source/drain region and the second source/drain region,

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wherein the one or more channel layers and the gate structure are overlapped in the stacking direction.

3. The semiconductor device of claim 1, wherein the second contact plug has a first portion and a second portion disposed on and connected to the first portion, and a first lateral size of the first portion is less than a second lateral size of the second portion, wherein the first portion is between the second portion and the second source/drain feature.

4. The semiconductor device of claim 1, wherein a surface of the second contact plug disposed with the conductive line is partially exposed by the nitride capping layer.

5. The semiconductor device of claim 1, wherein a surface of the second contact plug disposed with the conductive line is entirely covered by the nitride capping layer.

6. The semiconductor device of claim 1, wherein a material of the nitride capping layer comprises tungsten nitride, ruthenium nitride, cobalt nitride, molybdenum nitride, titanium nitride, tantalum nitride, or a combination of thereof.

7. The semiconductor device of claim 1, further comprising:

a dielectric layer, laterally covering the second contact plug, wherein the nitride capping layer further extends onto a surface of the dielectric layer.

8. The semiconductor device of claim 1, wherein the nitride capping layer further extends into a portion of the second contact plug by a non-zero distance.

9. A semiconductor device, comprising:

a device structure, having a front side and a rear side opposite to the front side, wherein the device structure comprises at least one transistor, a first contact connected to the at least one transistor and exposed by the front side, a second contact connected to the at least one transistor and exposed by the rear side, and a capping layer disposed on the rear side and covering the second contact exposed by the rear side;

a first interconnection structure, disposed on the front side of the device structure and comprising a first conductive feature electrically coupled to the at least one transistor through the first contact; and

a second interconnection structure, disposed on the rear side of the device structure and comprising a second conductive feature electrically coupled to the at least one transistor through the second contact, wherein the capping layer is interposed between the second contact and the second conductive feature, and sidewalls of the second conductive feature are free of the capping layer.

10. The semiconductor device of claim 9, wherein the at least one transistor comprises a finFET transistor, a GAA transistor, or a nanowire transistor.

11. The semiconductor device of claim 9, wherein a surface of the second contact disposed with the second conductive feature is partially exposed by the capping layer.

12. The semiconductor device of claim 9, wherein a surface of the second contact disposed with the second conductive feature is entirely covered by the capping layer.

13. The semiconductor device of claim 9, wherein a material of the capping layer comprises tungsten nitride, ruthenium nitride, cobalt nitride, molybdenum nitride, titanium nitride, tantalum nitride, or a combination of thereof.

14. The semiconductor device of claim 9, wherein a size of the capping layer is greater than a size of the second contact along a direction perpendicular to a stacking direction of the device structure, the first interconnection structure, and the second interconnection structure.

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15. A method of manufacturing a semiconductor device, comprising:

providing a device structure with a front side and a rear side opposite to the front side, the device structure comprising at least one transistor, a first contact connected to the at least one transistor and exposed by the front side, a second contact connected to the at least one transistor and exposed by the rear side, and a capping layer disposed on the rear side and covering the second contact exposed by the rear side;

disposing a first interconnection structure on the front side of the device structure, the first interconnection structure comprising a first conductive feature electrically coupled to the at least one transistor through the first contact; and

disposing a second interconnection structure on the rear side of the device structure, the second interconnection structure comprising a second conductive feature electrically coupled to the at least one transistor through the second contact, wherein the capping layer is interposed between the second contact and the second conductive feature, and sidewalls of the second conductive feature are free of the capping layer.

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16. The method of claim **15**, wherein the capping layer is formed by a nitridation treatment, wherein:

the nitridation treatment is performed by a thermal process, or

the nitridation treatment is performed by an implantation process.

17. The method of claim **15**, wherein the capping layer is formed by tungsten nitride, ruthenium nitride, cobalt nitride, molybdenum nitride, titanium nitride, tantalum nitride, or a combination of thereof.

18. The method of claim **15**, wherein the capping layer is formed to have a size greater than a size of the second contact in a direction perpendicular to a stacking direction of the first interconnection structure, the device structure, and the second interconnection structure.

19. The method of claim **15**, wherein the capping layer is formed to completely cover the second contact by the rear side.

20. The method of claim **15**, wherein the capping layer is formed to partially expose the second contact exposed by the rear side.

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